Model Structure (26.1_1Reg):

This is a panmictic model (do_spatial == 0).
There are 2 sexes, 2 fishing fleets, 3 survey fleets, and 1 regions being modeled.
The model has 67 years, starting in 1960 and ending in 2026.
There are 30 ages, where recruitment occurs at age−2 and the plus group occurs at age−31.
There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model estimates new growth curves for the post−1996 time block (upd_growth == 1, use_new_bio == 2) and uses the 2021 SAFE growth estimates for the pre−1996 time block (use_hist_grwth_blk == 1, do_rec_growth == 0). A new weight curve is estimated with a single block (do_rec_WAA == 0).
For fishery composition data aggregation catch−weighting, the newly estimated sex−disaggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 1).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) −− no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2026, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations were NOT included in the LLS RPN index (drop_WAI_LLS == 1).

The longline survey used carryover values for off−year BS (even years) or AI (odd years) area RPNs in all index years (do_LLS_carry == 1).

The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

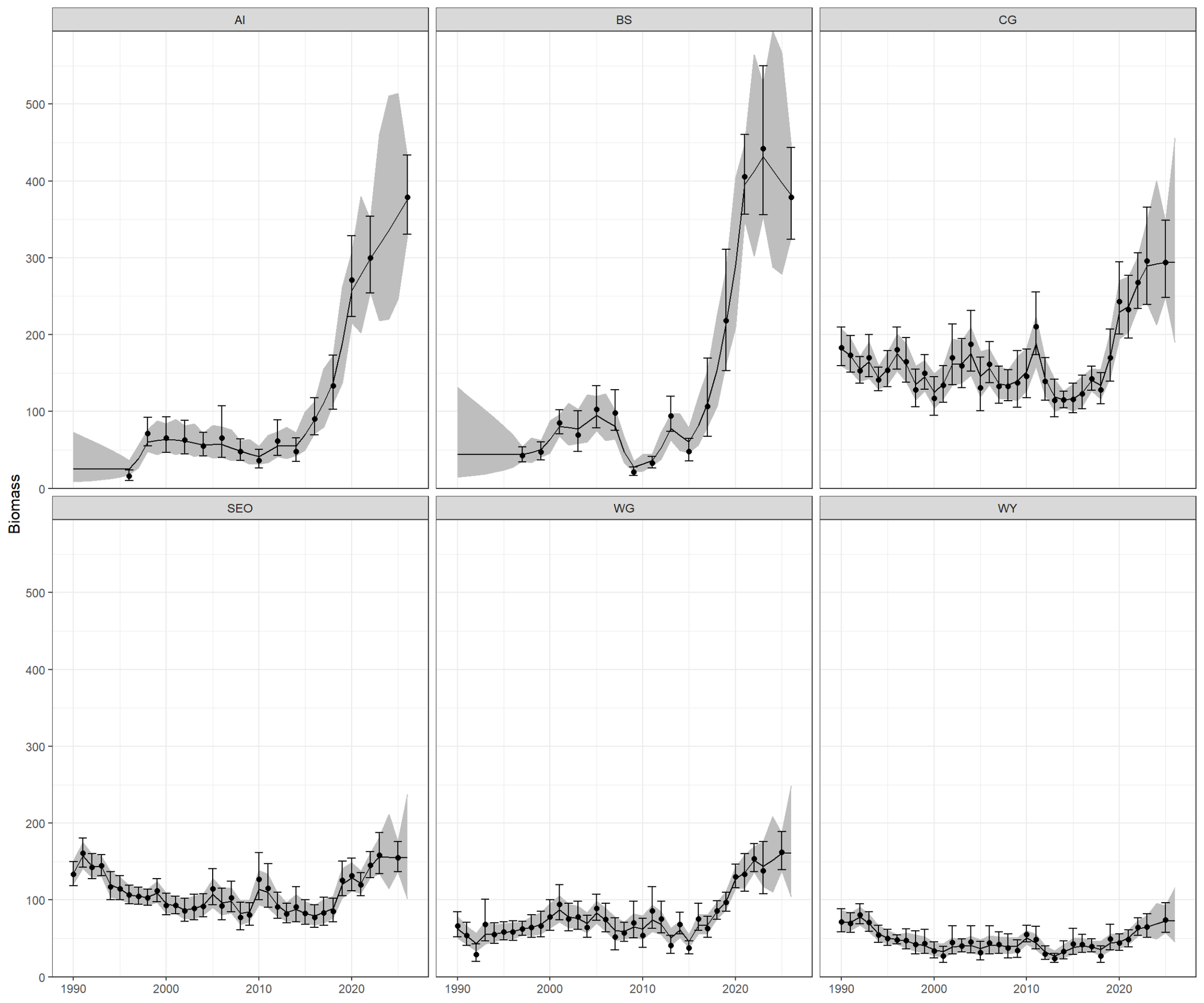
Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years) (do_spatial == 0 & sex_age_comps == 5).
The Input Sample Size (ISS) for composition data is scaled to a maximum value across years/sexes to address interannual variability (do_spatial == 0 & do_scale_ISS == 1).

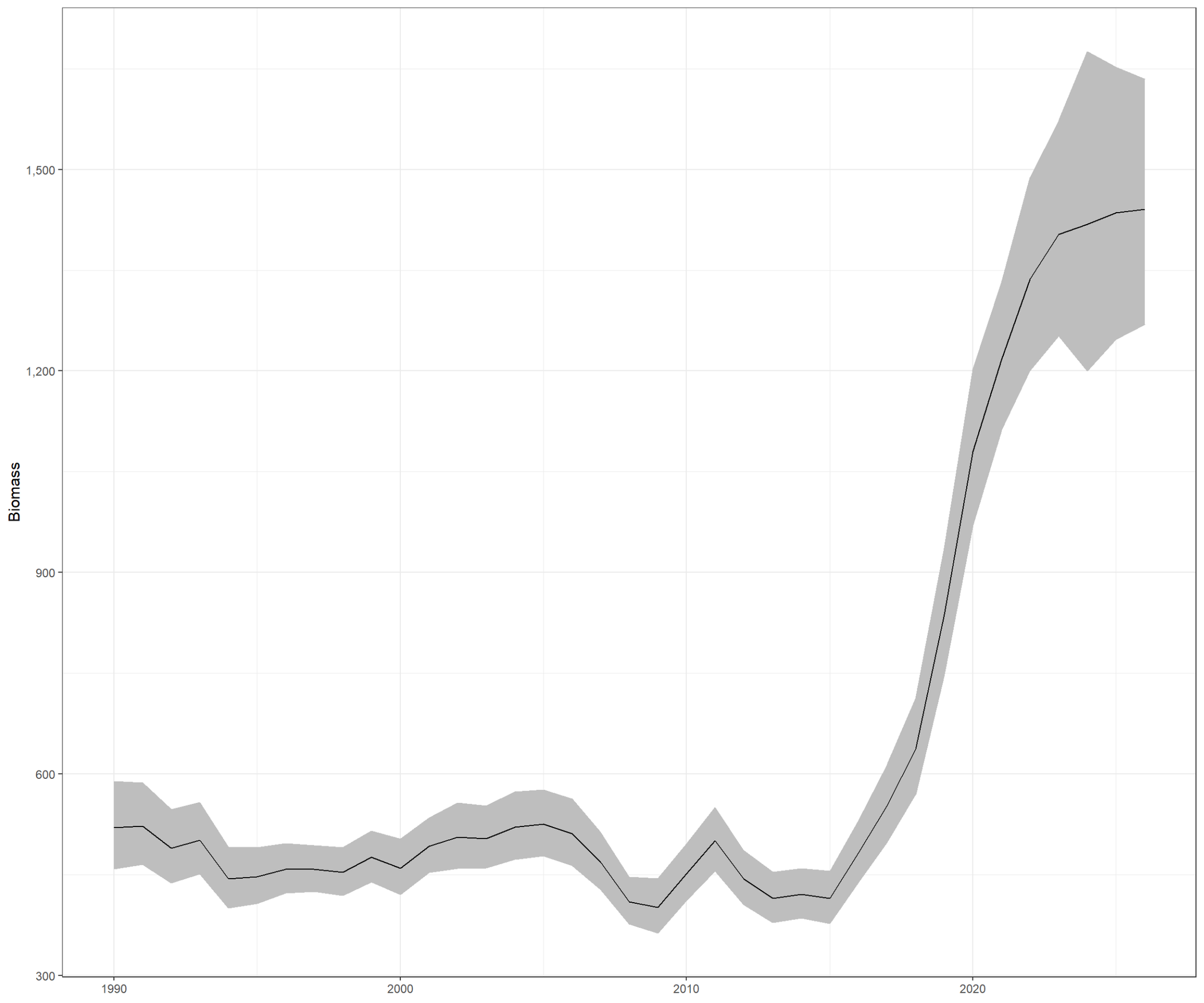
Length Composition Data Use:

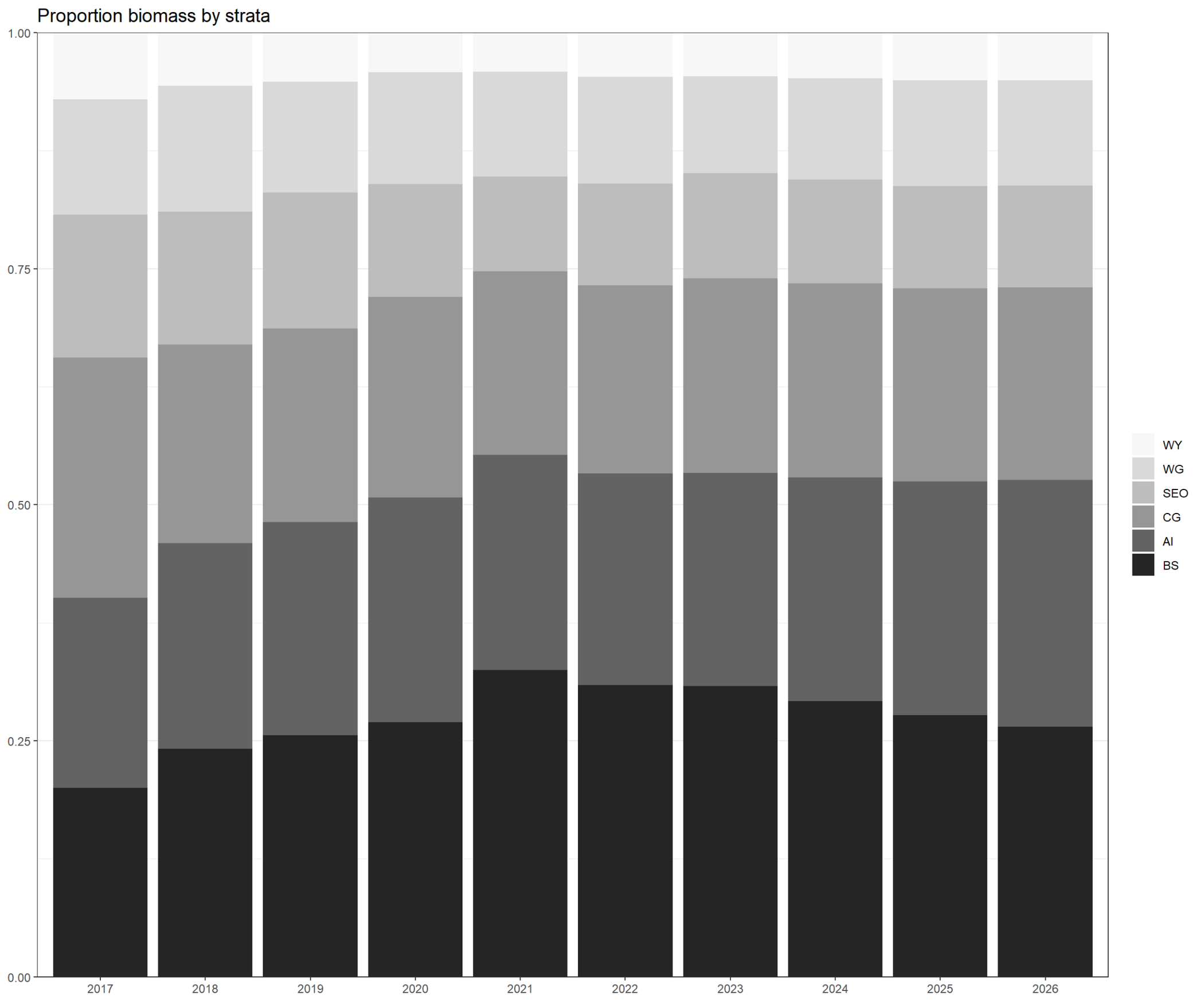
The longline survey length composition data is weighted/calculated using the raw length data and weighted by area RPNs consistent with the age composition data (LLS_length_comp_source == 1).
No longline survey length data are fit in the model (this does not apply to JPN LLS lengths) −− only age data are used (fit_LLS_len == 0).
No fixed gear fishery length data are fit in the model (this does not apply to trawl fishery lengths) −− only age data are used (fit_LLF_len == 0).

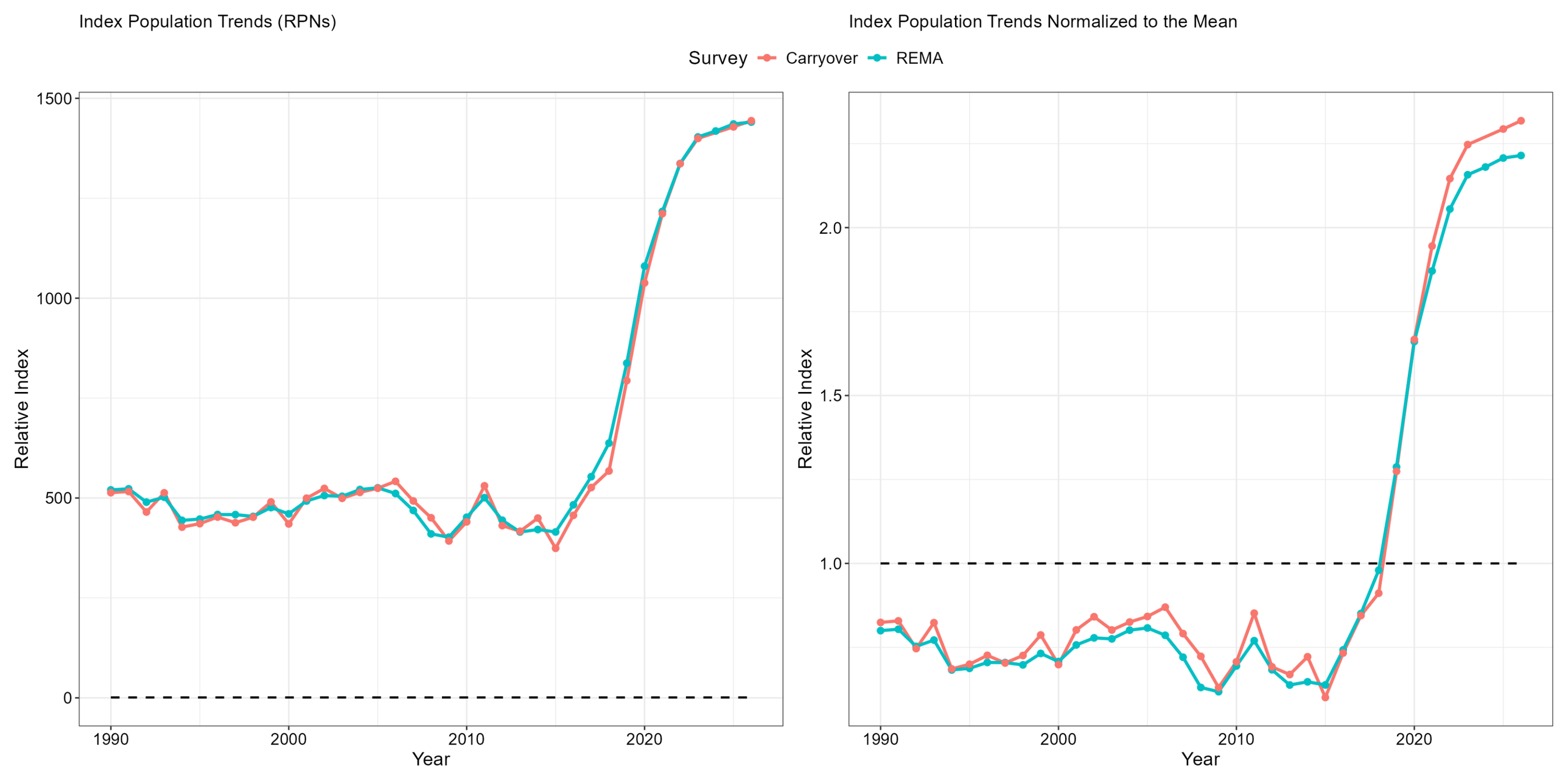
General Data Fitting Notes:

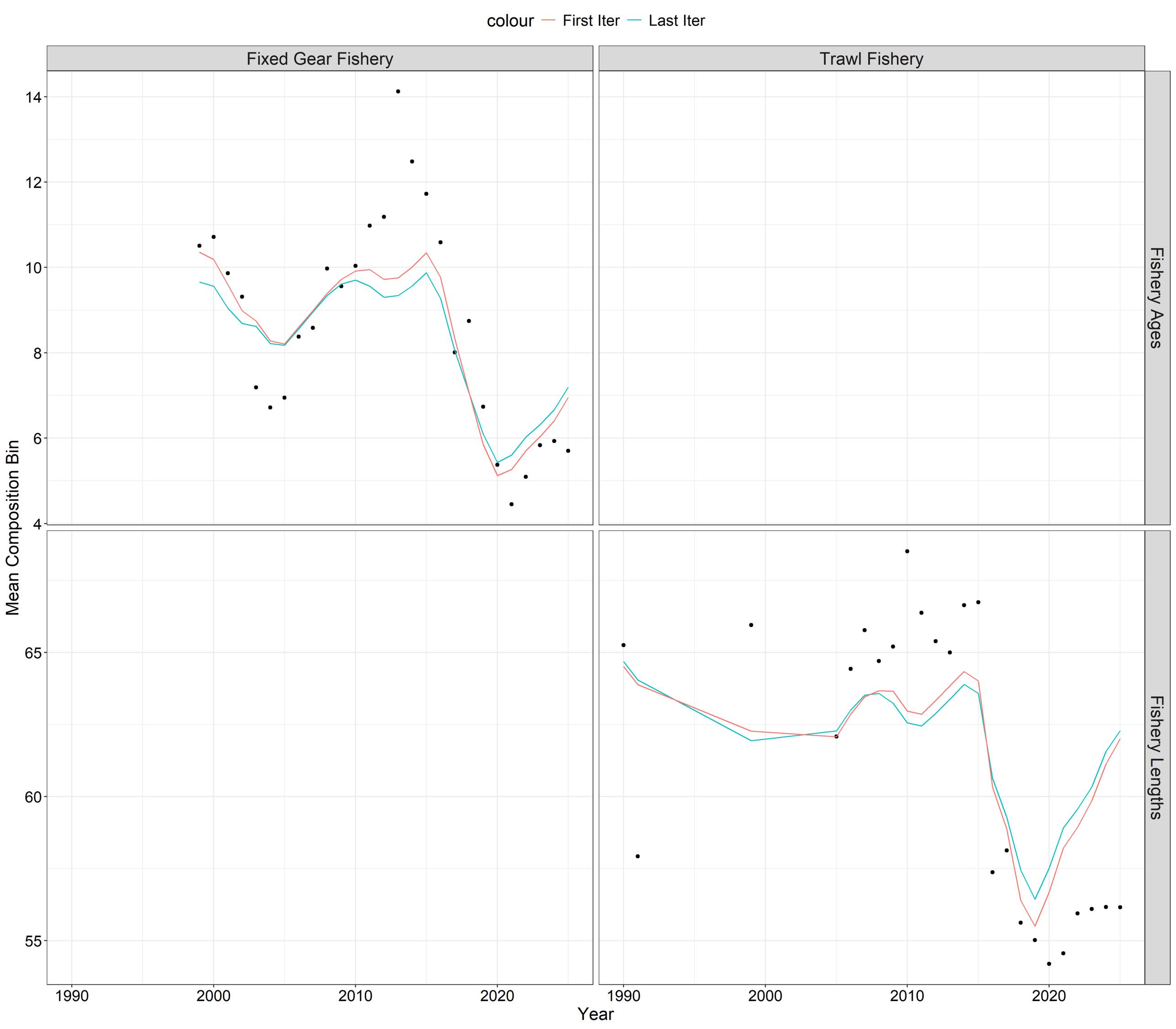
Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).
The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).
The fishery CPUE index was NOT fit in this model (use_CPUE == 0).
Non−fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where all sources of removals were available for terminal year (add_non_fish_catch == 1 & non_fsh_catch_updated == 1).

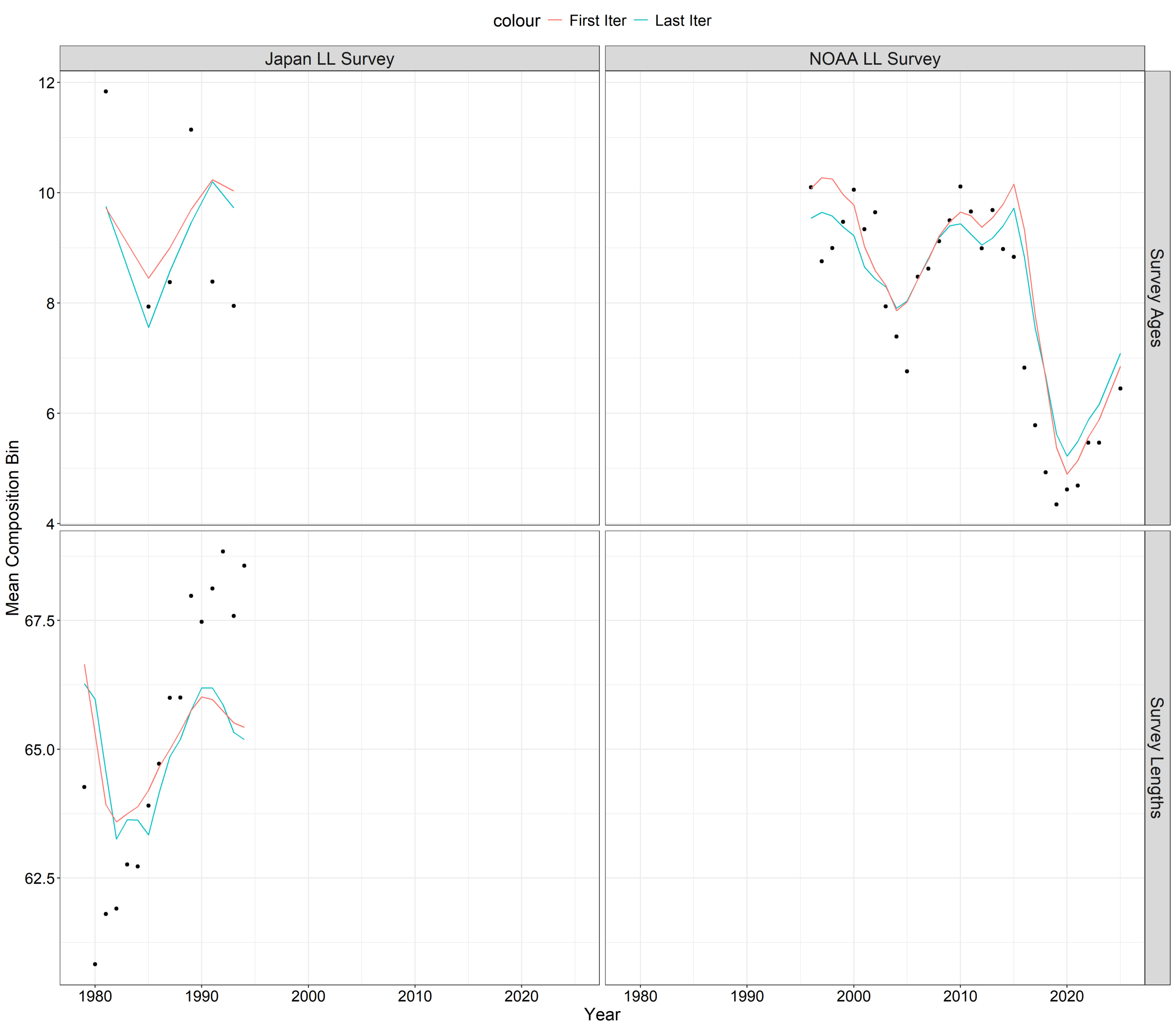

AI
BS
CG
SEO
WG
WY
Biomass
0
100
200
300
400
500
1990
2000
2010
2020

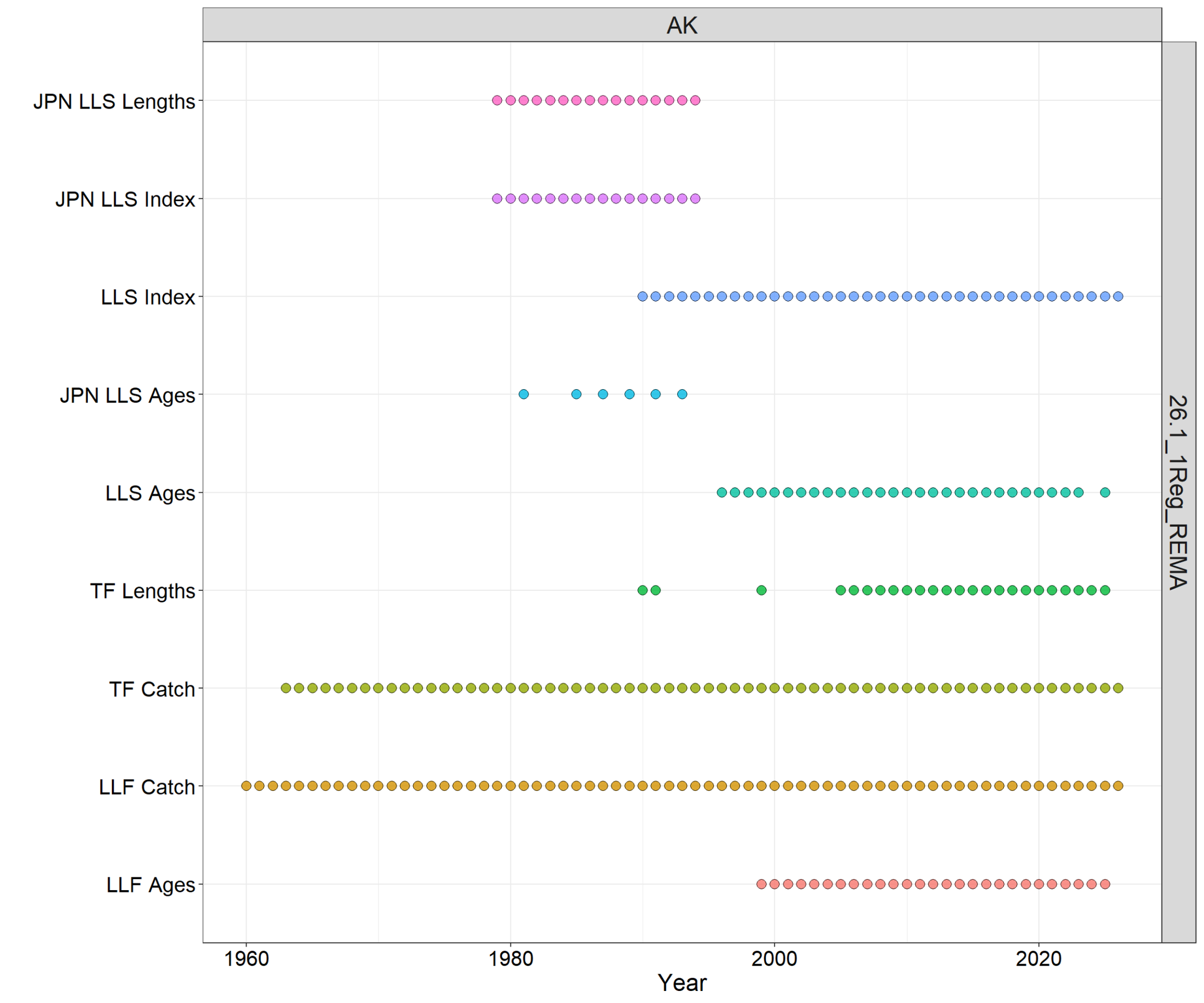

Biomass
1,500
1,200
900
600
300
1990
2000
2010
2020

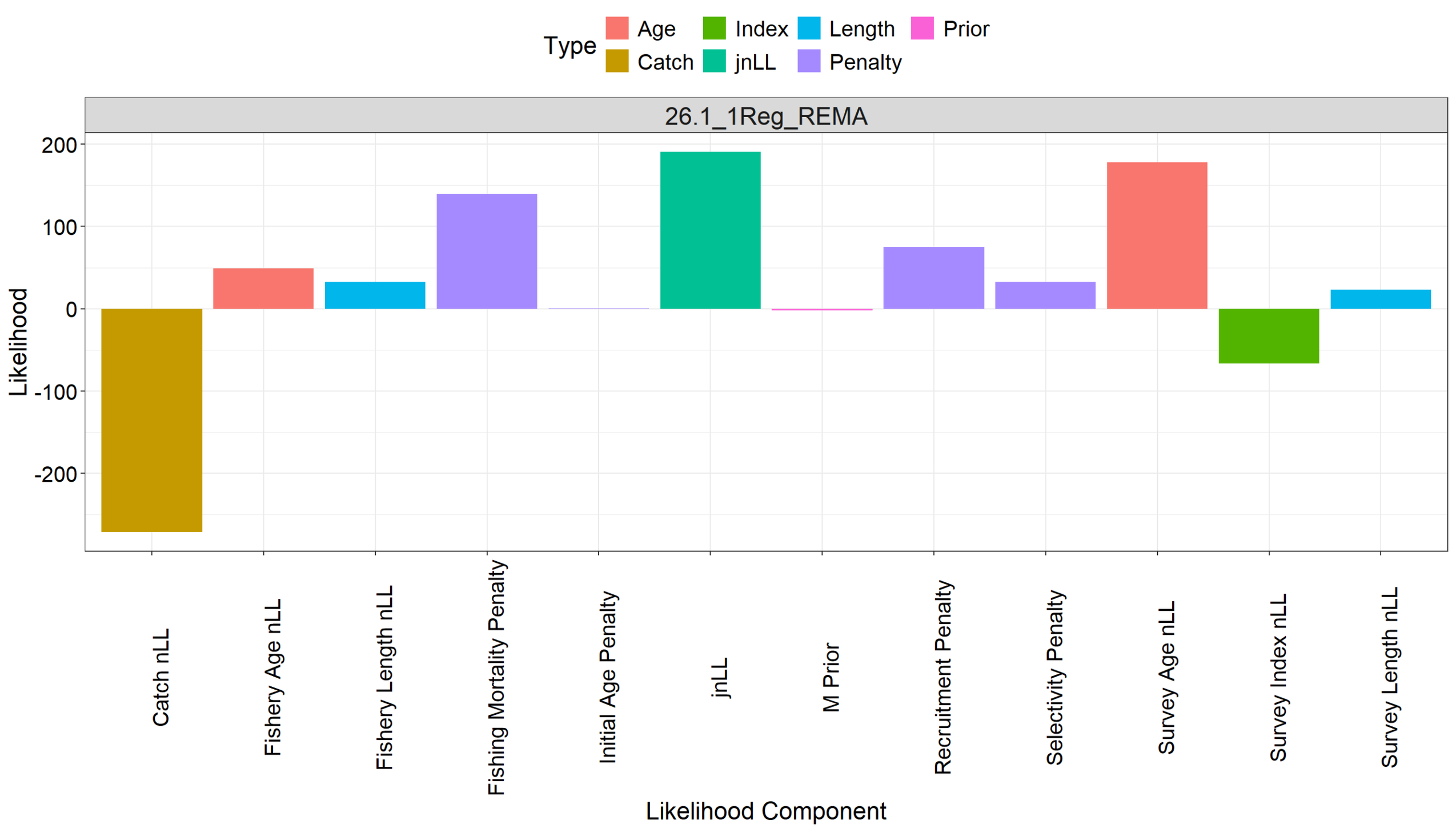

Proportion biomass by strata
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
WY
WG
SEO
CG
AI
BS


Index Population Trends (RPNs)
Index Population Trends Normalized to the Mean
Survey
Carryover
REMA
Relative Index
Year
1500
1000
500
0
2.0
1.5
1.0
1990
2000
2010
2020


colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year


colour
First Iter
Last Iter
Japan LL Survey
NOAA LL Survey
Survey Ages
Survey Lengths
Mean Composition Bin
12
10
8
6
4
67.5
65.0
62.5
1980
1990
2000
2010
2020
Year


AK
JPN LLS Lengths
JPN LLS Index
LLS Index
JPN LLS Ages
LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages
26.1_1Reg_REMA
1960
1980
2000
2020
Year


Type
Age
Index
Length
Prior
Catch
jnLL
Penalty
26.1_1Reg_REMA
200
100
0
-100
-200
Likelihood
Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL
Likelihood Component

Numbers-at-Age

Sex — Female ····· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

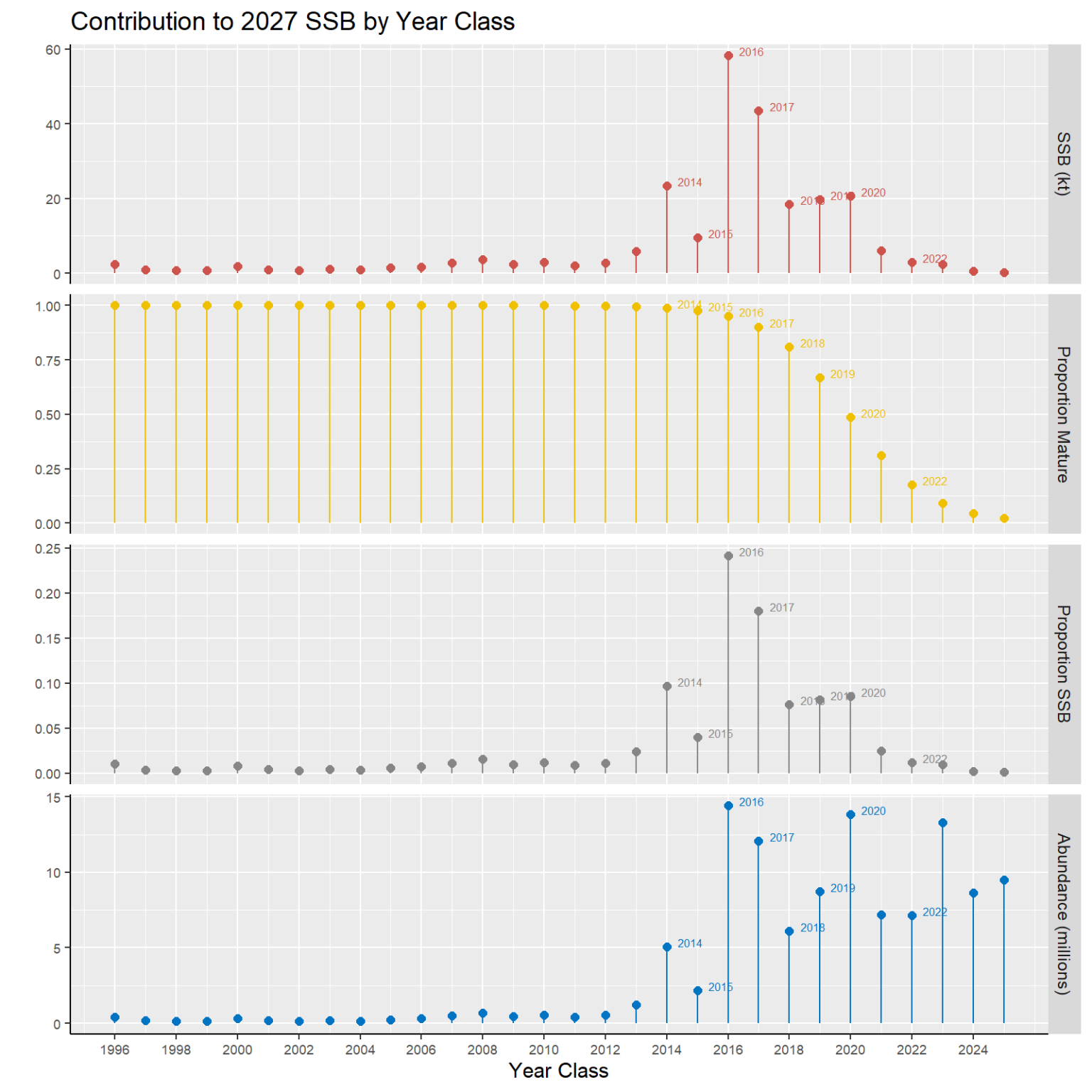
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

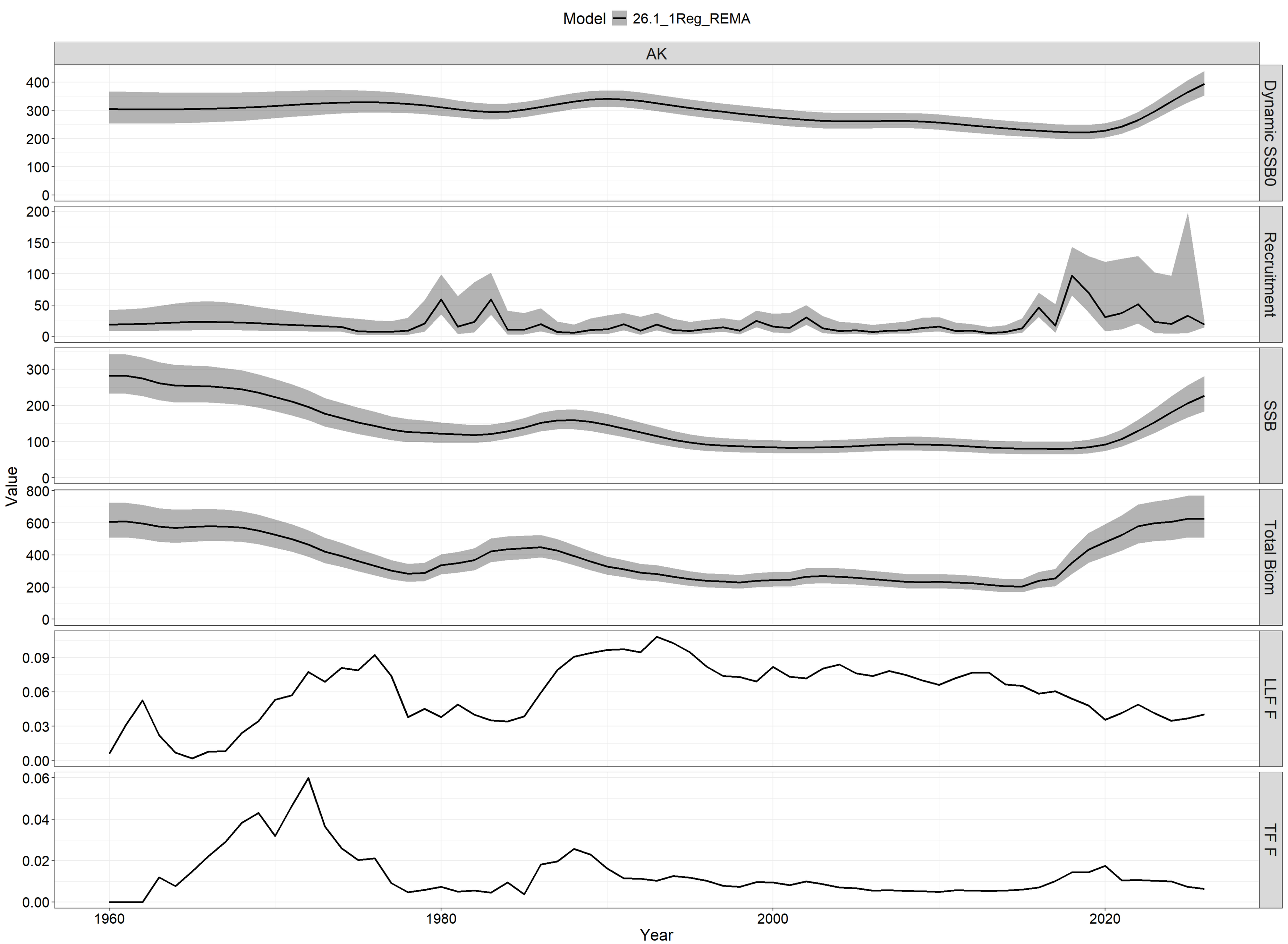
Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

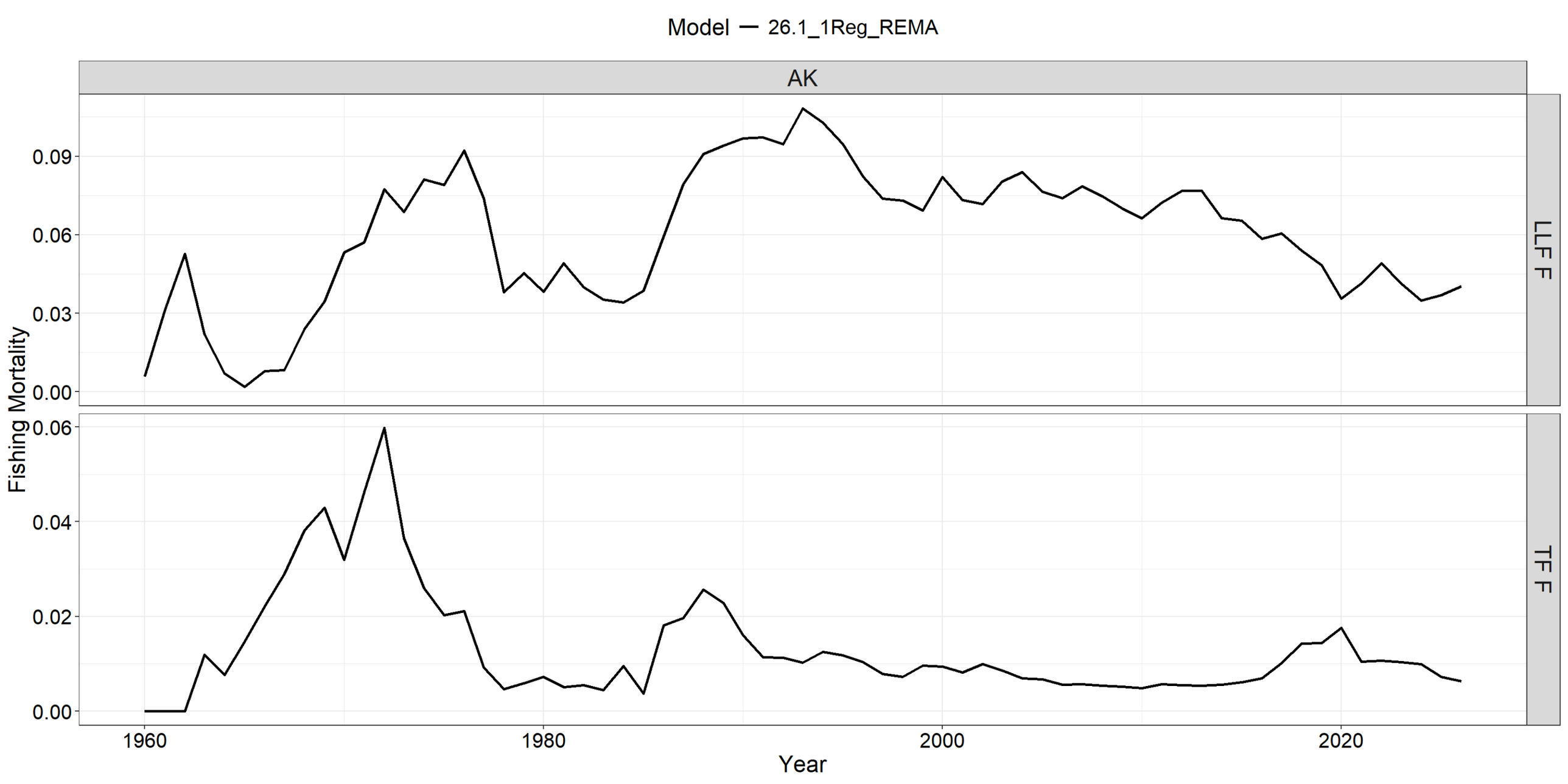
Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

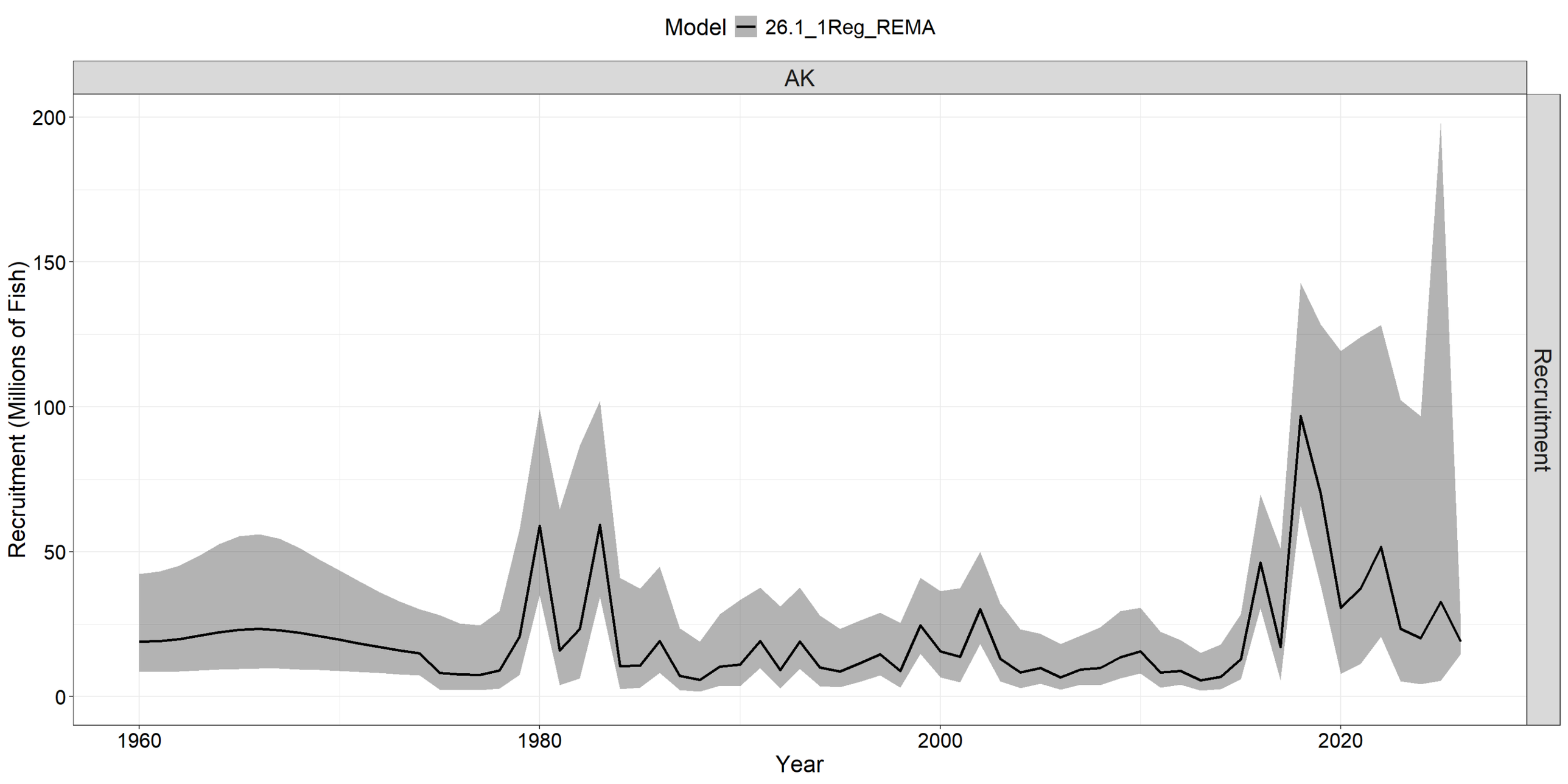
Numbers-at-Age (millions)

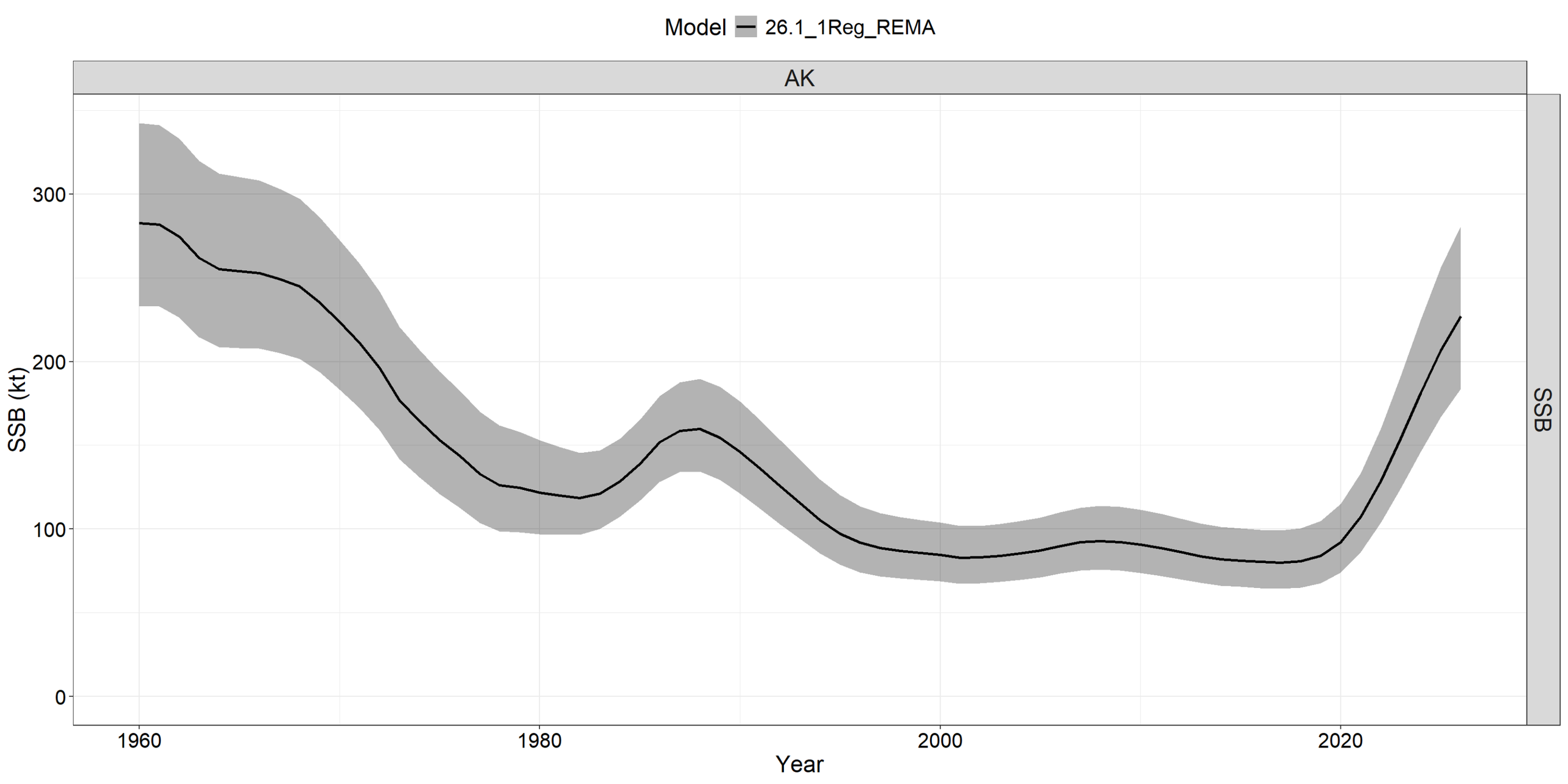
Year

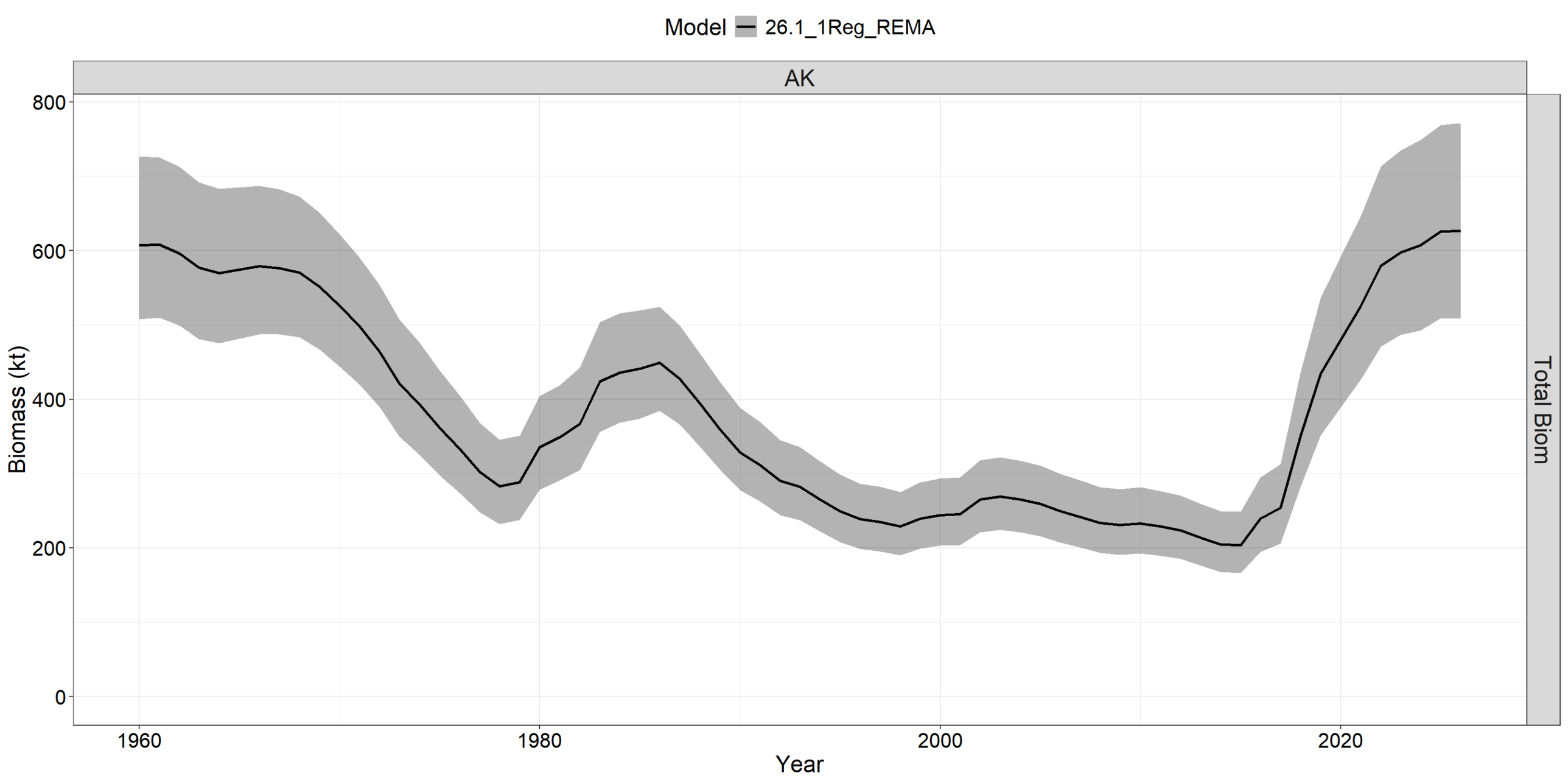

Contribution to 2027 SSB by Year Class
SSB (kt)
Proportion Mature
Proportion SSB
Abundance (millions)
Year Class
2016
2017
2014
2015
2018
2019
2020
2022

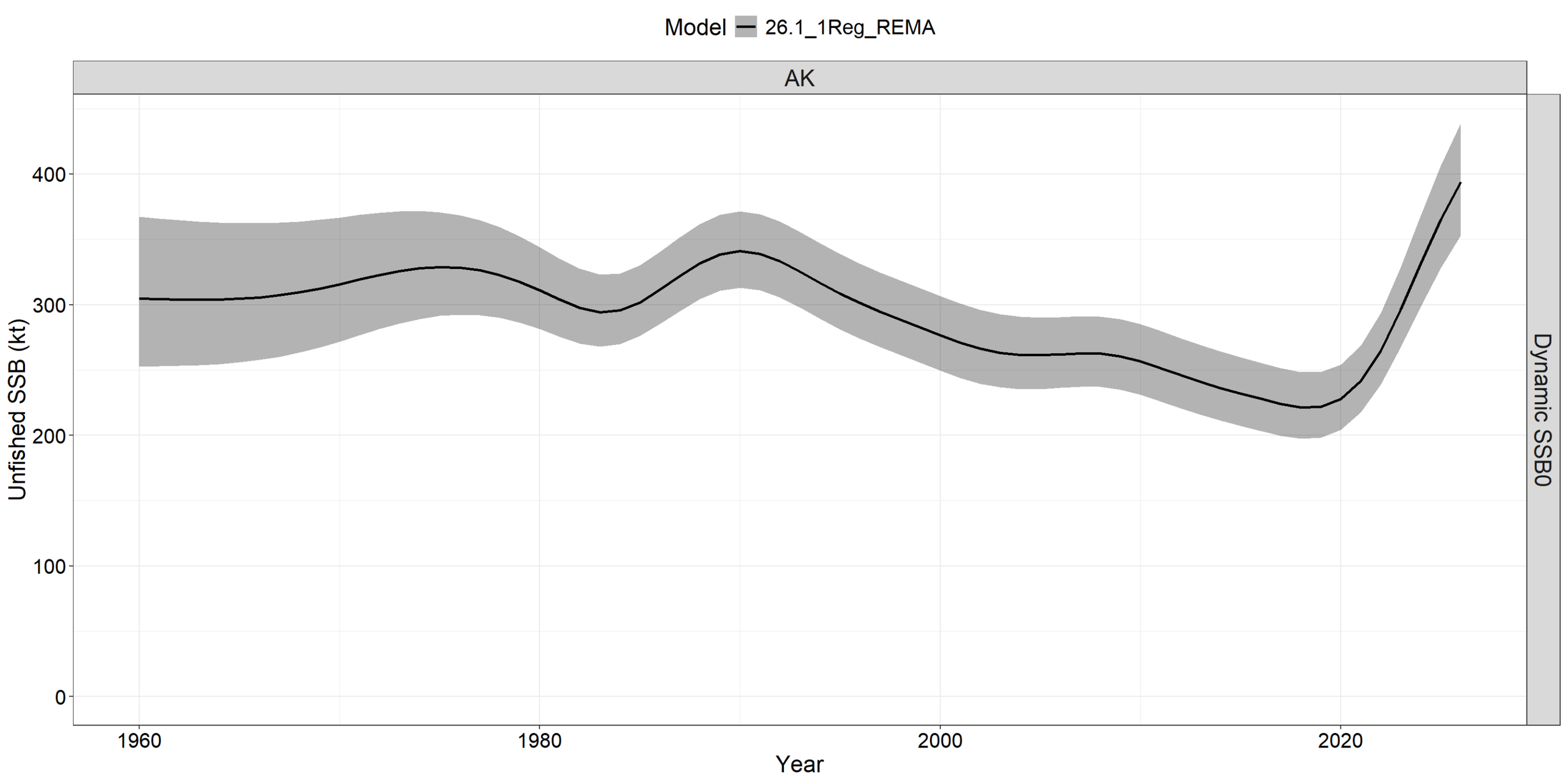

Model 26.1_1Reg_REMA
AK
Dynamic SSB0
Recruitment
SSB
Total Biom
LLF F
TF F
Value
Year
0
100
200
300
400
50
150
0.00
0.03
0.06
0.09
0.02
0.04
600
800
1960
1980
2000
2020

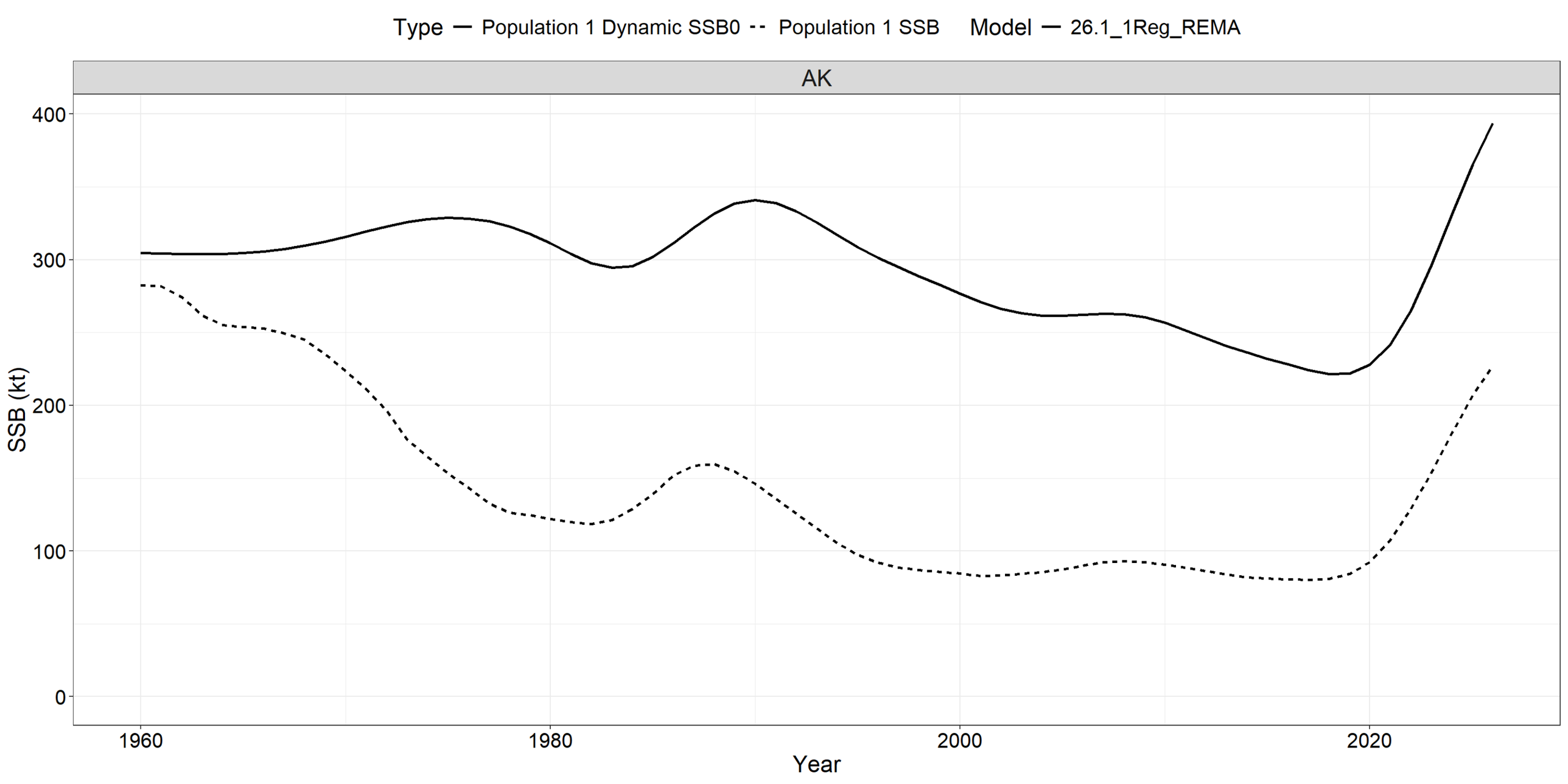

Model — 26.1_1Reg_REMA
AK
LLF F
TF F
Fishing Mortality
0.00
0.03
0.06
0.09
0.00
0.02
0.04
0.06
1960
1980
2000
2020
Year

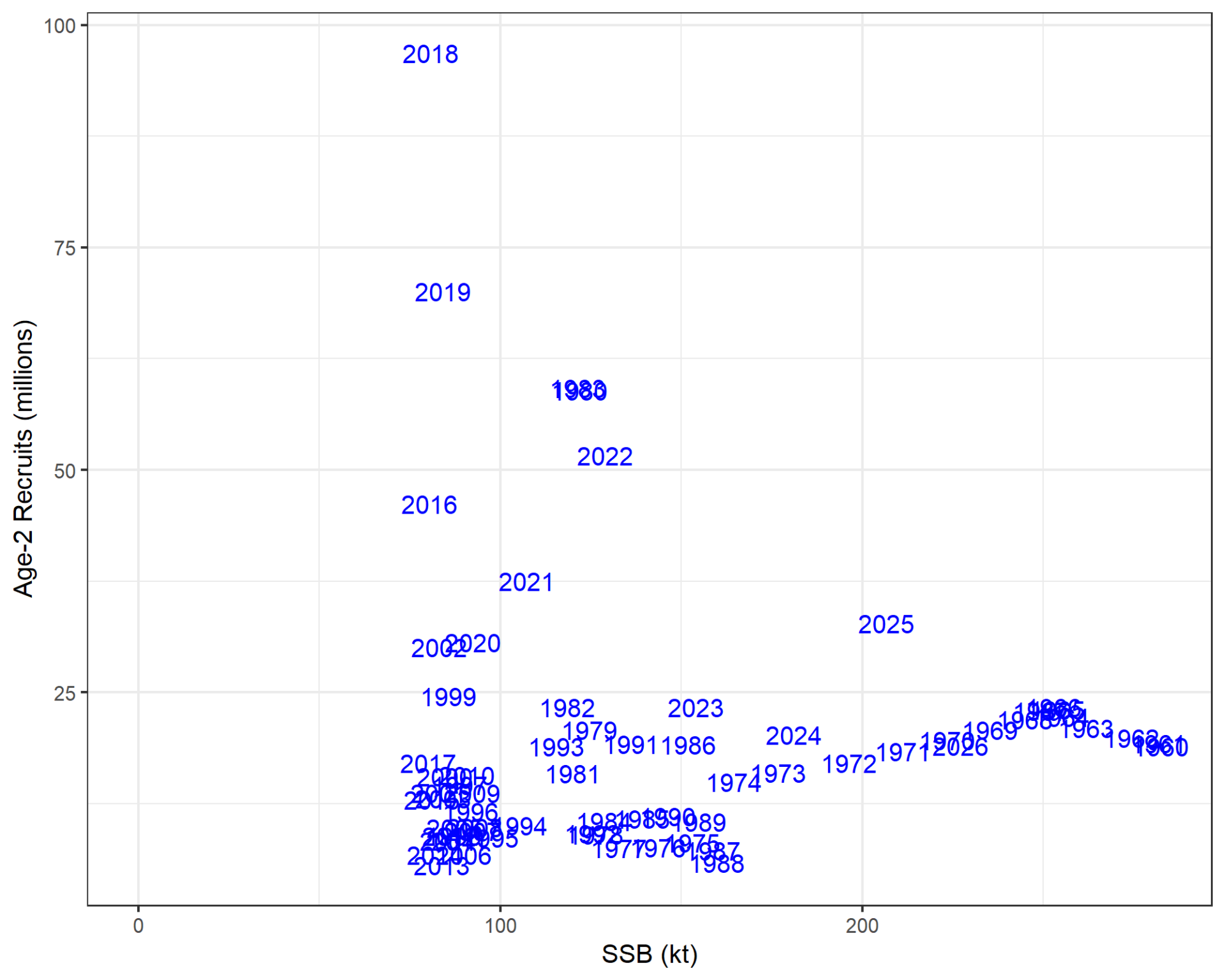

Model
26.1_1Reg_REMA
AK
Recruitment
Recruitment (Millions of Fish)
200
150
100
50
0
1960
1980
2000
2020
Year

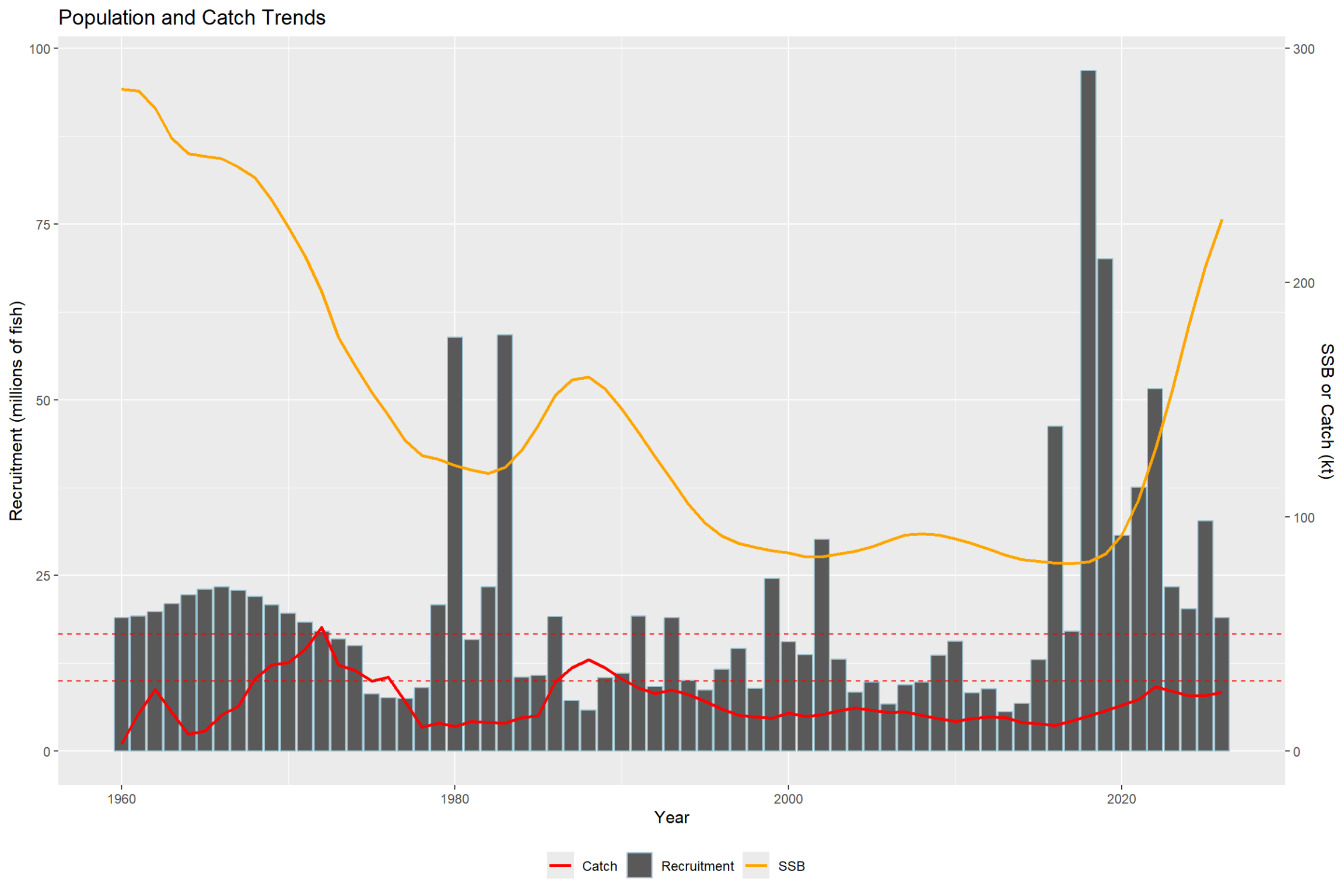

Model
26.1_1Reg_REMA
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year

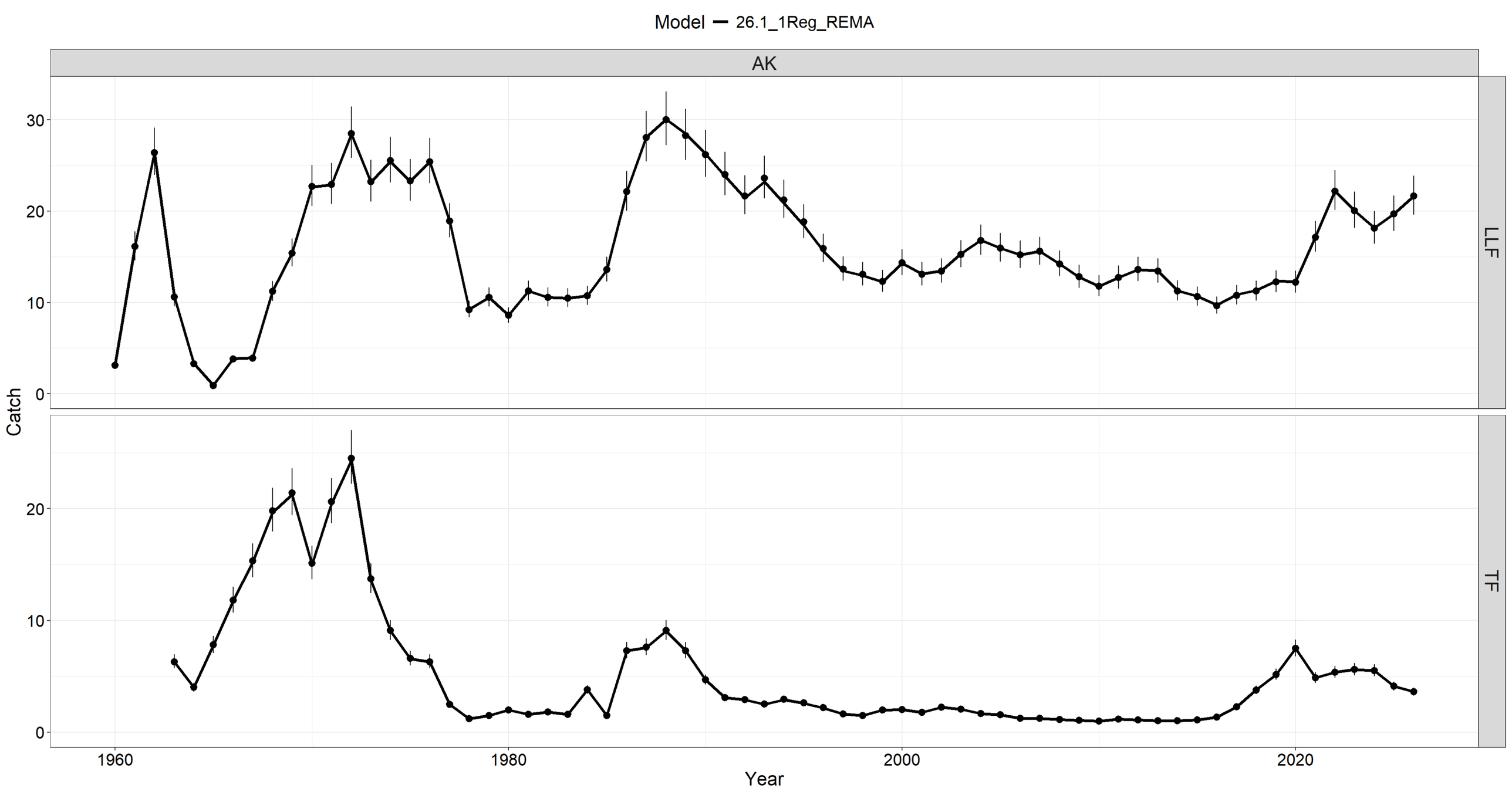

Model
26.1_1Reg_REMA
AK
Total Biom
Biomass (kt)
800
600
400
200
0
1960
1980
2000
2020
Year

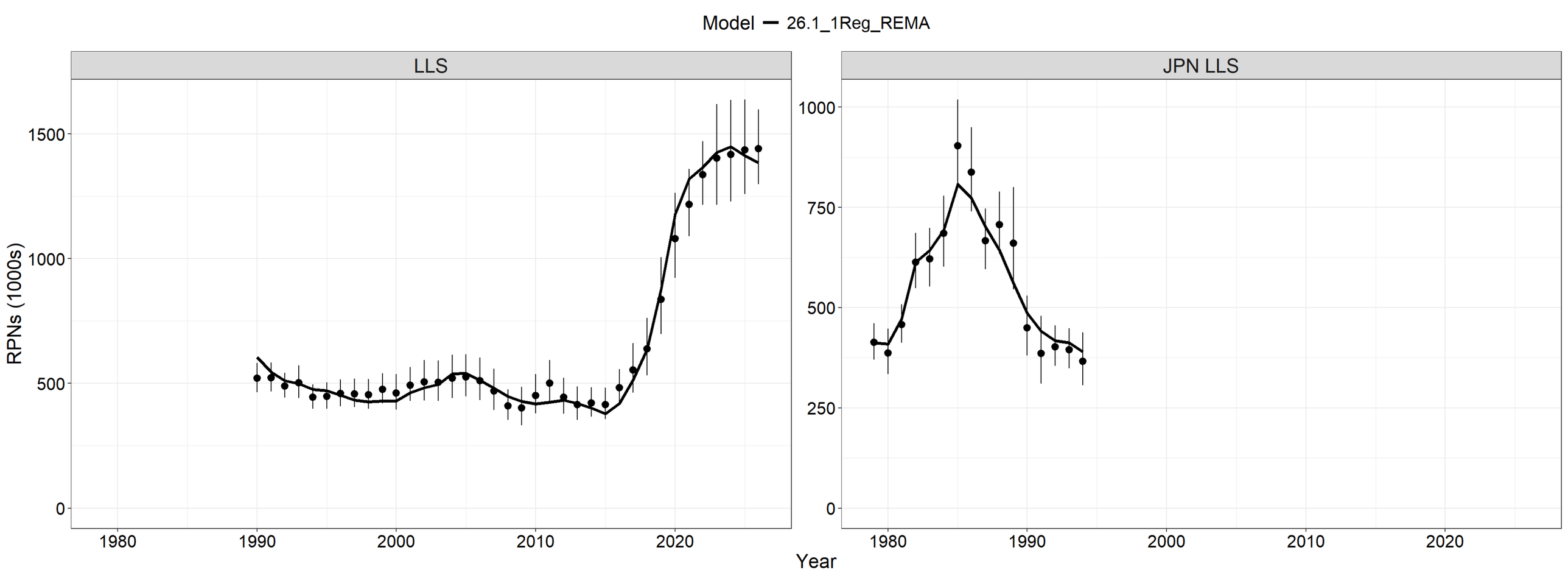

Model
26.1_1Reg_REMA
AK
Dynamic SSB0
Unfished SSB (kt)
0
100
200
300
400
1960
1980
2000
2020
Year

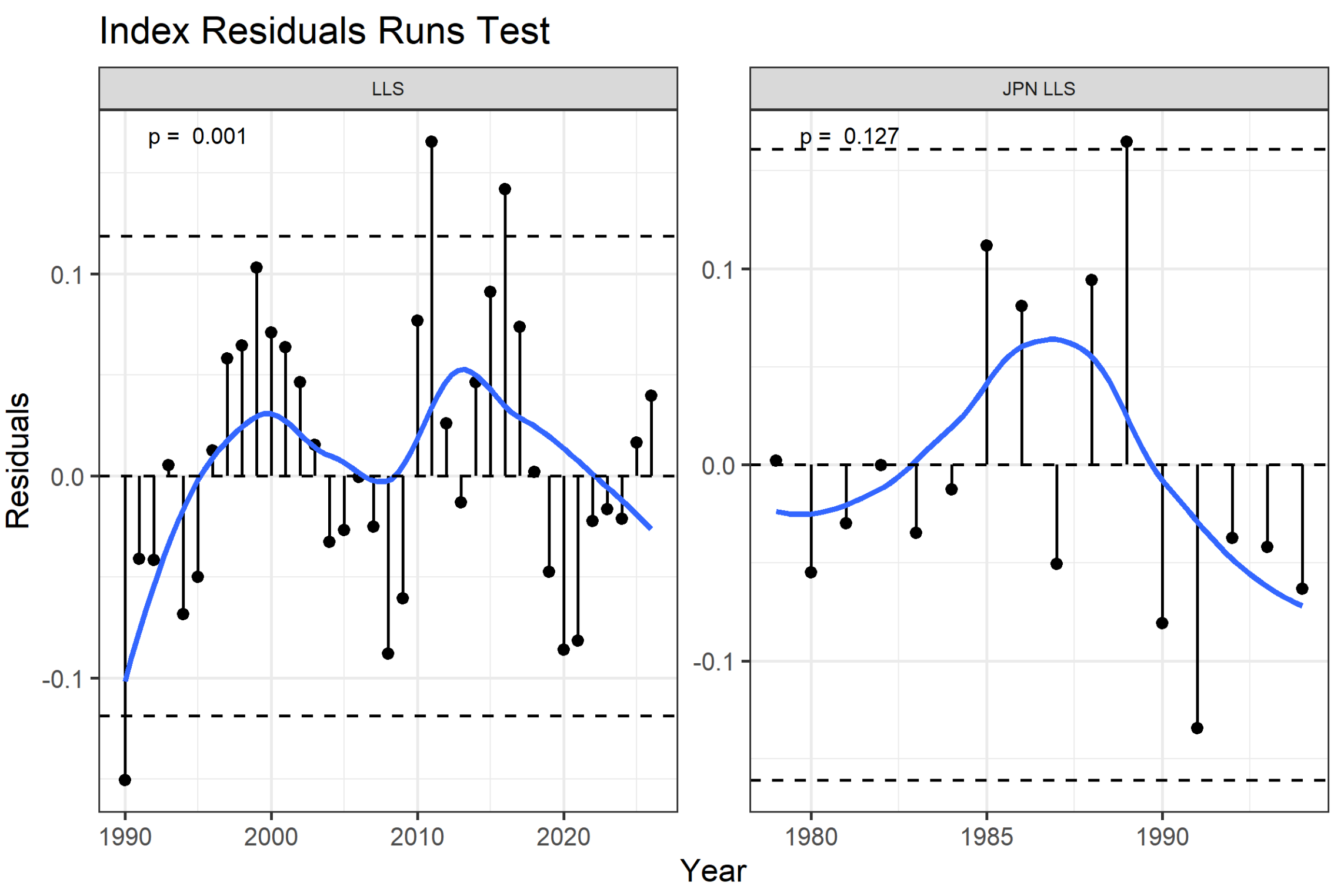

Type — Population 1 Dynamic SSB0 -- Population 1 SSB
Model — 26.1_1Reg_REMA
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year

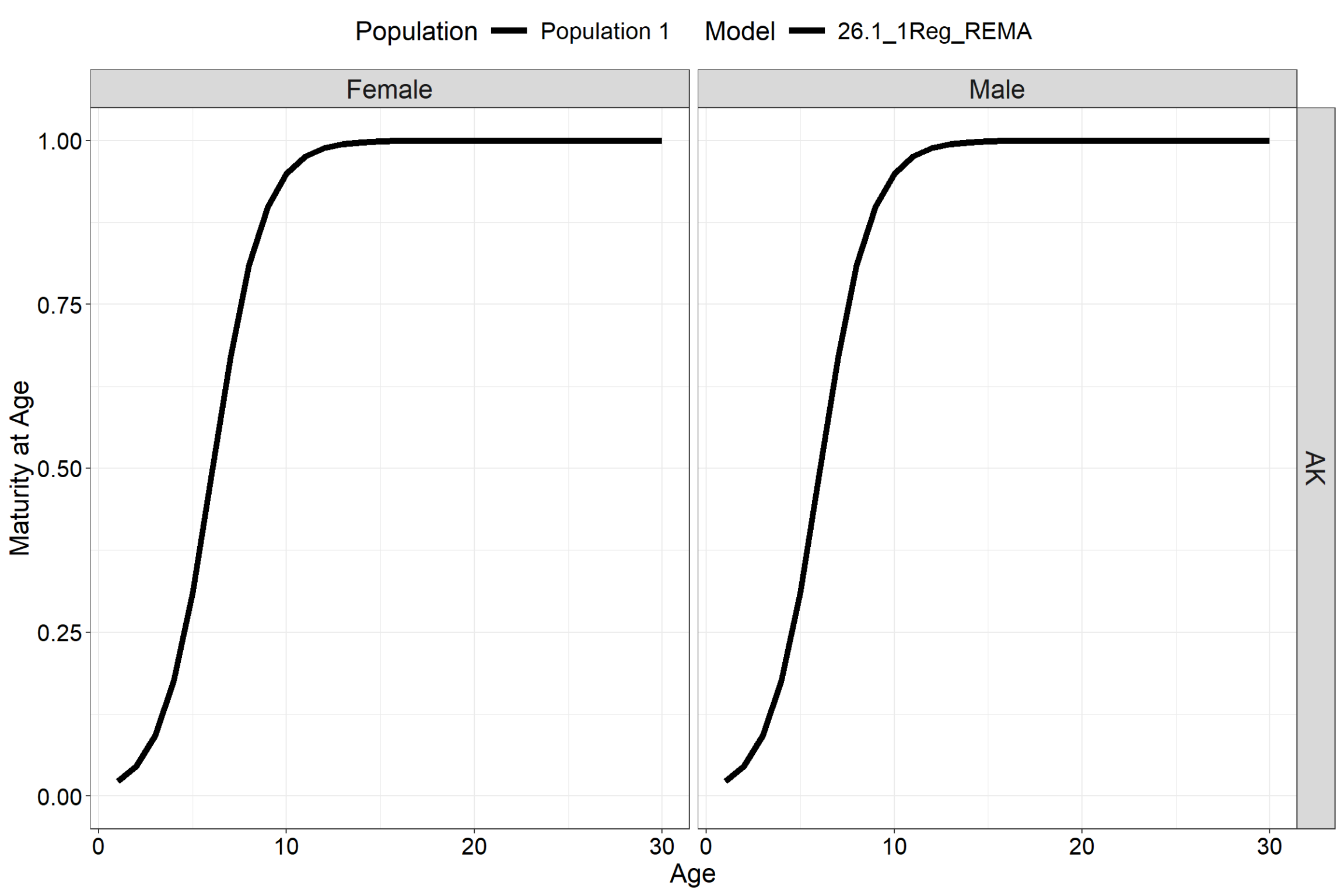

Age-2 Recruits (millions)
100
75
50
25
0
100
200
SSB (kt)
2018
2019
2022
2016
2021
2025
2020
1999
1982
2023
1979
1993
1991
1986
2024
1972
1971
1973
1974
1981
1989
1988
2013

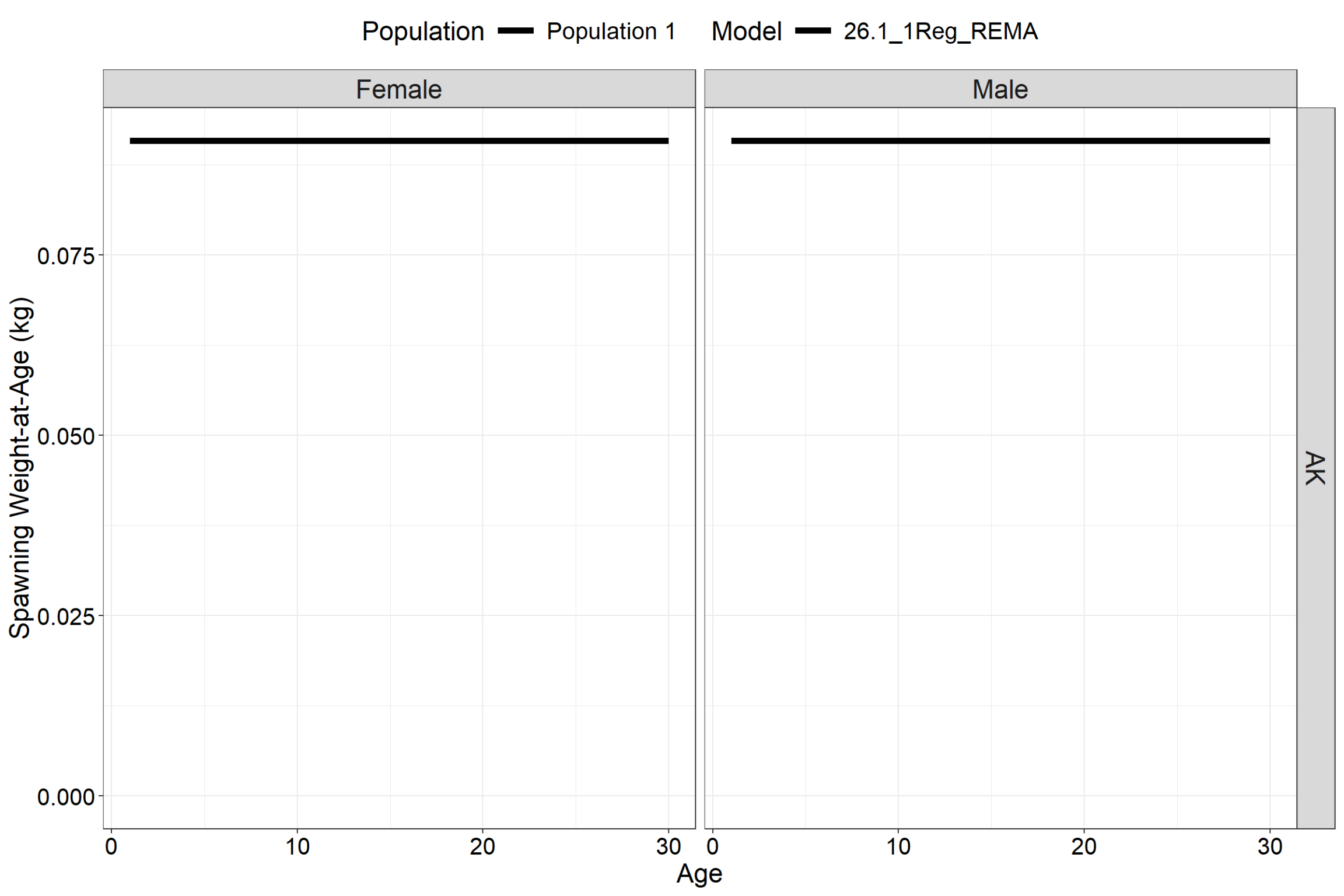

Population and Catch Trends
Recruitment (millions of fish)
SSB or Catch (kt)
Year
100
75
50
25
0
300
200
100
0
1960
1980
2000
2020
Catch
Recruitment
SSB

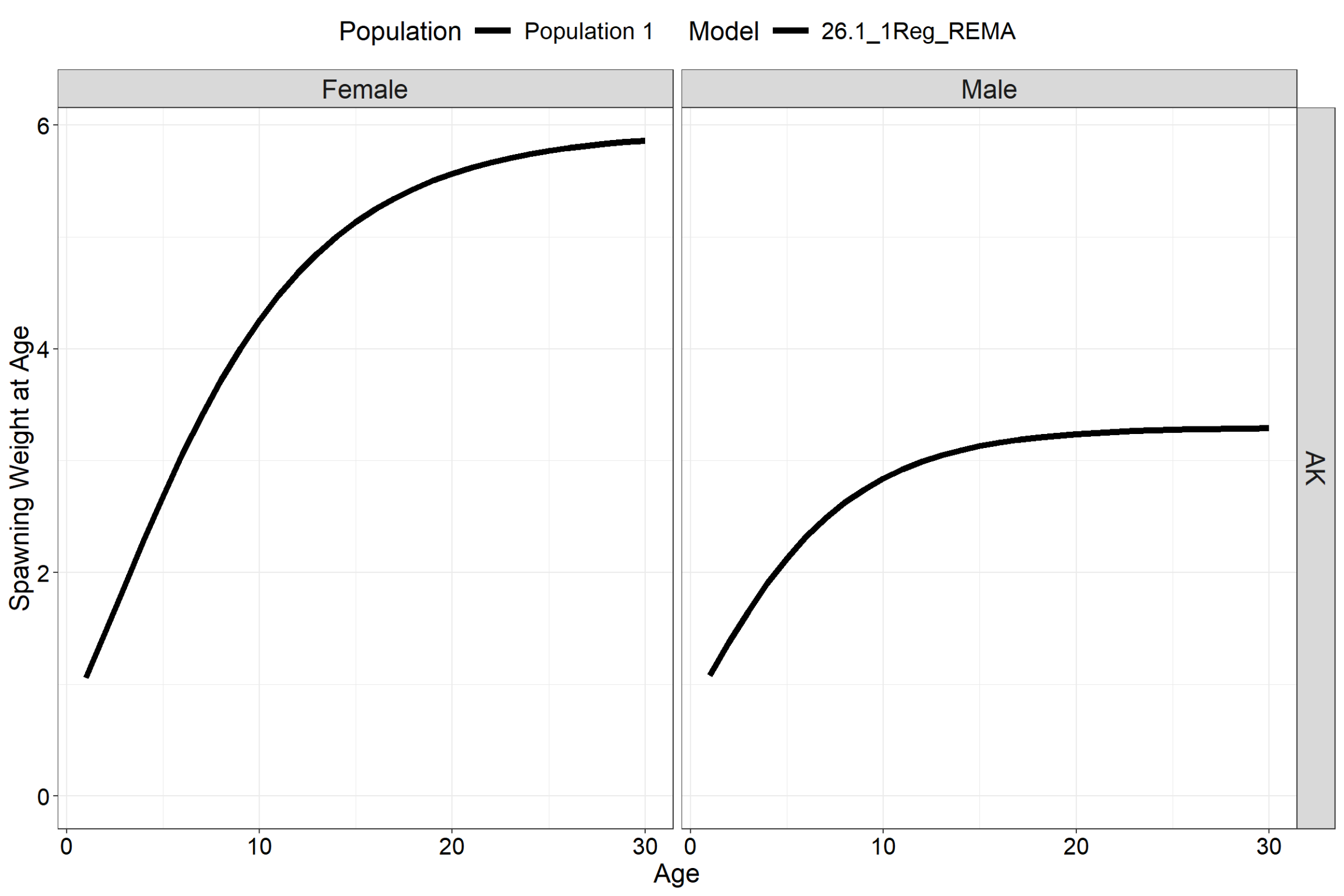

Model — 26.1_1Reg_REMA
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

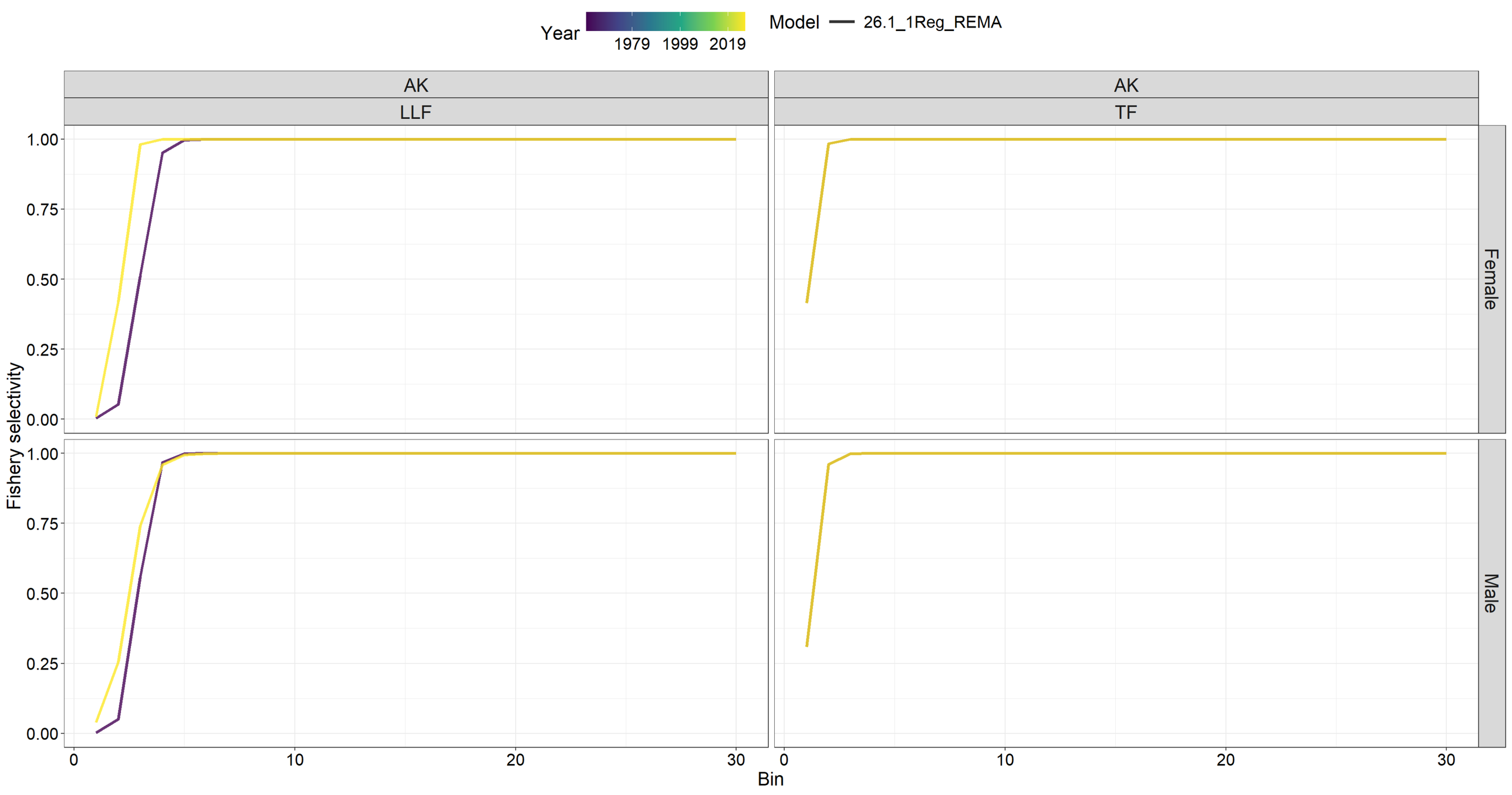

Model — 26.1_1Reg_REMA
LLS
JPN LLS
RPNs (1000s)
Year
0
500
1000
1500
250
750
1980
1990
2000
2010
2020

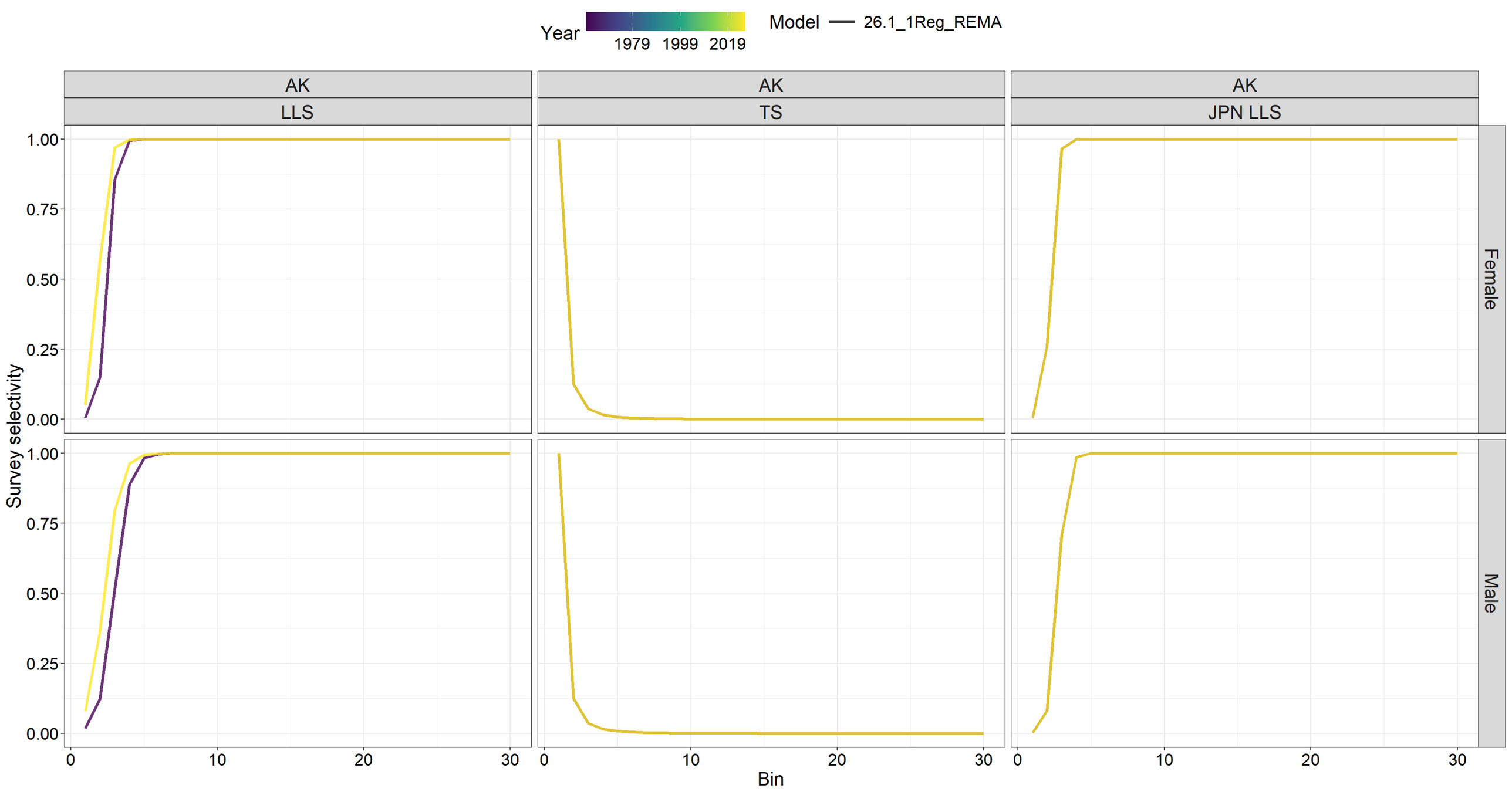

Index Residuals Runs Test
LLS
p = 0.001
JPN LLS
p = 0.127
Residuals
0.1
0.0
-0.1
1990
2000
2010
2020
1980
1985
1990
Year

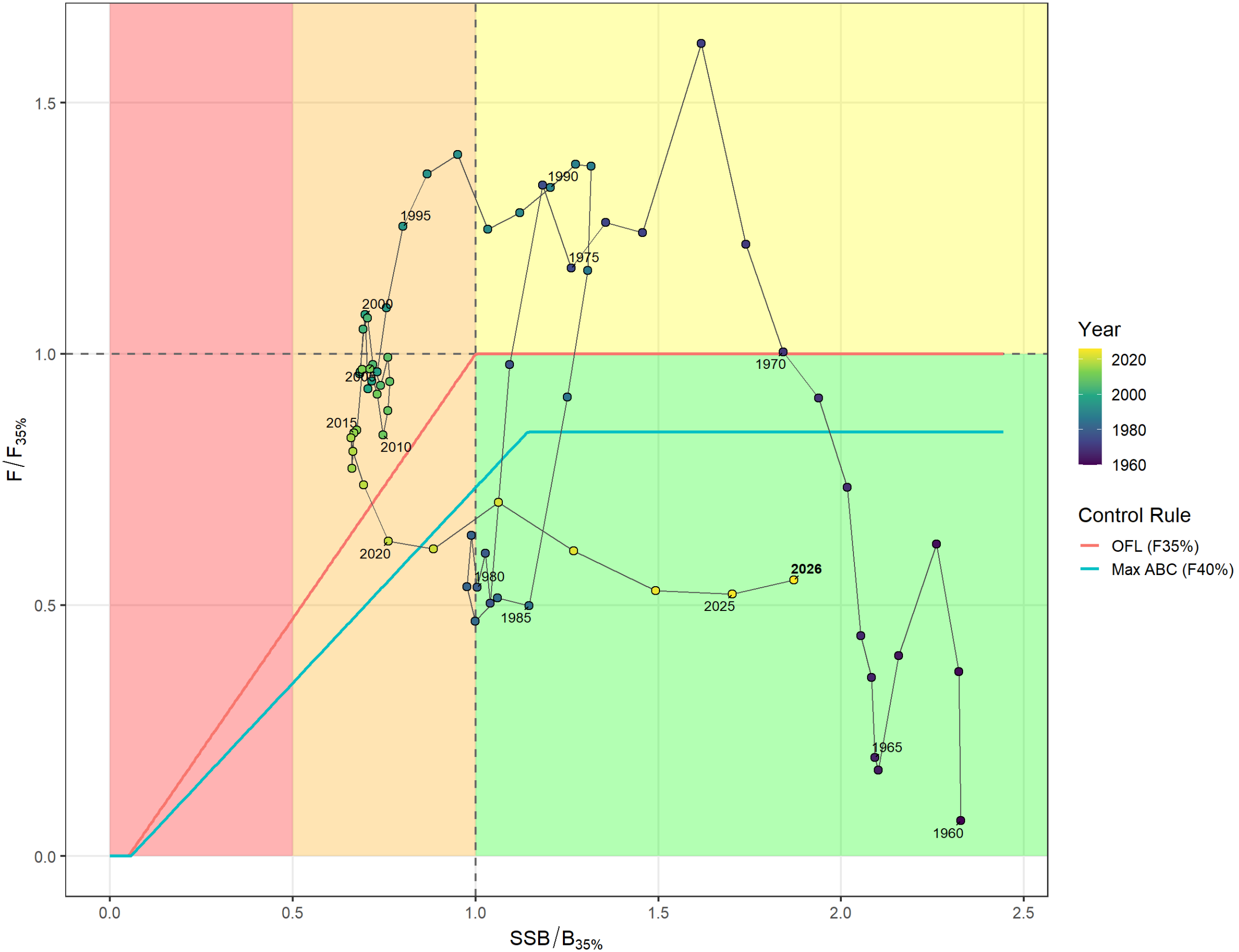

Population
Population 1
Model
26.1_1Reg_REMA
Female
Male
AK
Maturity at Age
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Age


Population
Population 1
Model
26.1_1Reg_REMA
Female
Male
AK
Spawning Weight-at-Age (kg)
0.075
0.050
0.025
0.000
0
10
20
30
Age

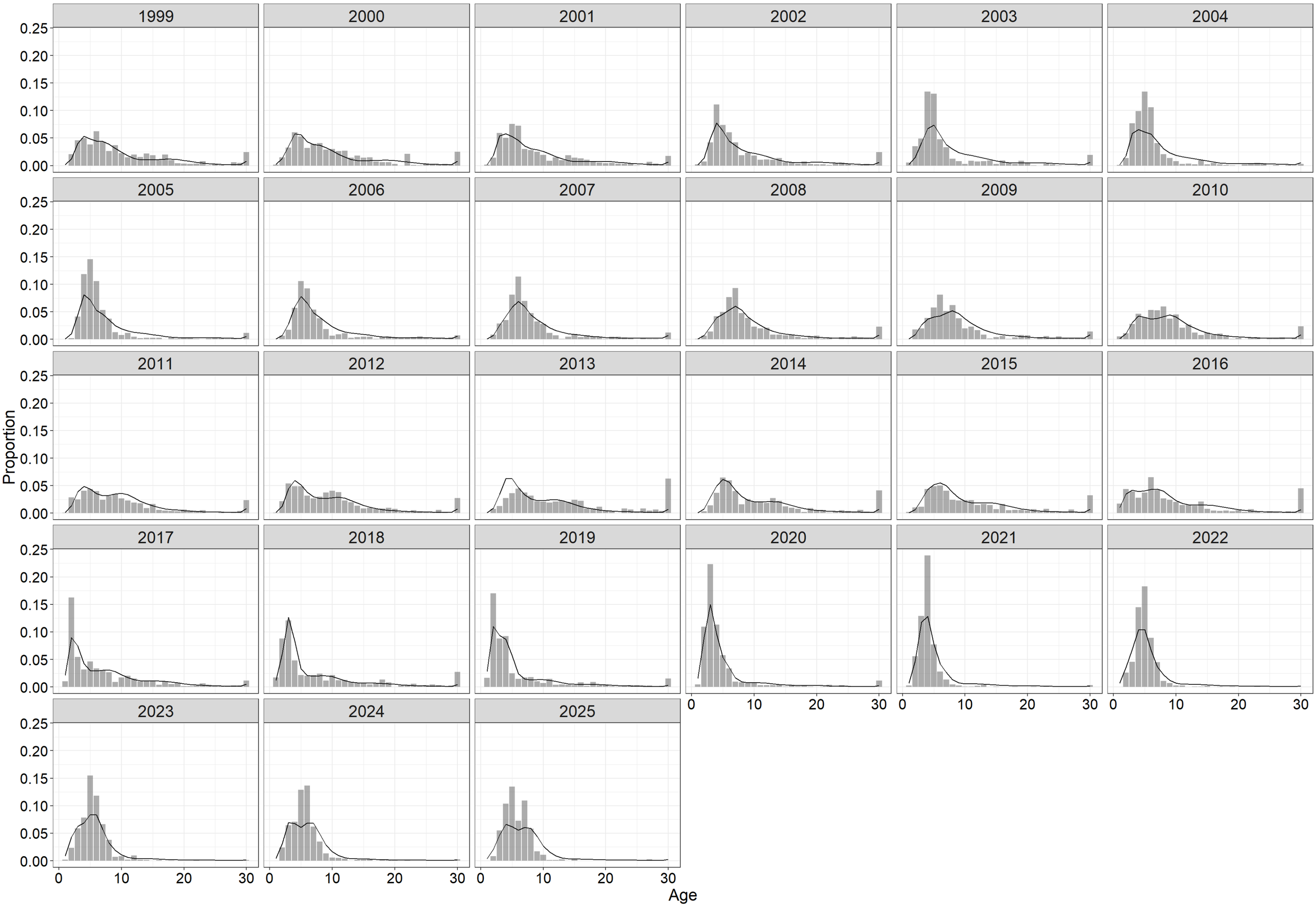

Population
Population 1
Model
26.1_1Reg_REMA
Female
Male
AK
Spawning Weight at Age
Age
0
2
4
6
0
10
20
30


Year
1979
1999
2019
Model
26.1_1Reg_REMA
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin

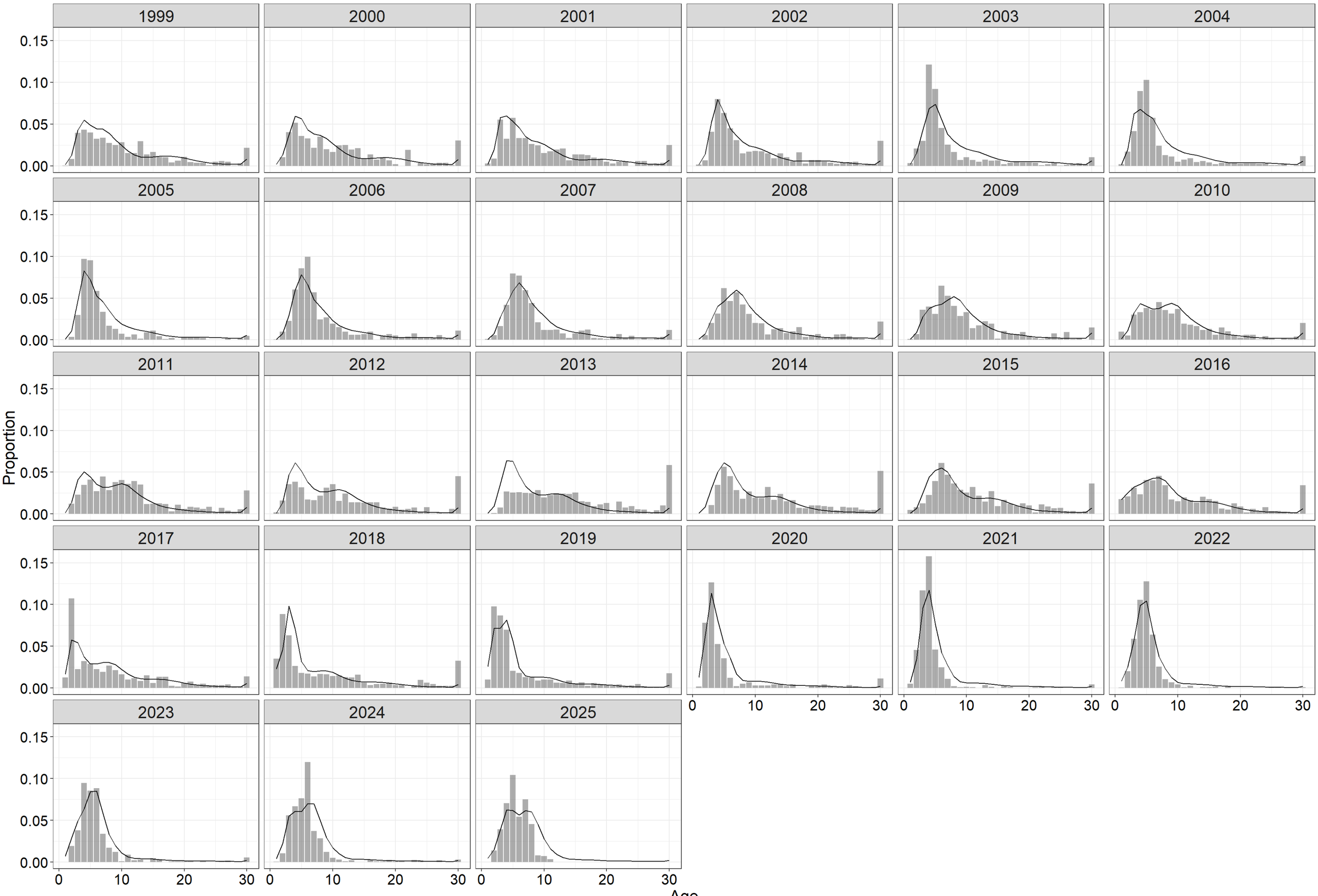

Year
1979
1999
2019
Model
26.1_1Reg_REMA
AK
LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


F/F35%
SSB/B35%
0.0
0.5
1.0
1.5
2.0
2.5
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2026
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

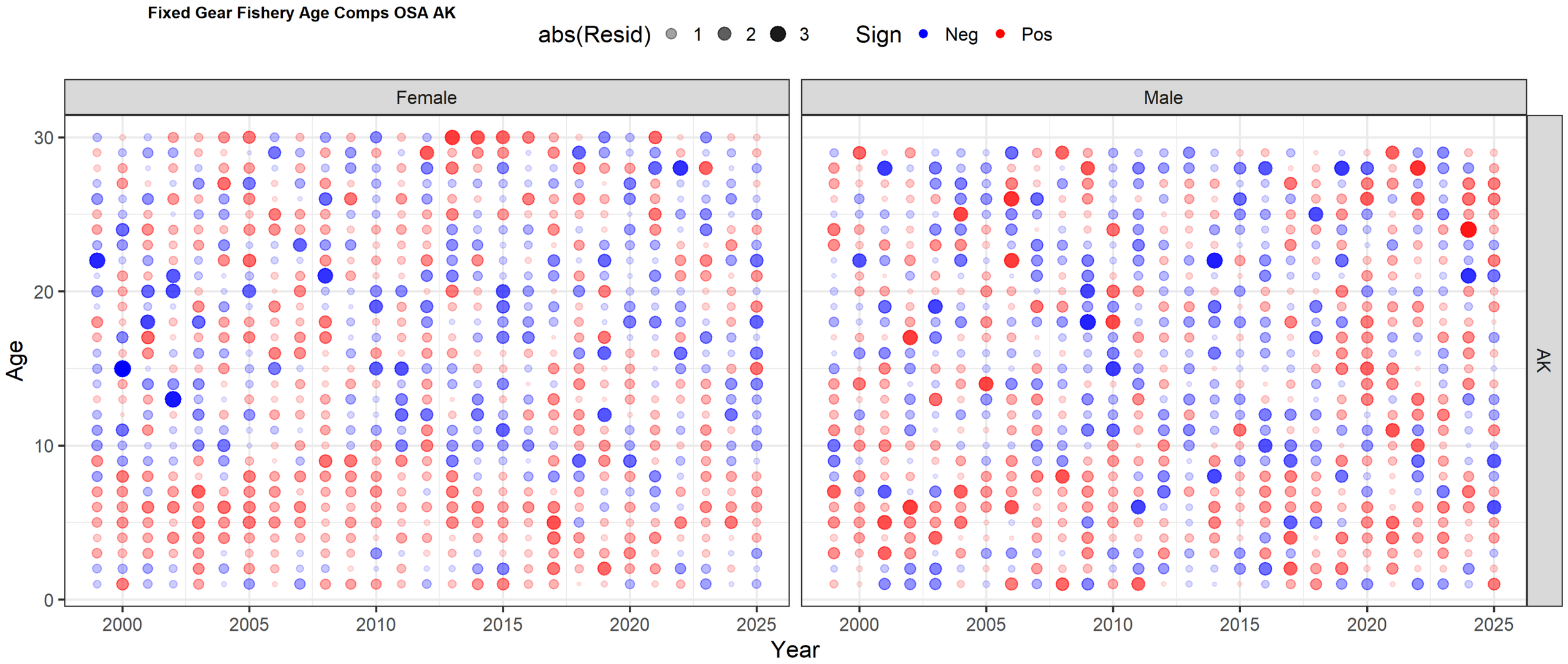
Fixed Gear Fishery Age Comps Female AK

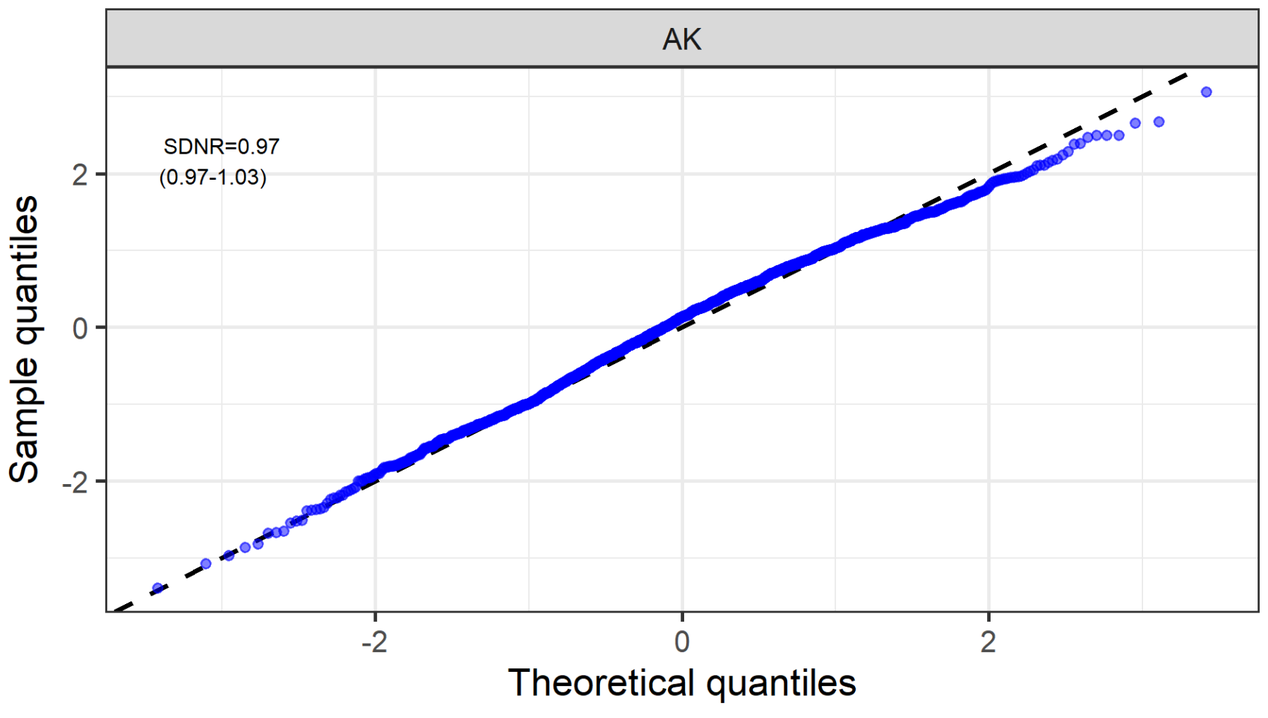


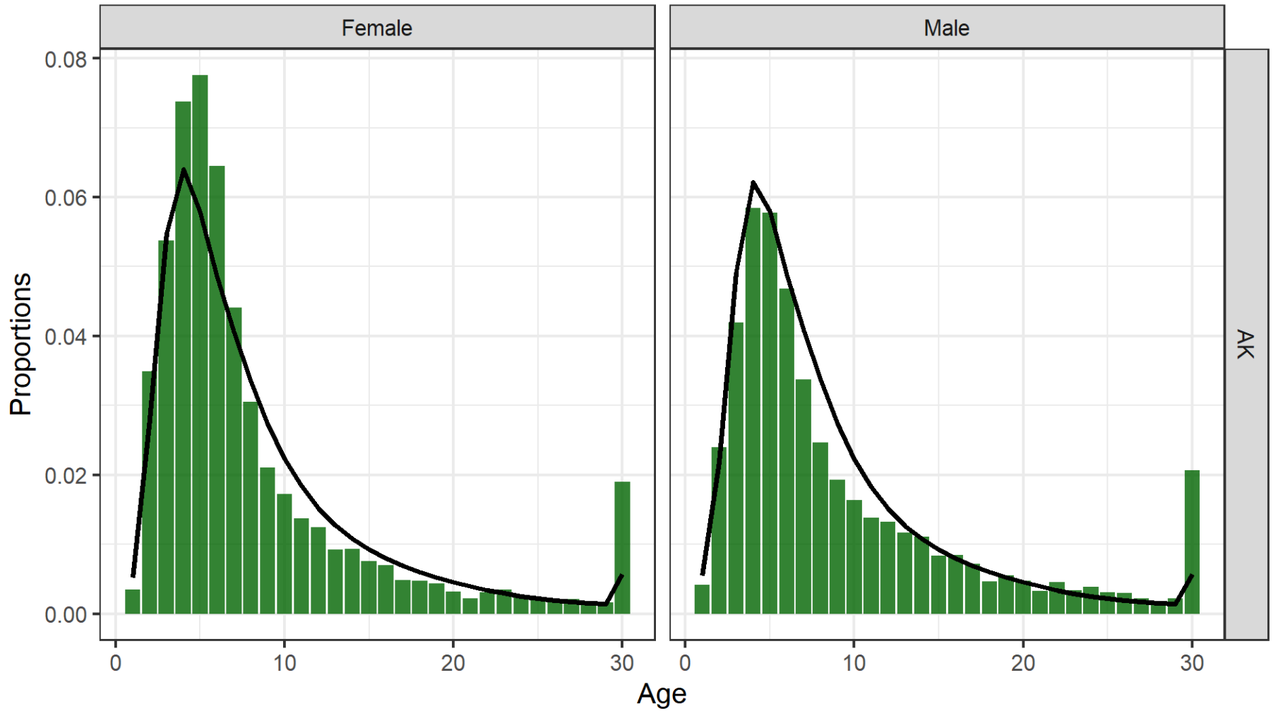
Fixed Gear Fishery Age Comps Male AK



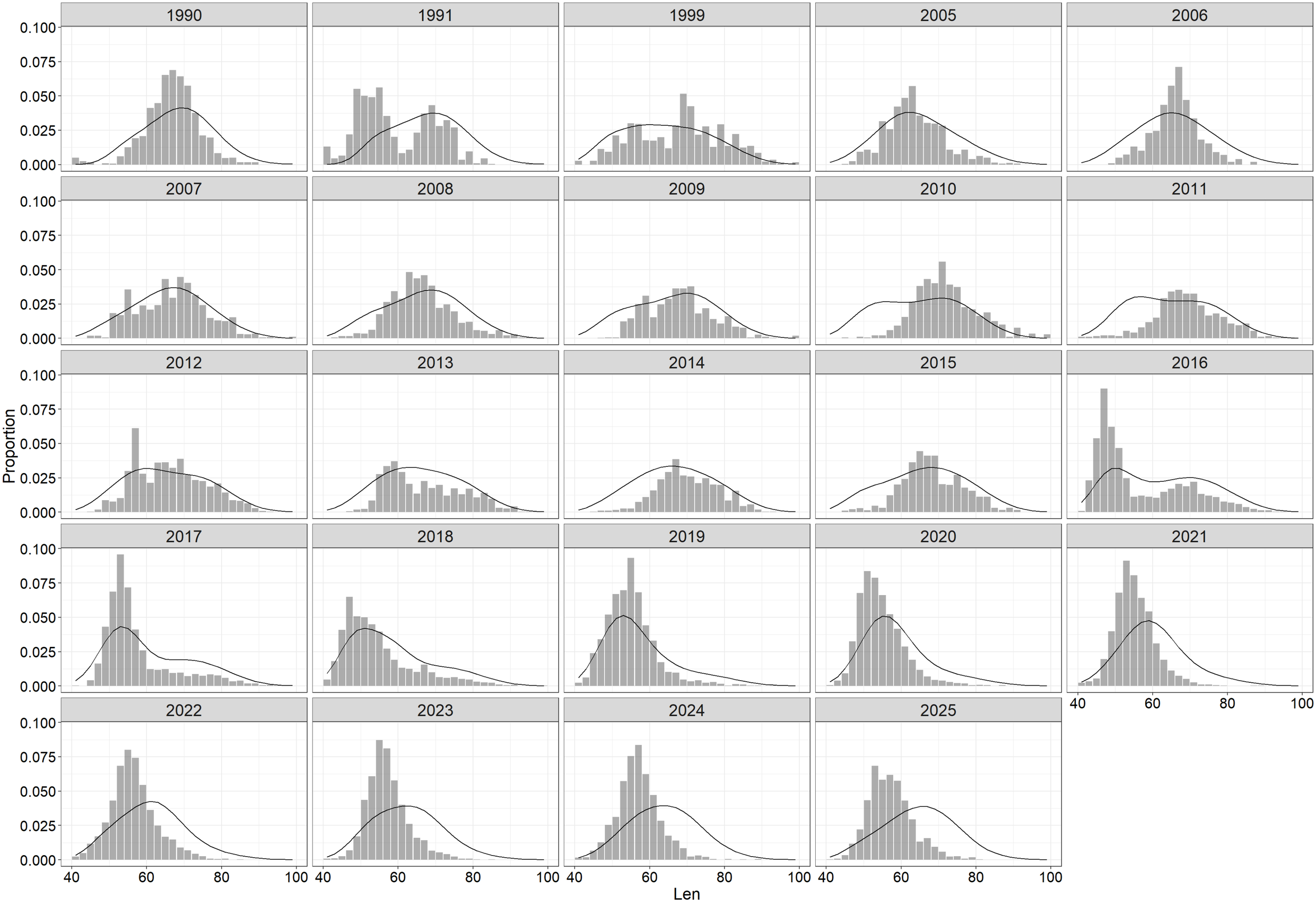
Fixed Gear Fishery Age Comps OSA AK



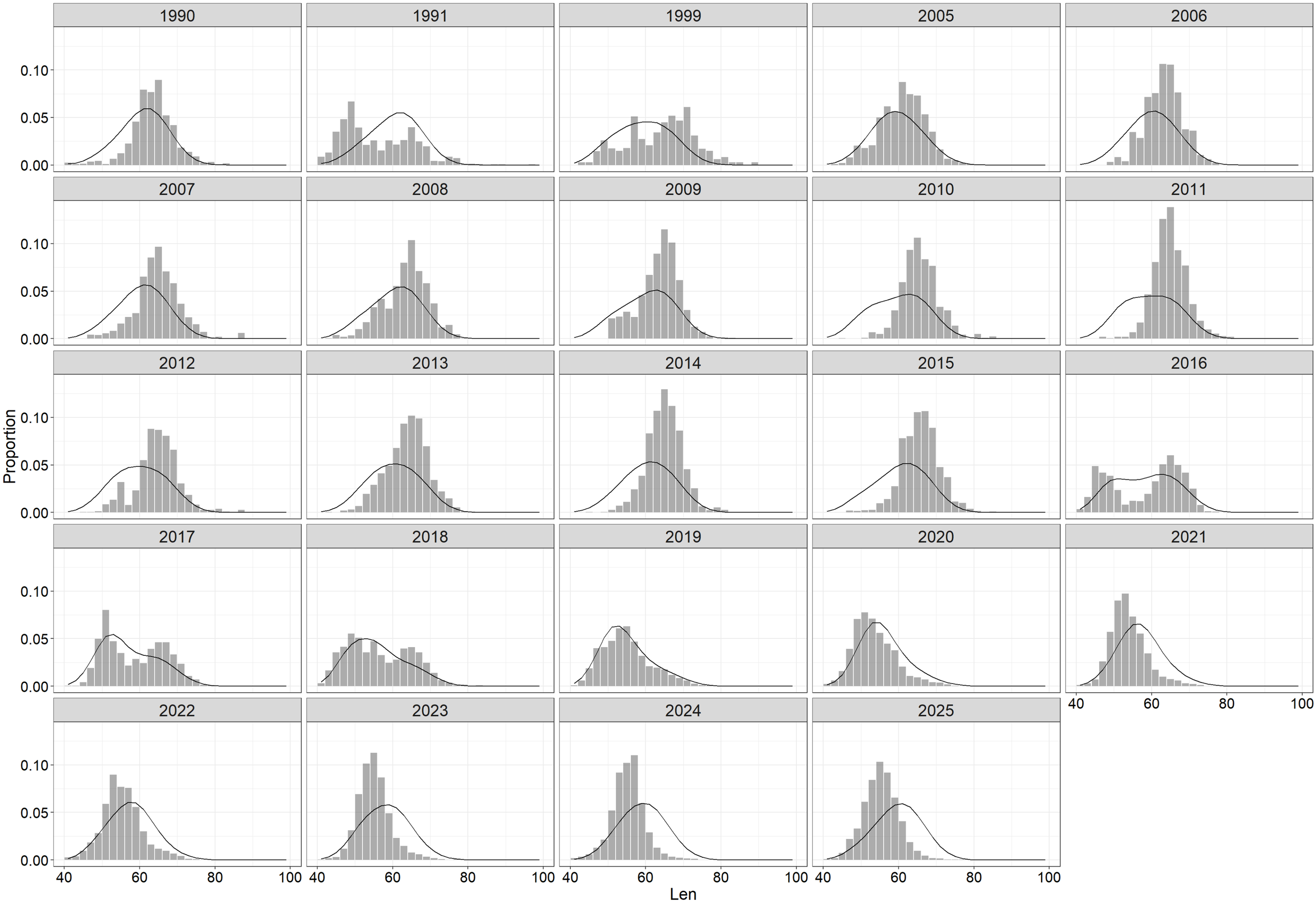




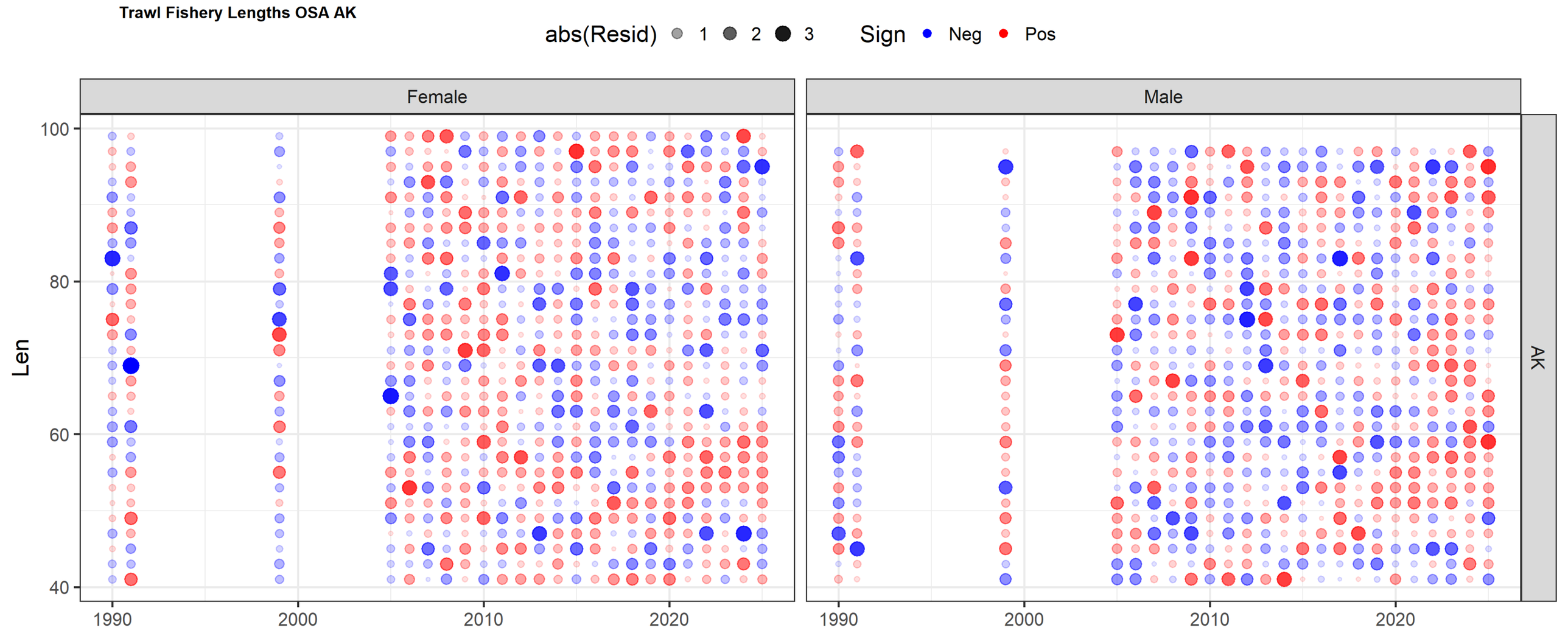
Trawl Fishery Lengths Female AK

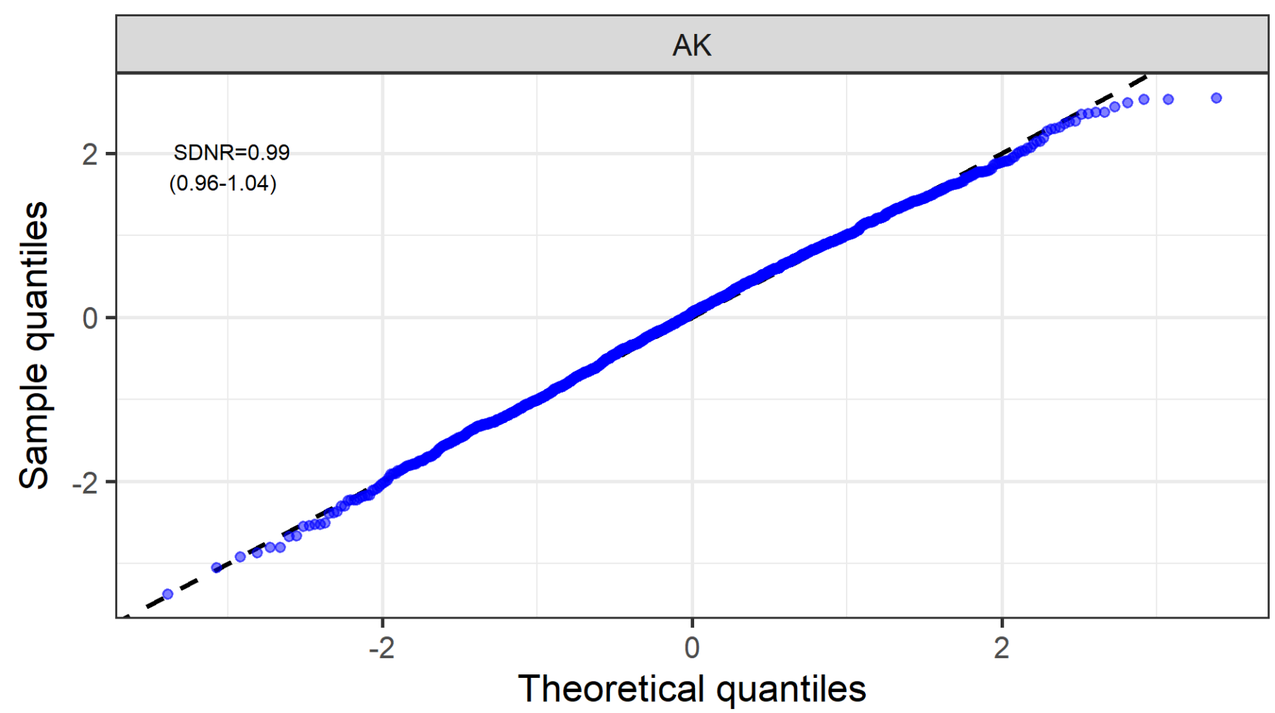


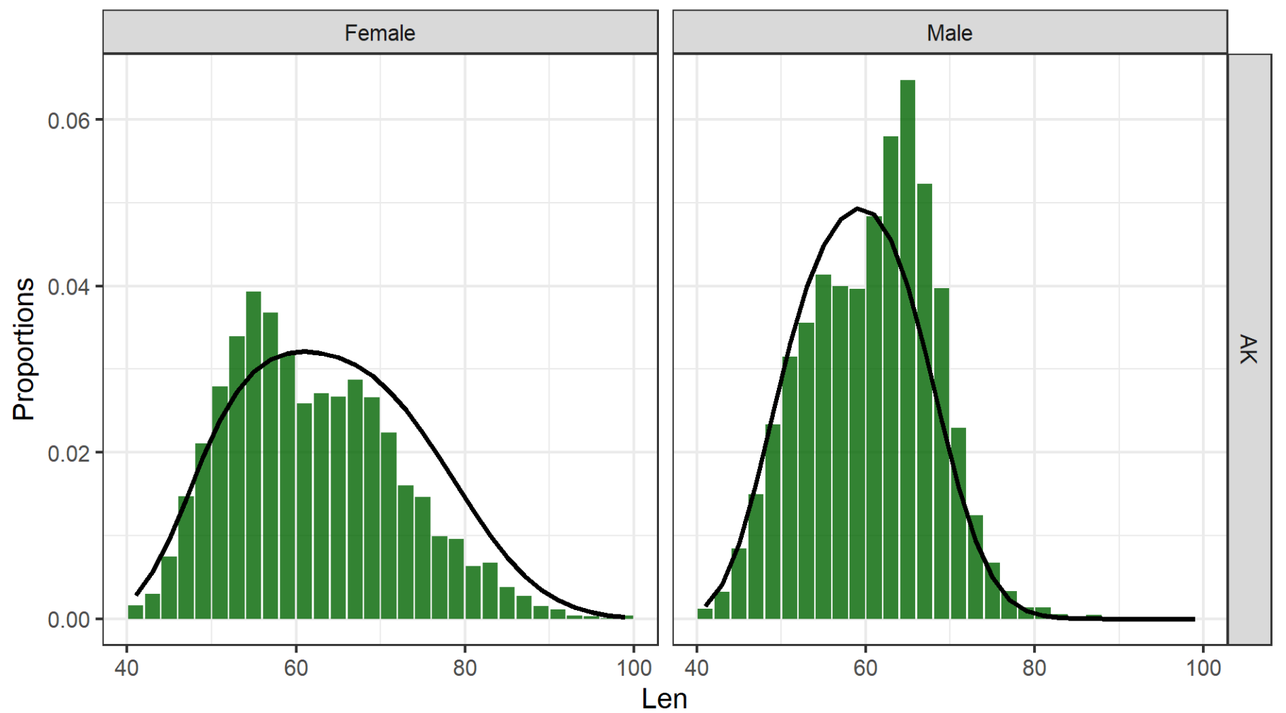
Trawl Fishery Lengths Male AK



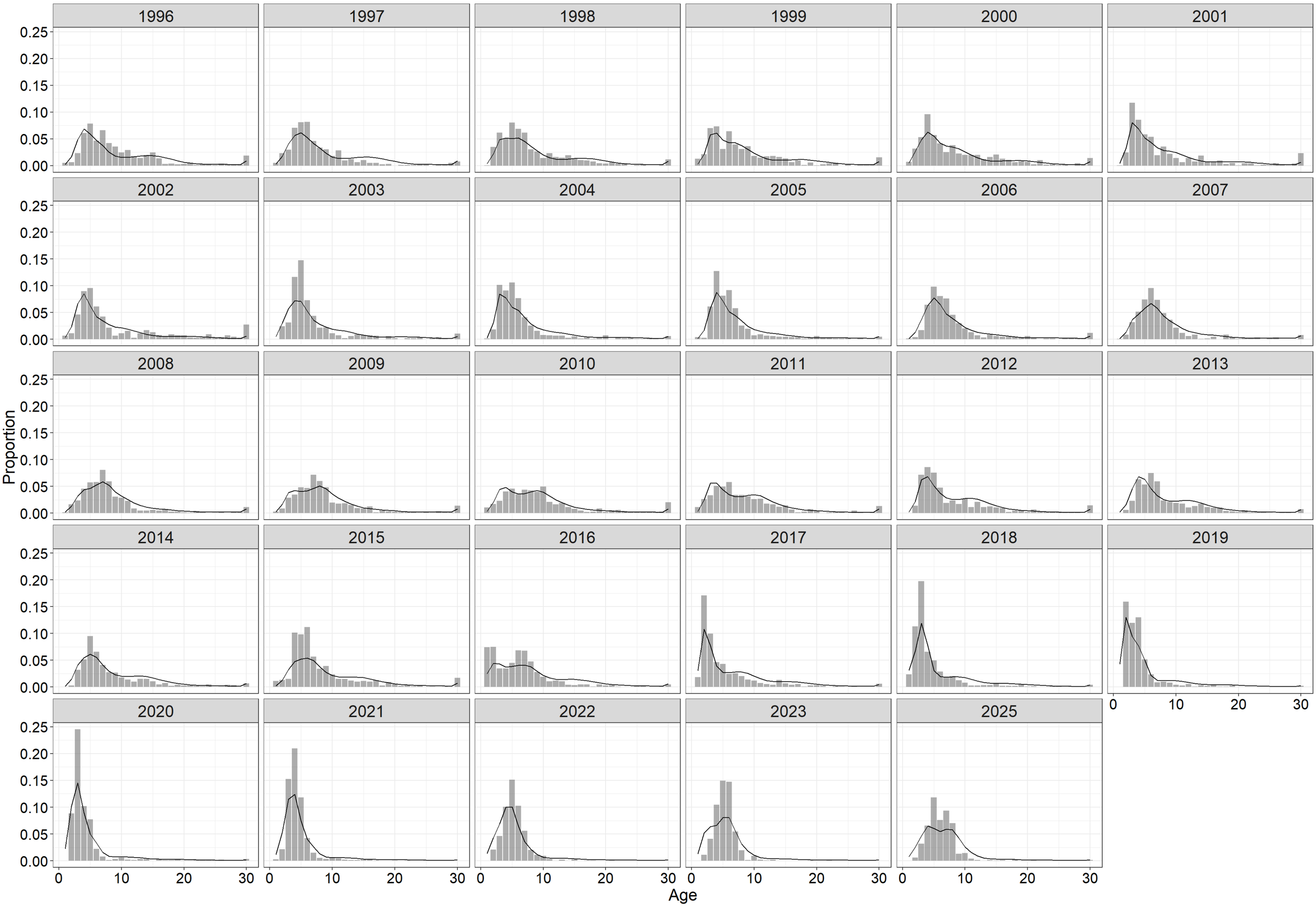
Trawl Fishery Lengths OSA AK



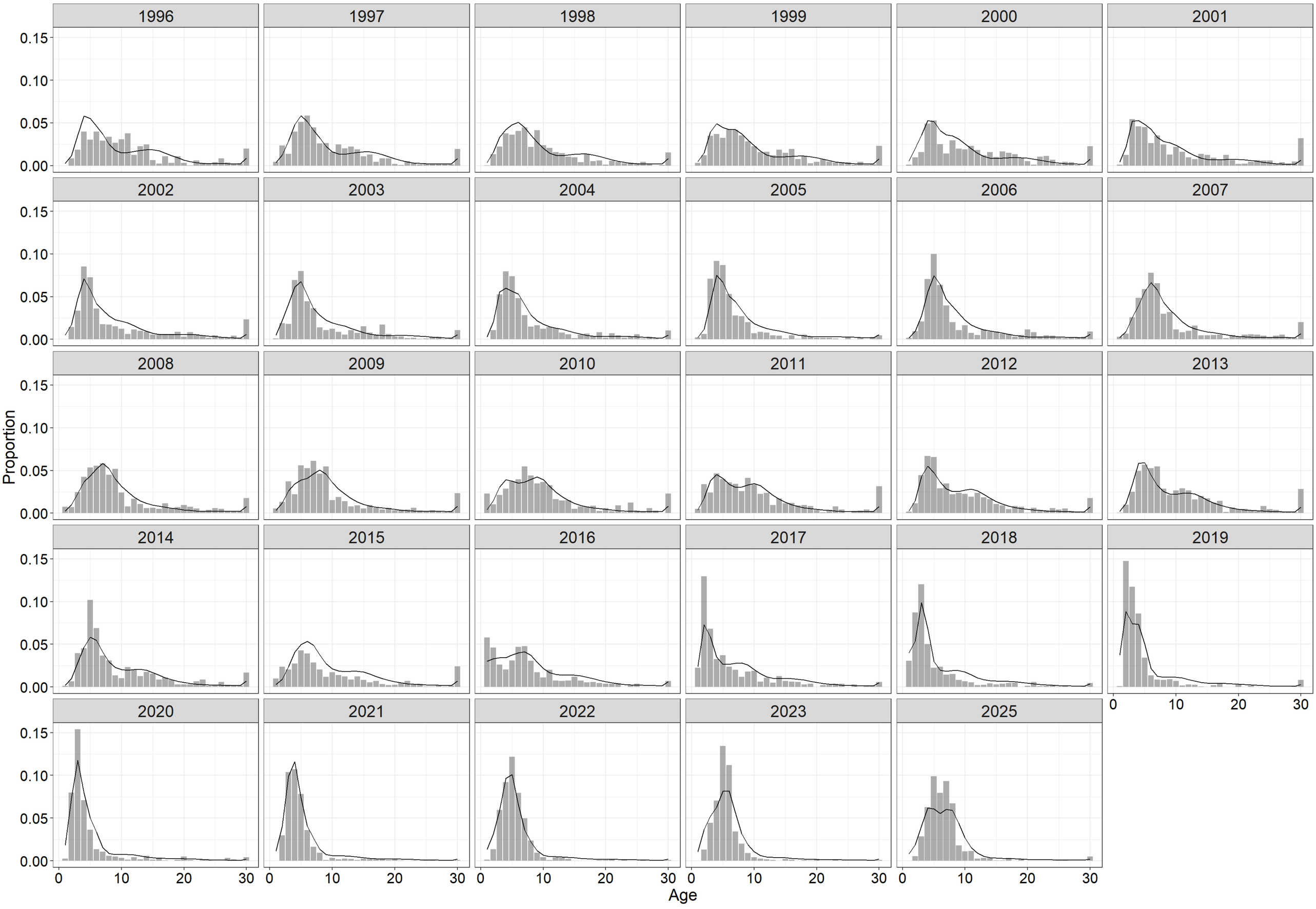




LL Survey Age Comps Female AK



LL Survey Age Comps Male AK



LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=0.88
(0.97-1.03)

Sample quantiles

Theoretical quantiles

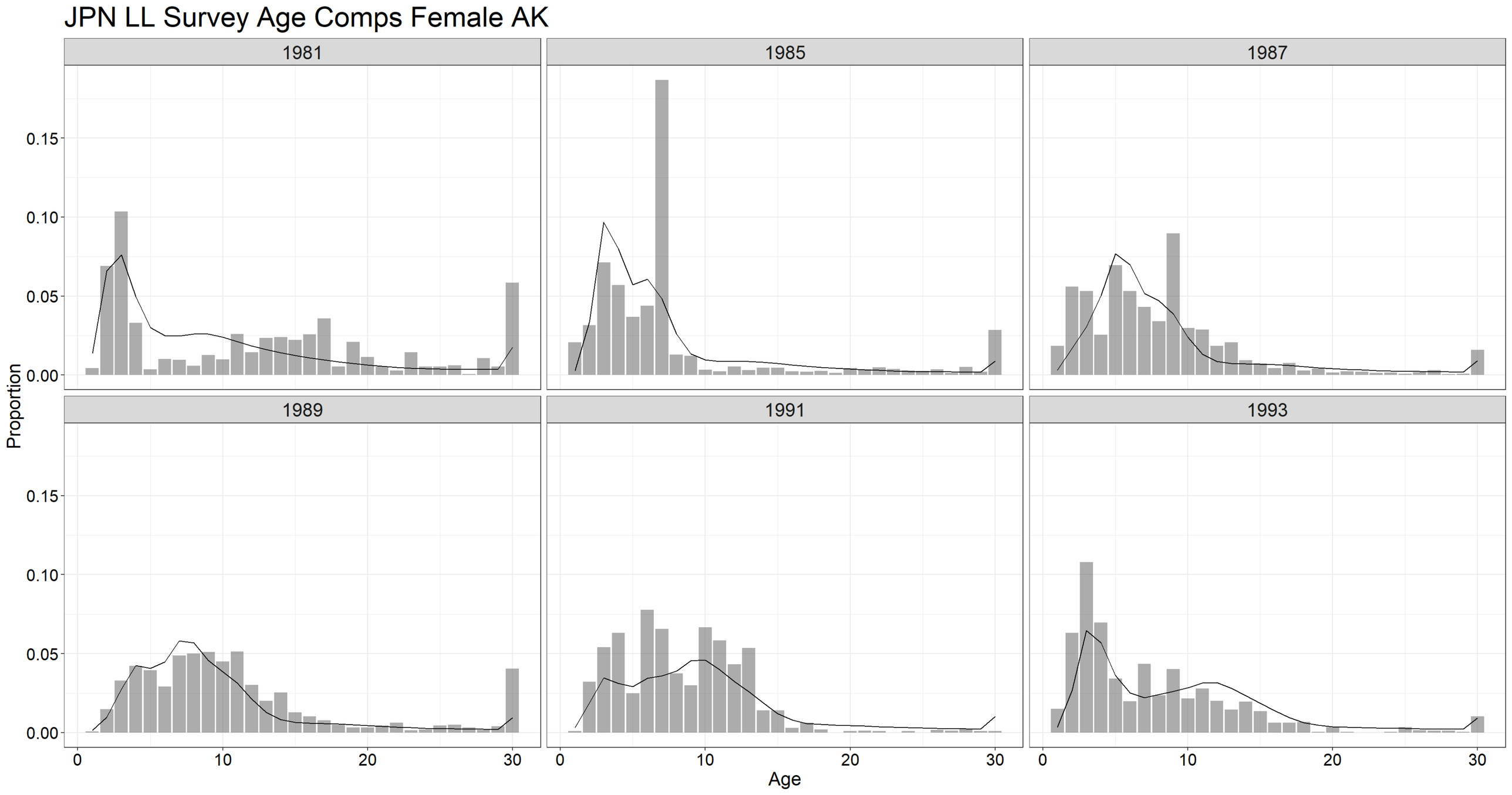
Female

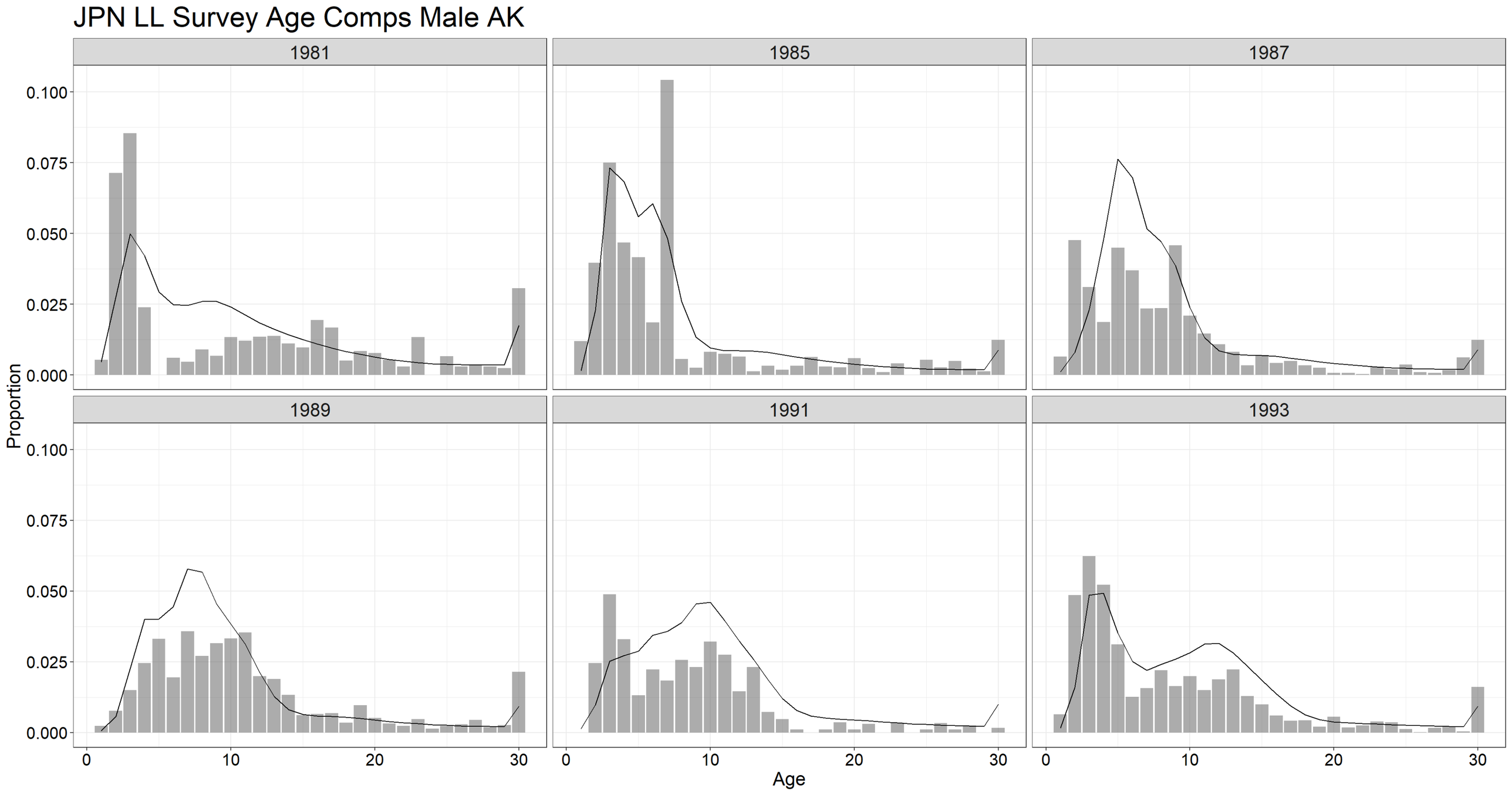
Male

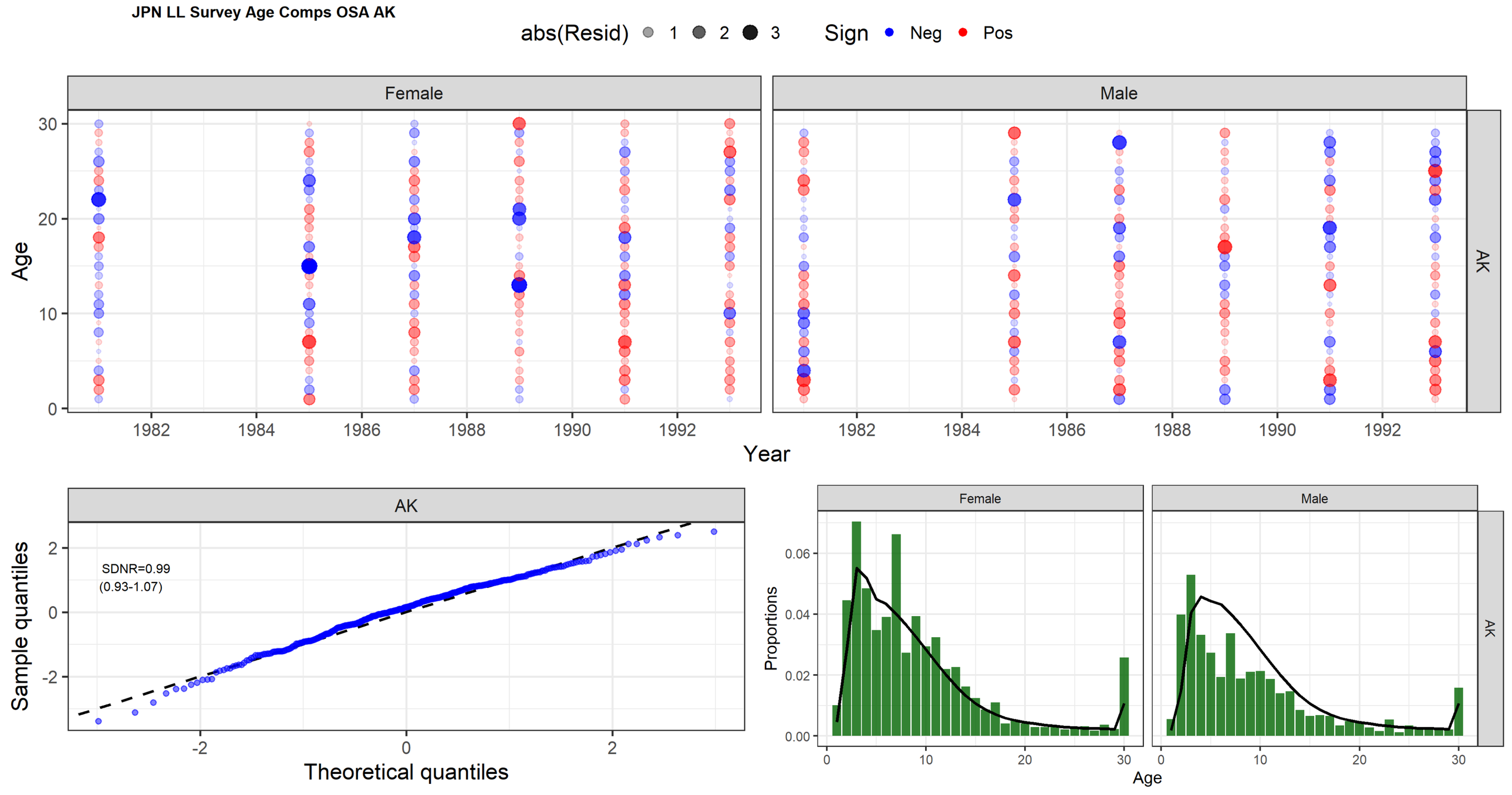
AK

Proportions

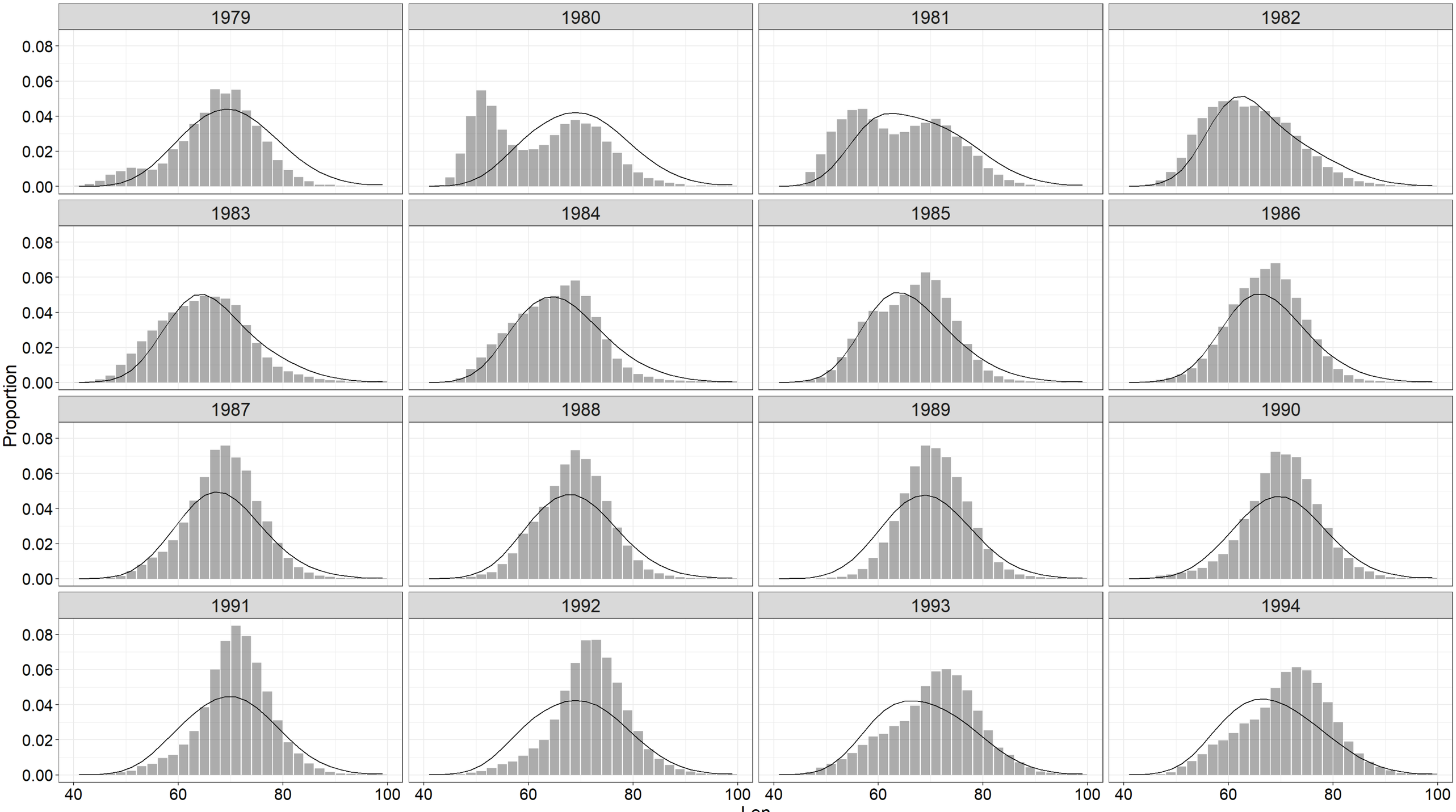
Age


JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age


JPN LL Survey Age Comps OSA AK
abs(Resid) 1 2 3
Sign Neg Pos
Female
Male
AK
Age
Year
SDNR=0.99
(0.93-1.07)
Sample quantiles
Theoretical quantiles
Proportions

JPN LL Survey Length Comps Female AK

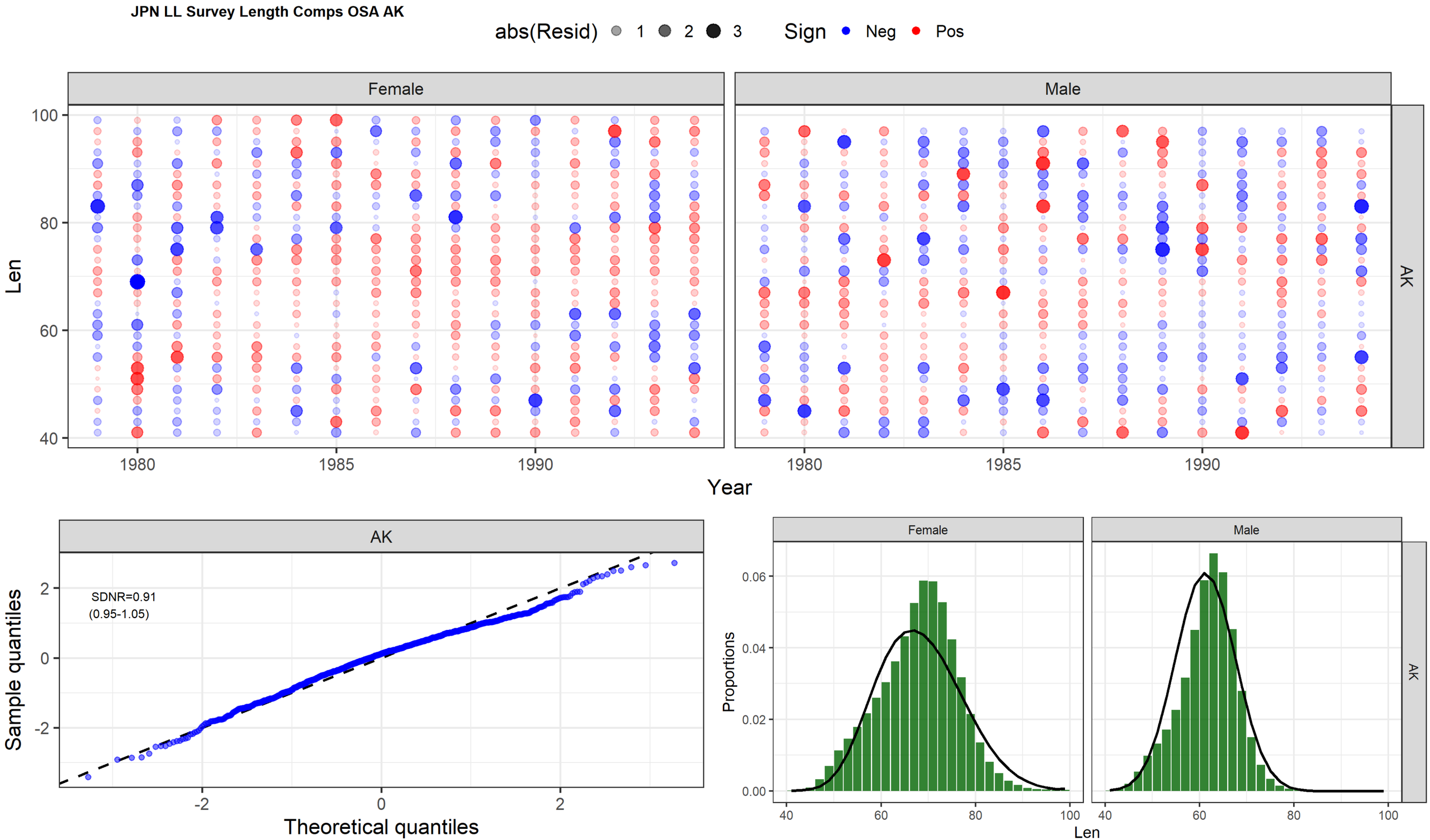


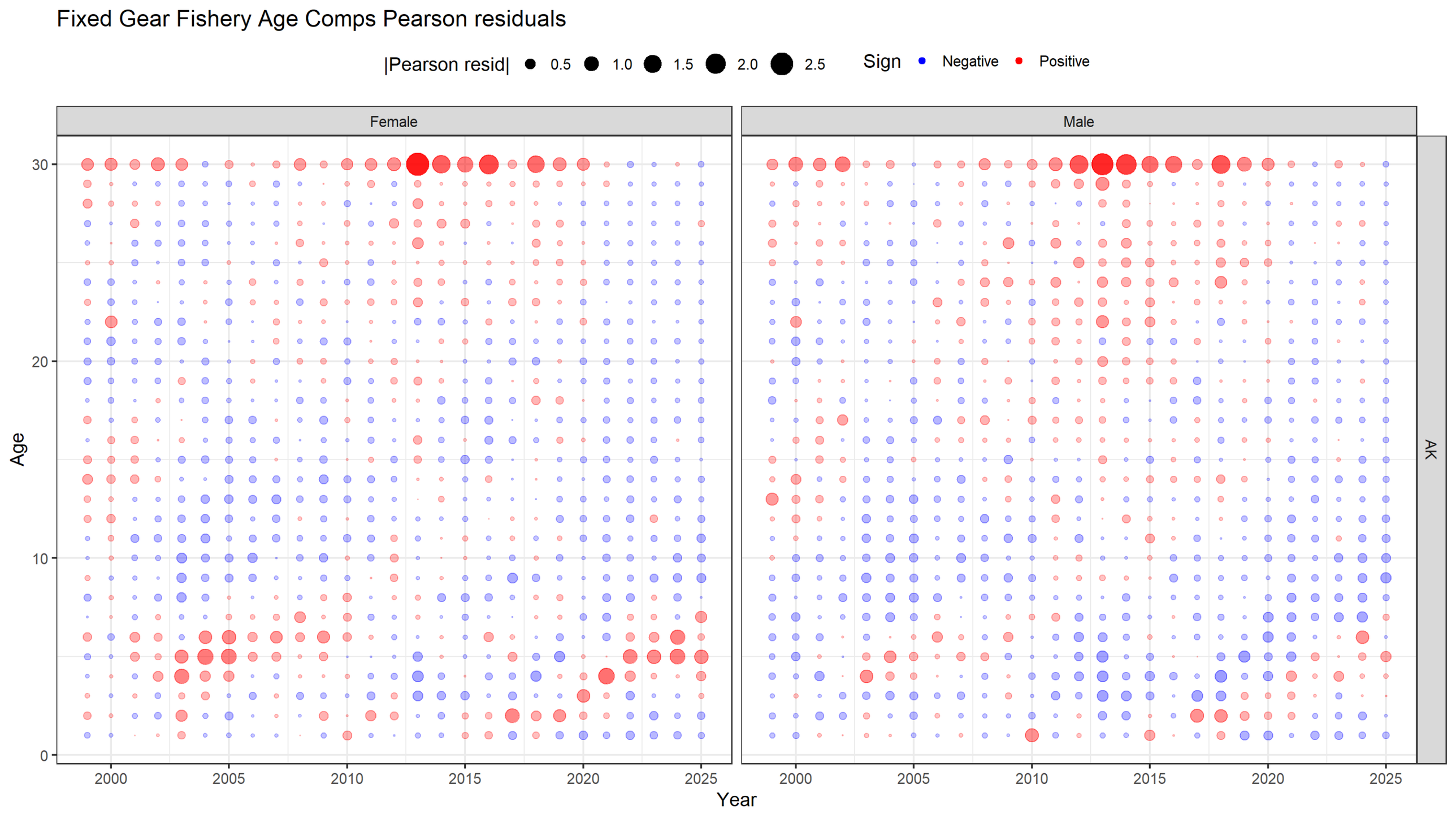
JPN LL Survey Length Comps Male AK

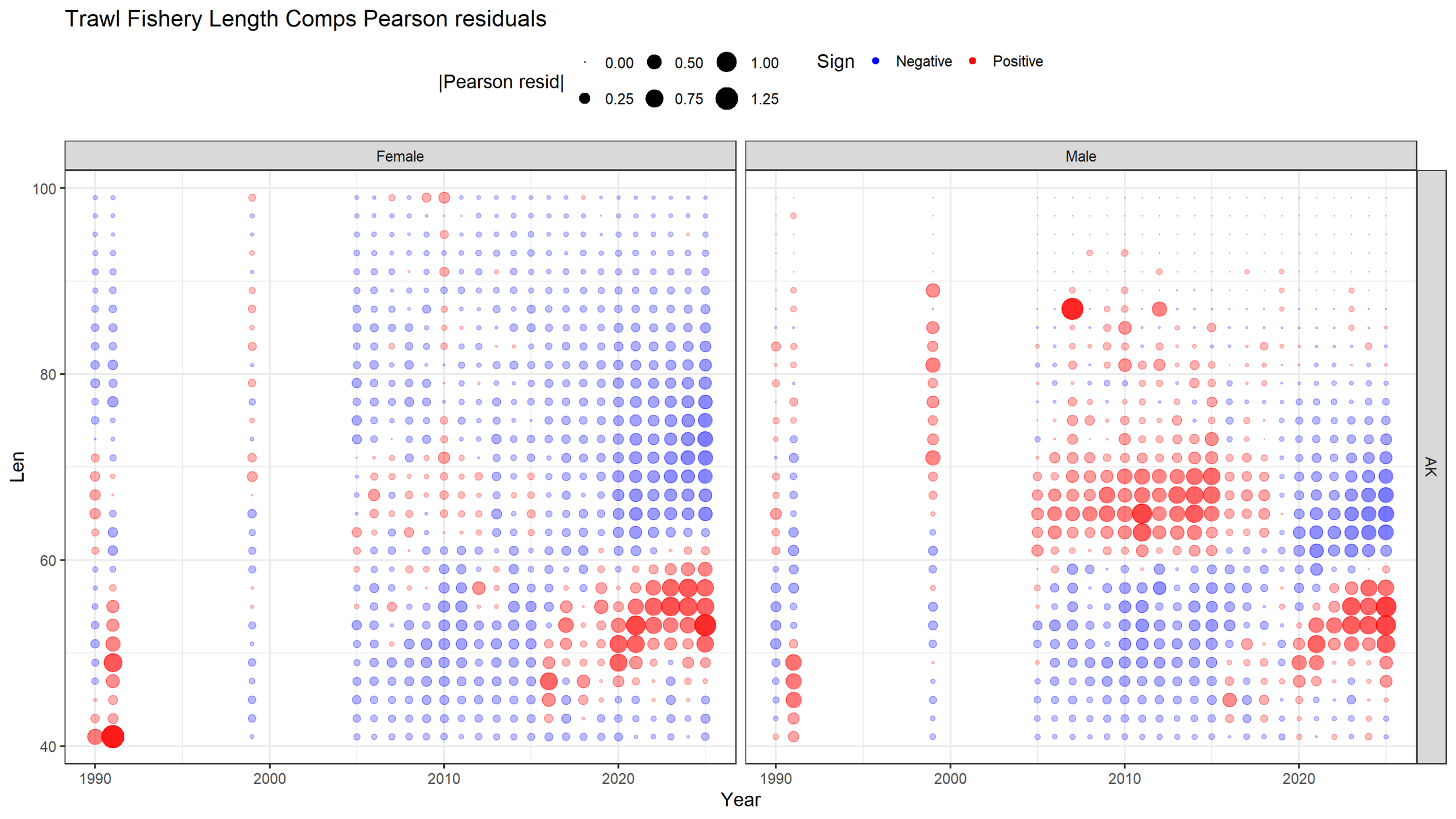
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

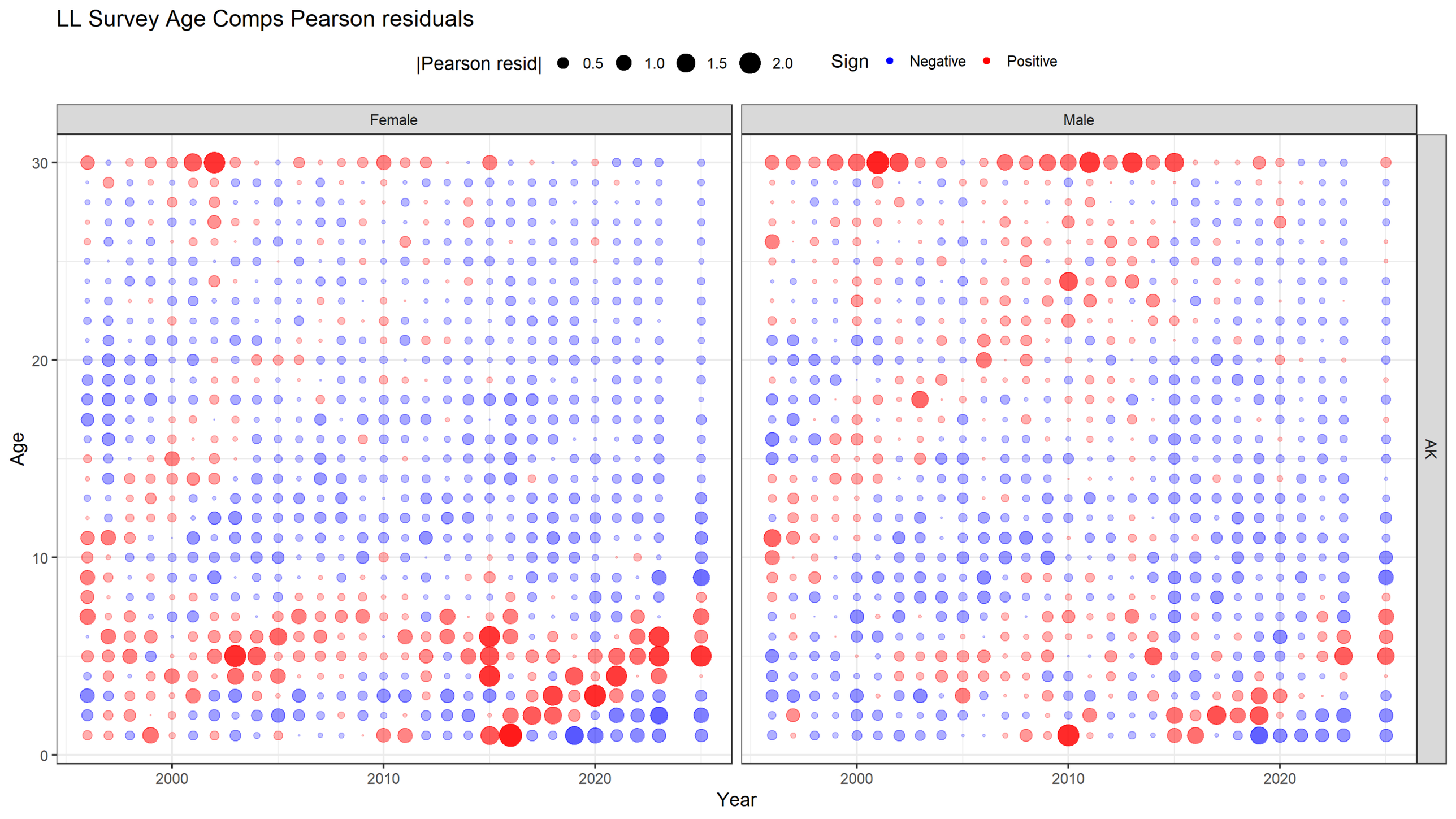
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

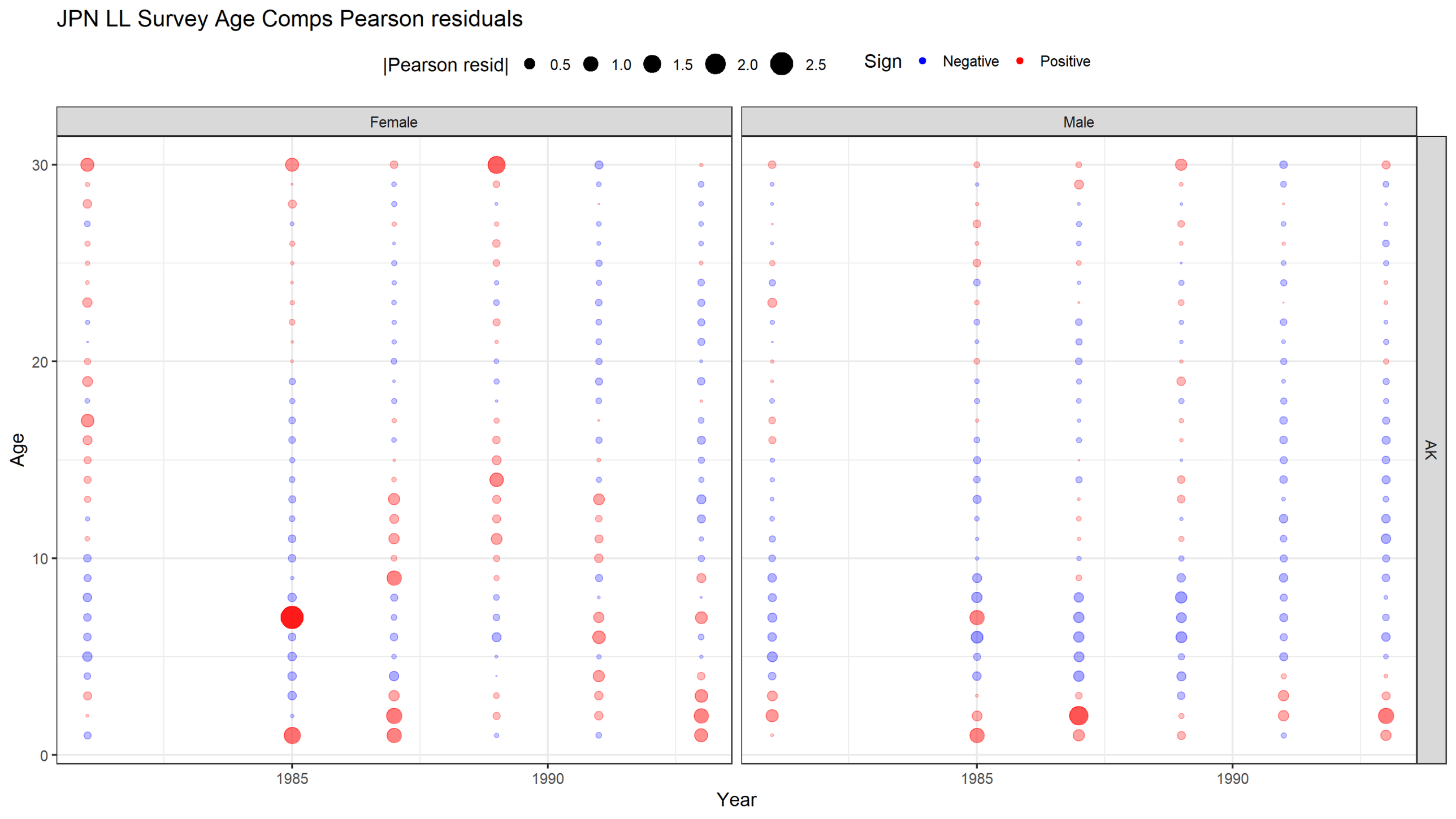
JPN LL Survey Length Comps OSA AK

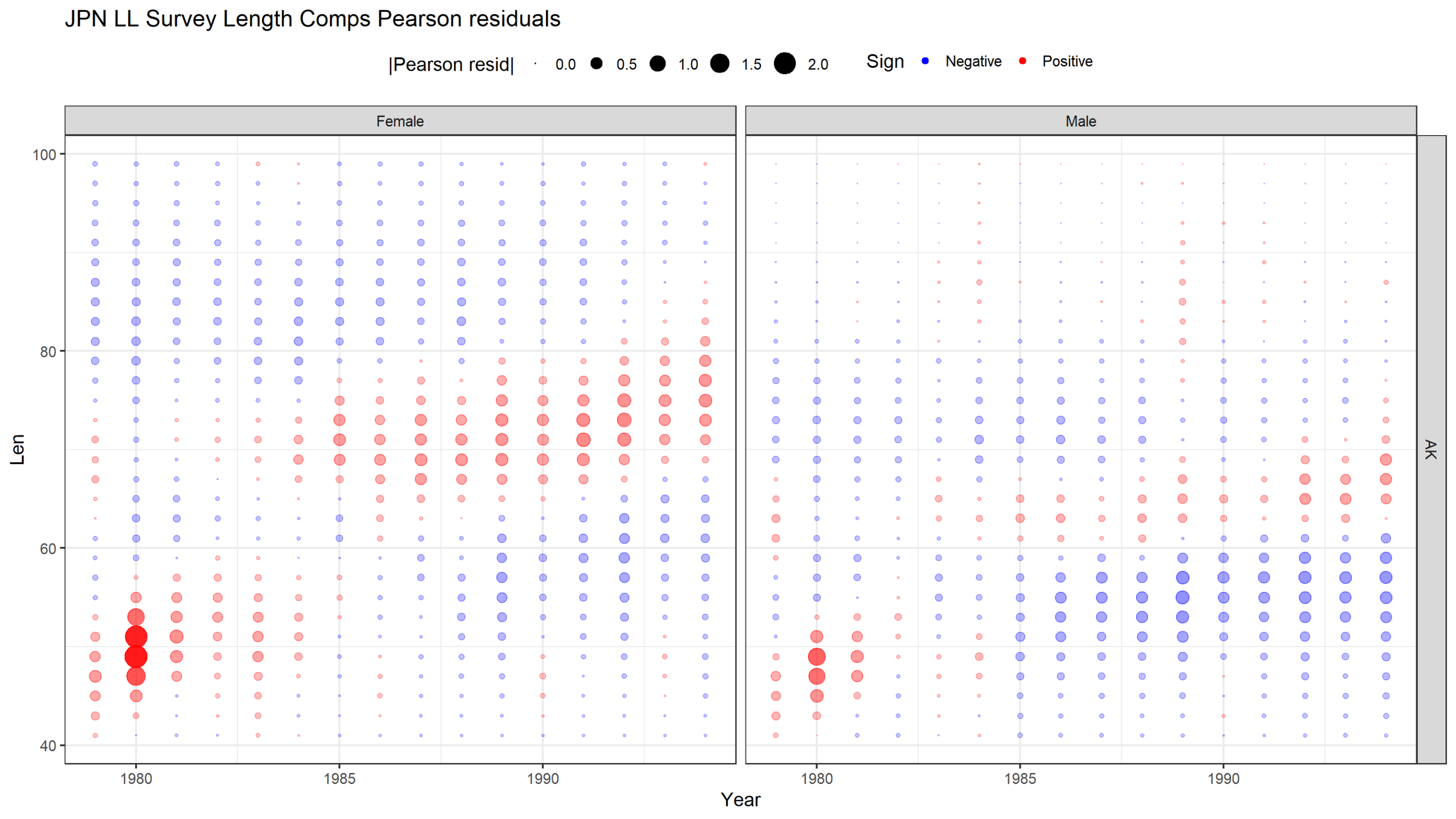


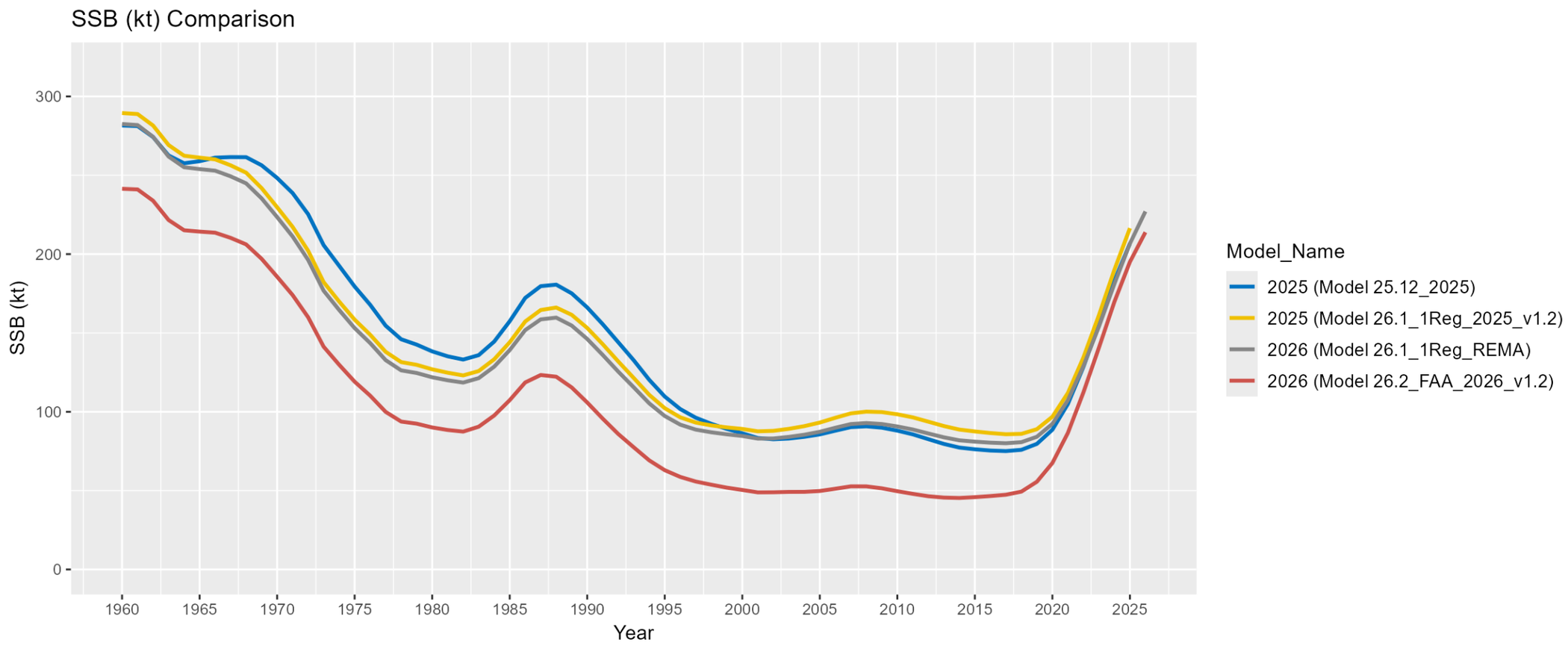

Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
2.5
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

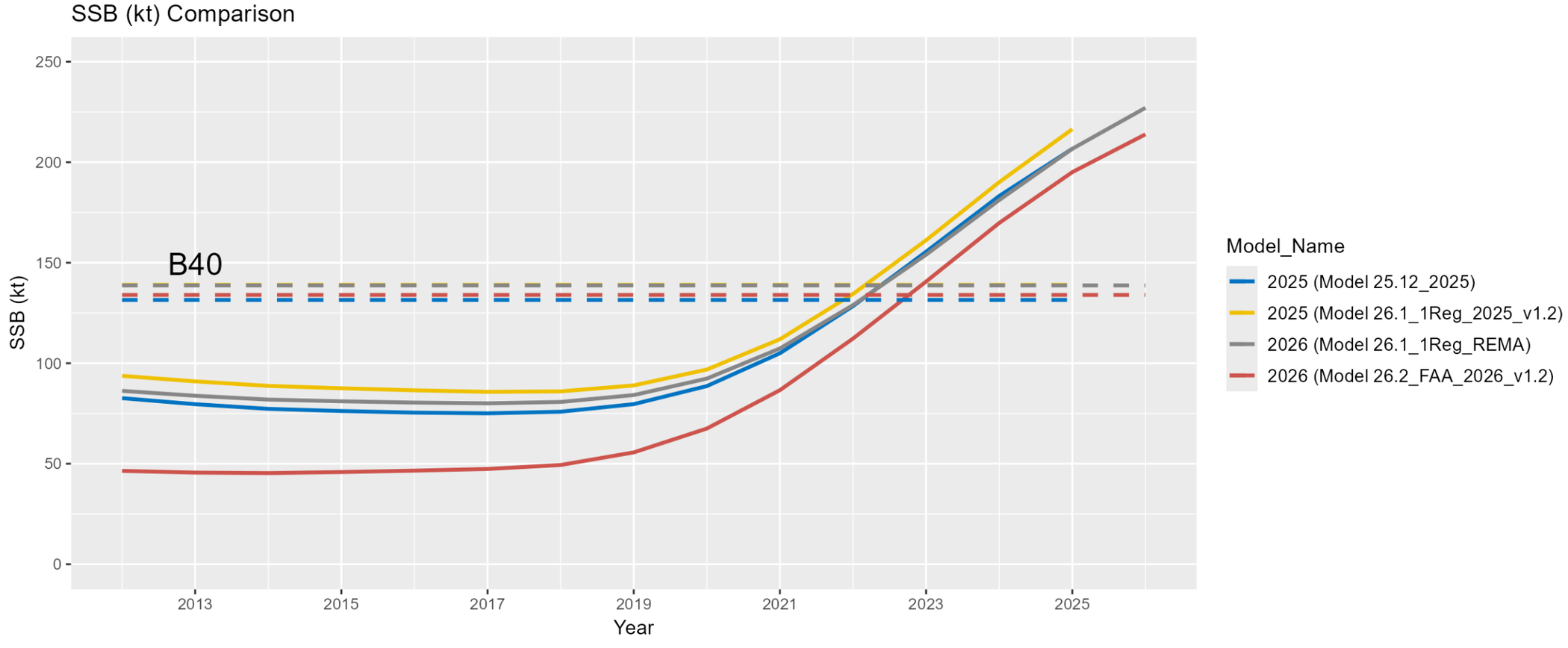

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.00
0.25
0.50
0.75
1.00
1.25
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

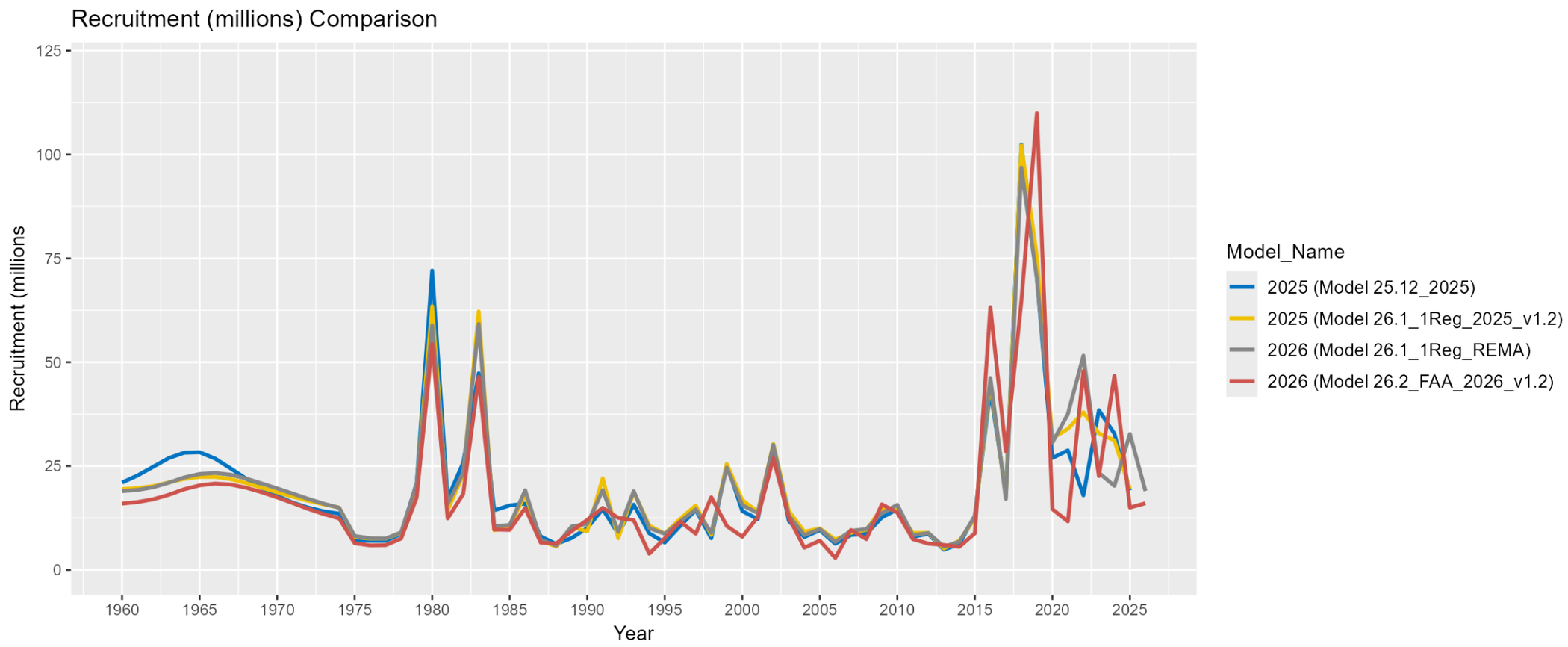

LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

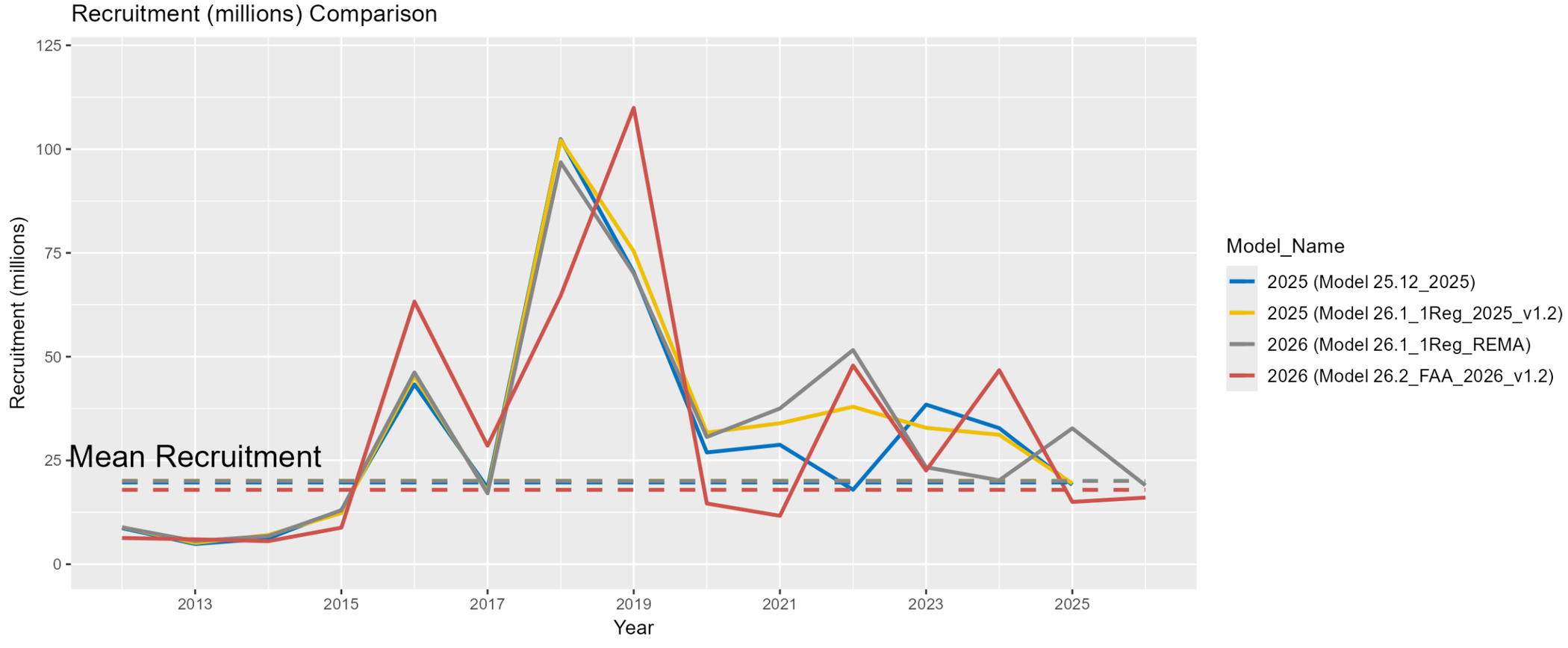

JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
2.5
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year

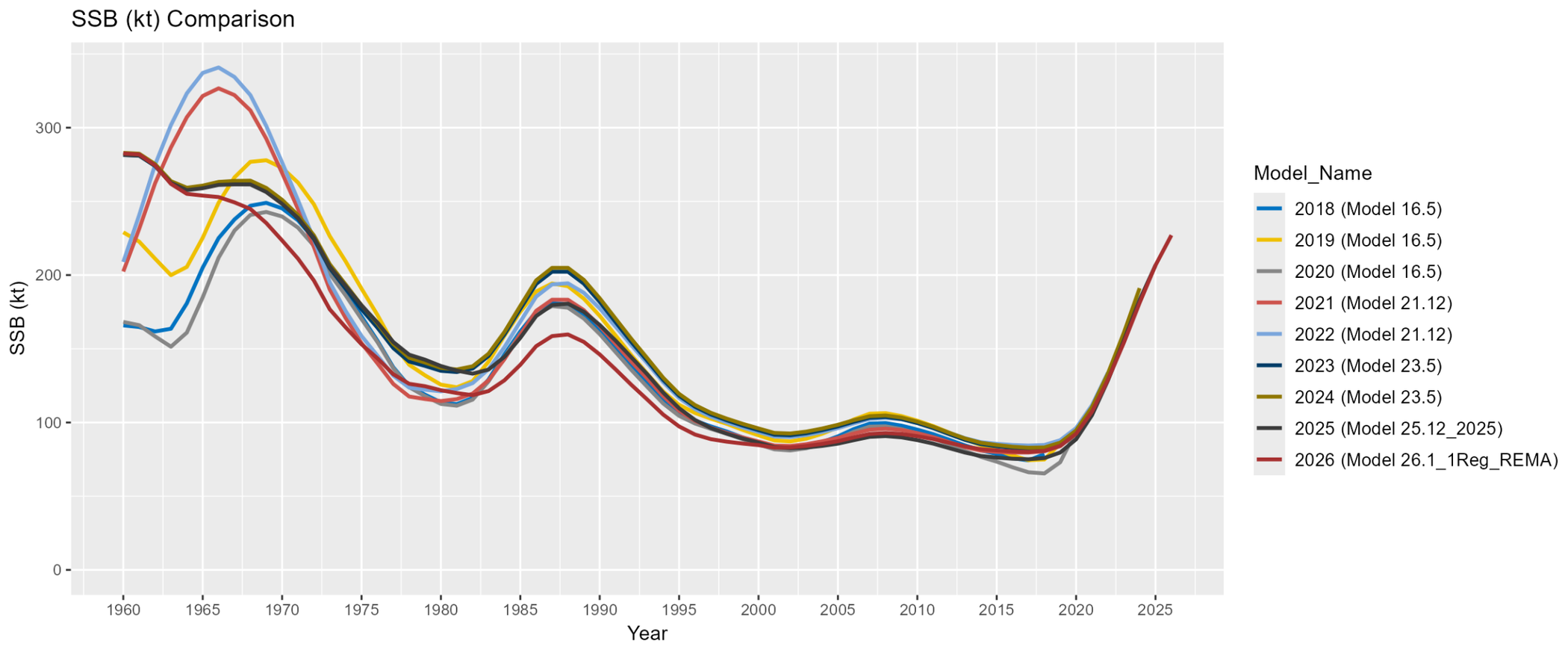

JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

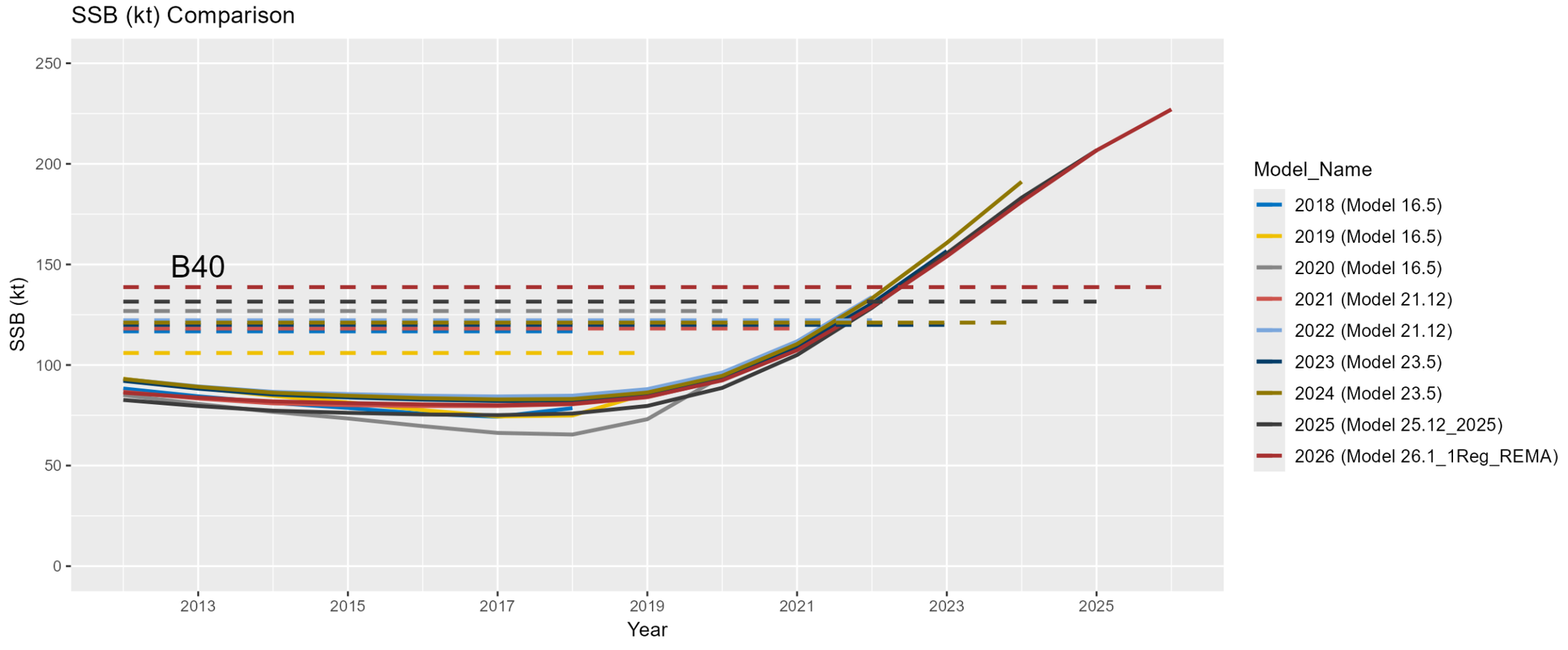

SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg_REMA)
2026 (Model 26.2_FAA_2026_v1.2)

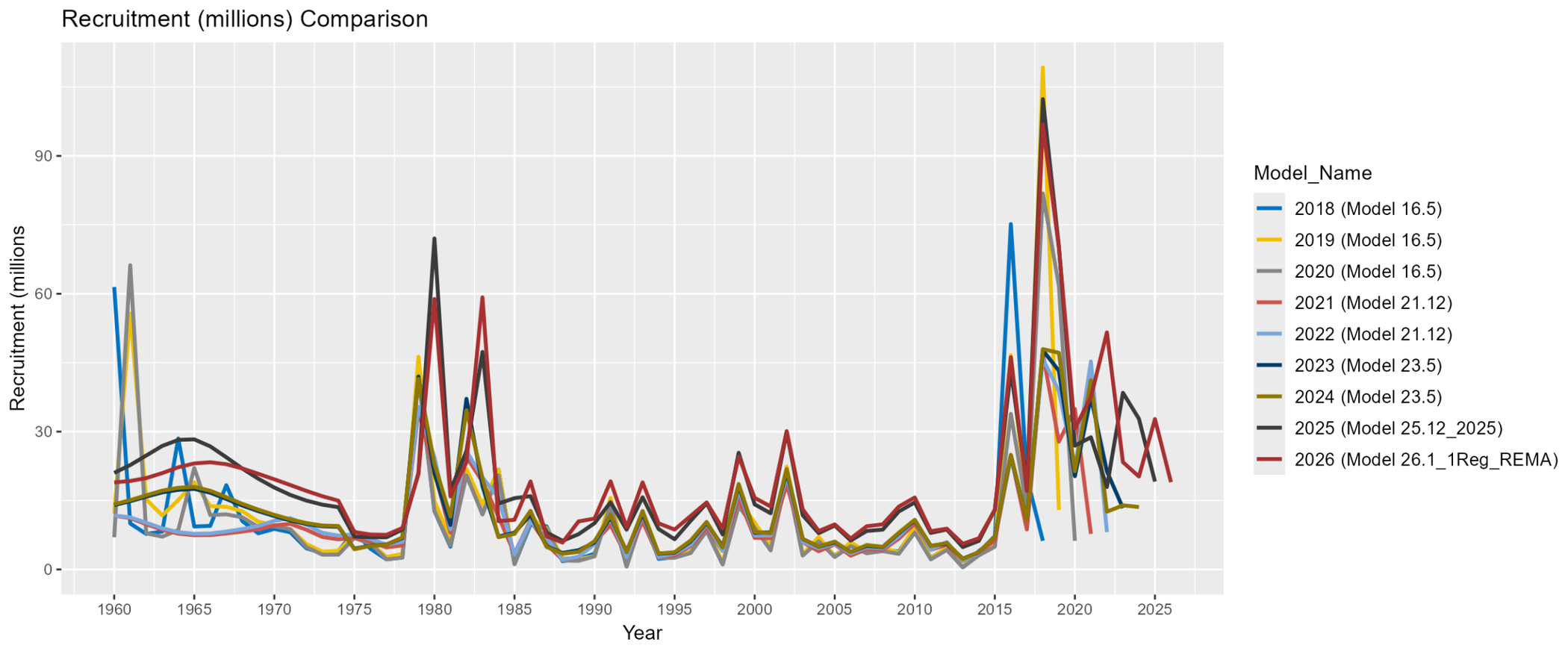

SSB (kt) Comparison
B40
SSB (kt)
250
200
150
100
50
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg_REMA)
2026 (Model 26.2_FAA_2026_v1.2)

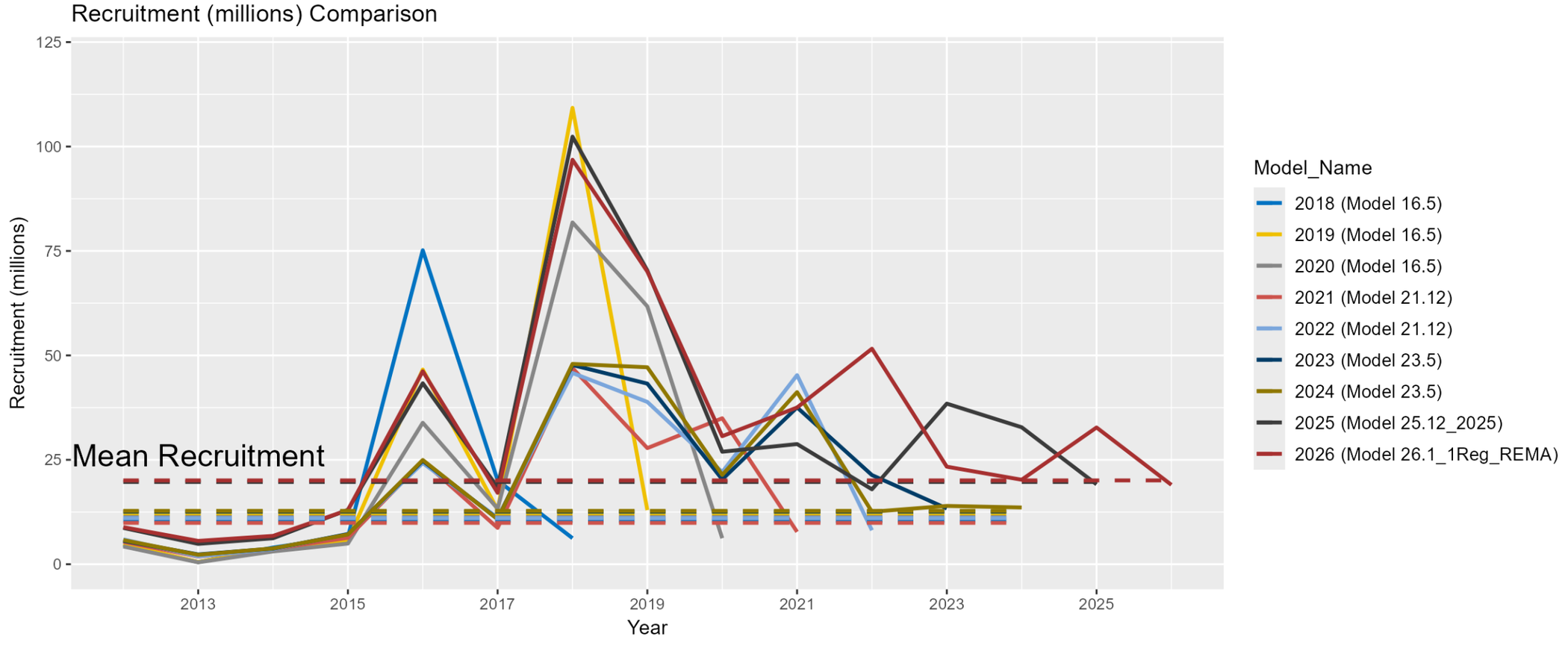

Recruitment (millions) Comparison
Recruitment (millions
125
100
75
50
25
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg_REMA)
2026 (Model 26.2_FAA_2026_v1.2)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg_REMA)
2026 (Model 26.2_FAA_2026_v1.2)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg_REMA)


SSB (kt) Comparison
B40
SSB (kt)
250
200
150
100
50
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg_REMA)


Recruitment (millions) Comparison
Recruitment (millions
90
60
30
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg_REMA)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg_REMA)

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.431

Mohns Rho: 0.0224

Relative Difference from Terminal Year

0.8
0.4
0.0
-0.4
-0.8

1960 1980 2000 2020

1960 1980 2000 2020

Year

AK

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

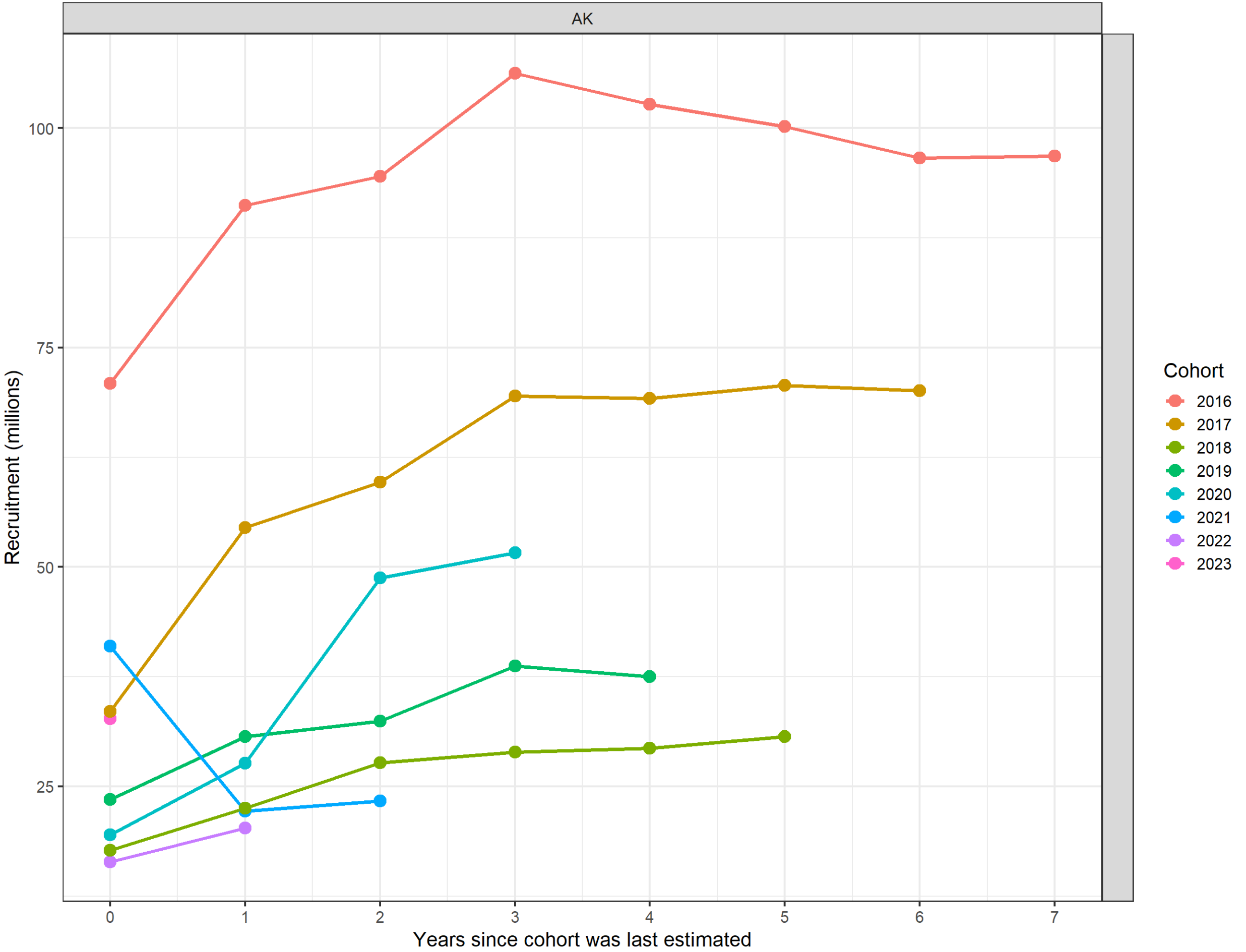
Value

90
60
30
0

200
100
0

1960
1980
2000
2020

Year


AK
100
75
50
25
Recruitment (millions)
0
1
2
3
4
5
6
7
Years since cohort was last estimated
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.4297

Mohns Rho: 0.0098

AK

Relative Difference from Terminal Year

1.5
1.0
0.5
0.0
-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

100
75
50
25
0

200
100
0

1960
1980
2000
2020

Year

AK

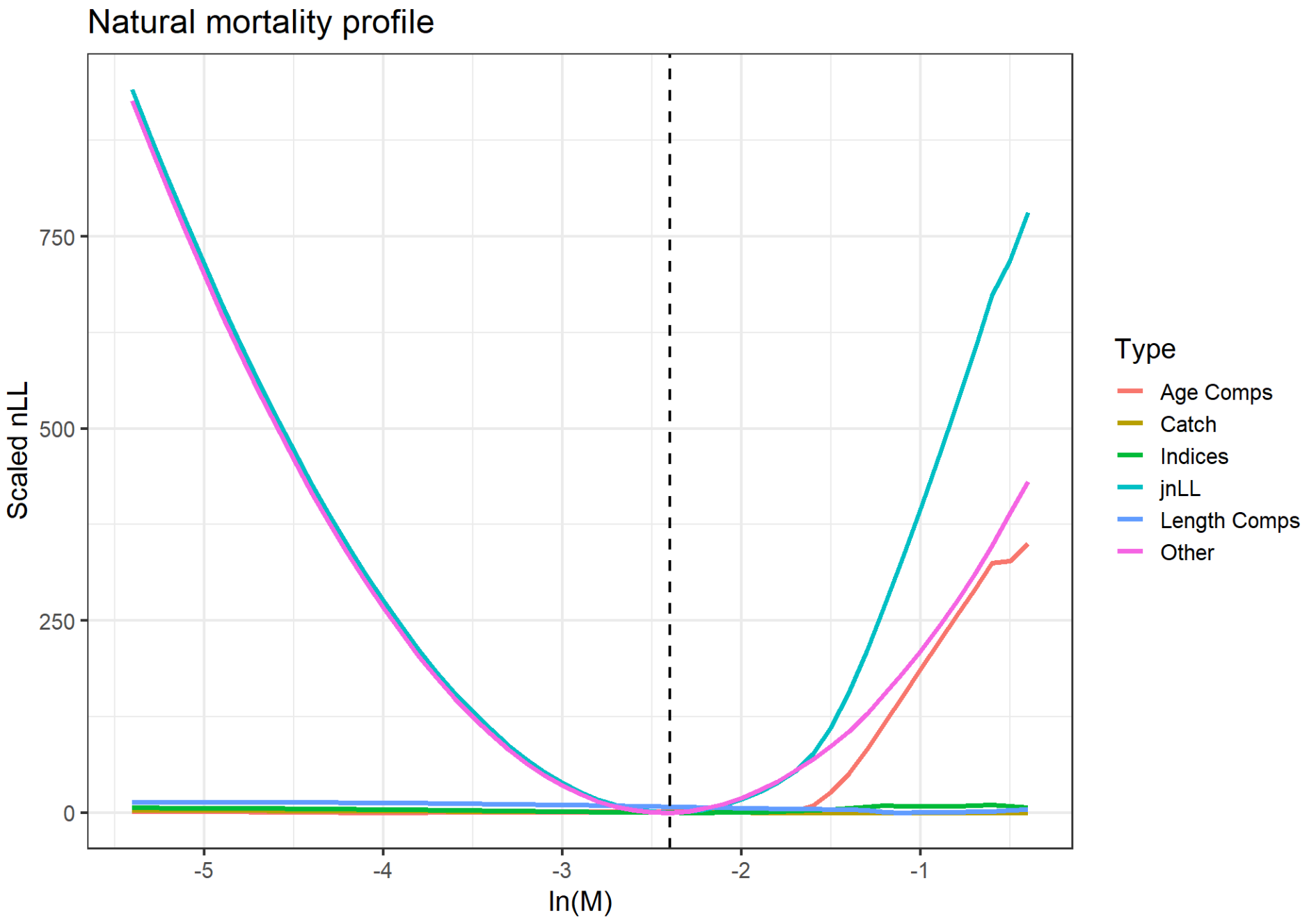
Recruitment (millions)

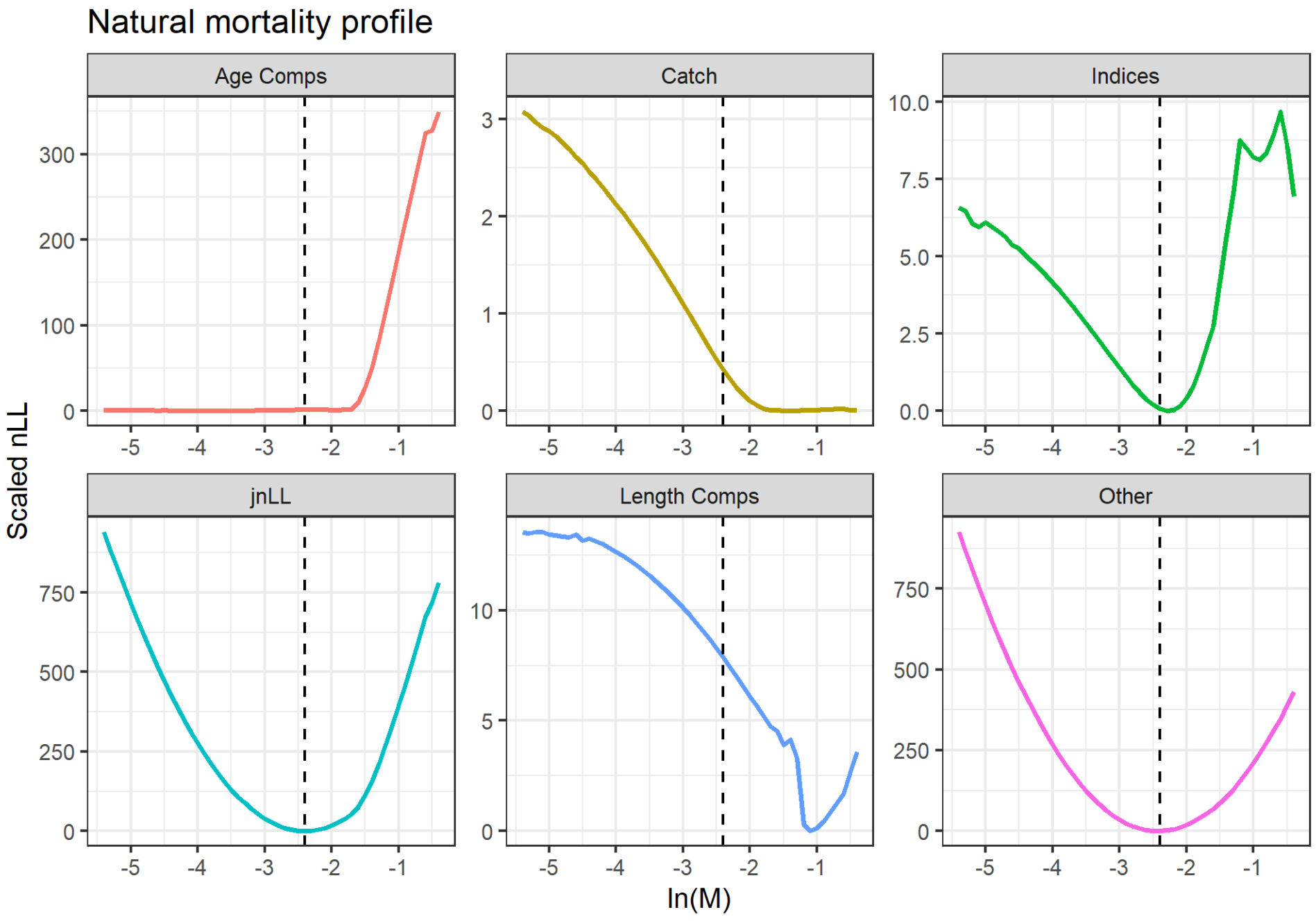
100
75
50
25

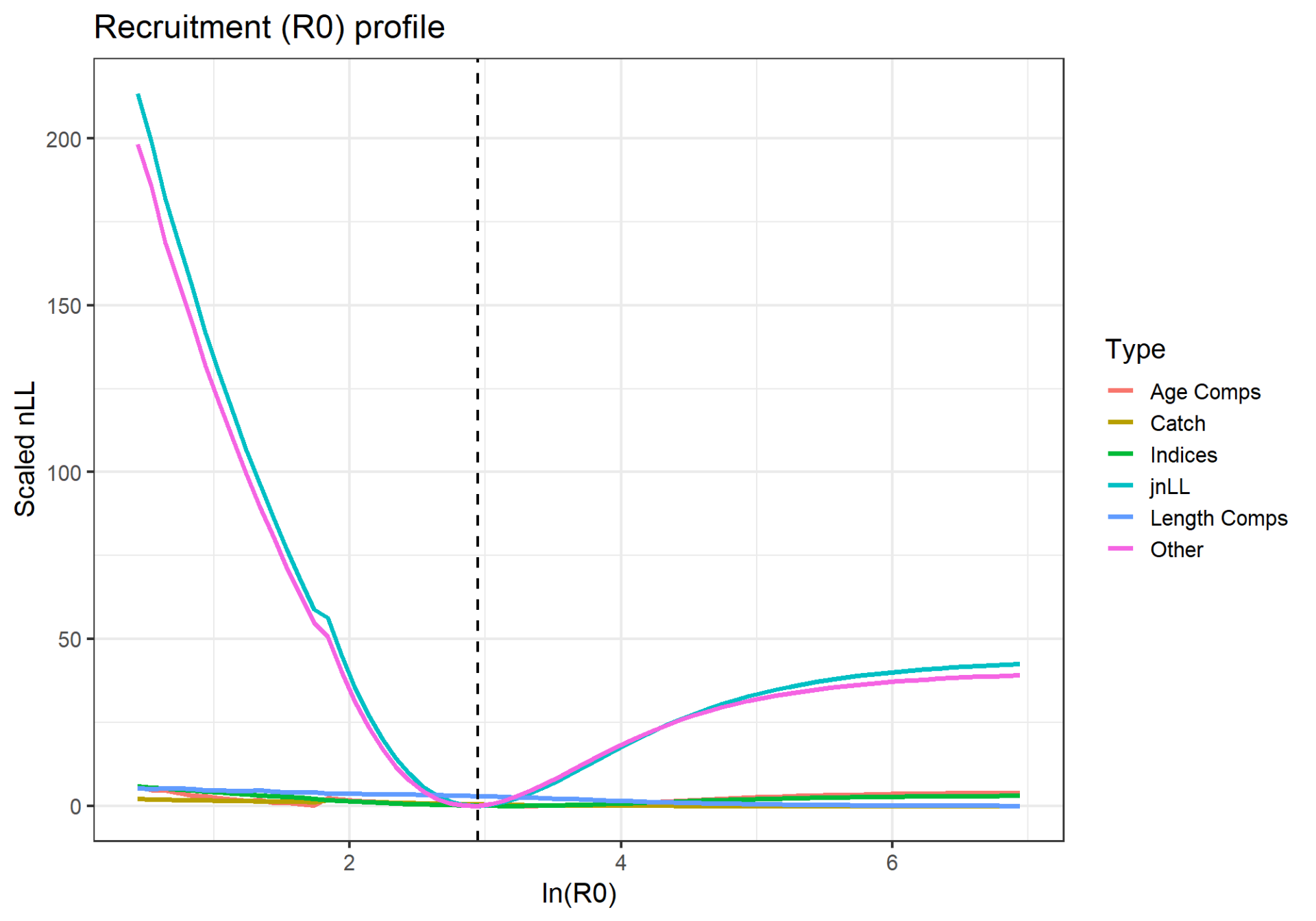
0
1
2
3
4
5
6
7

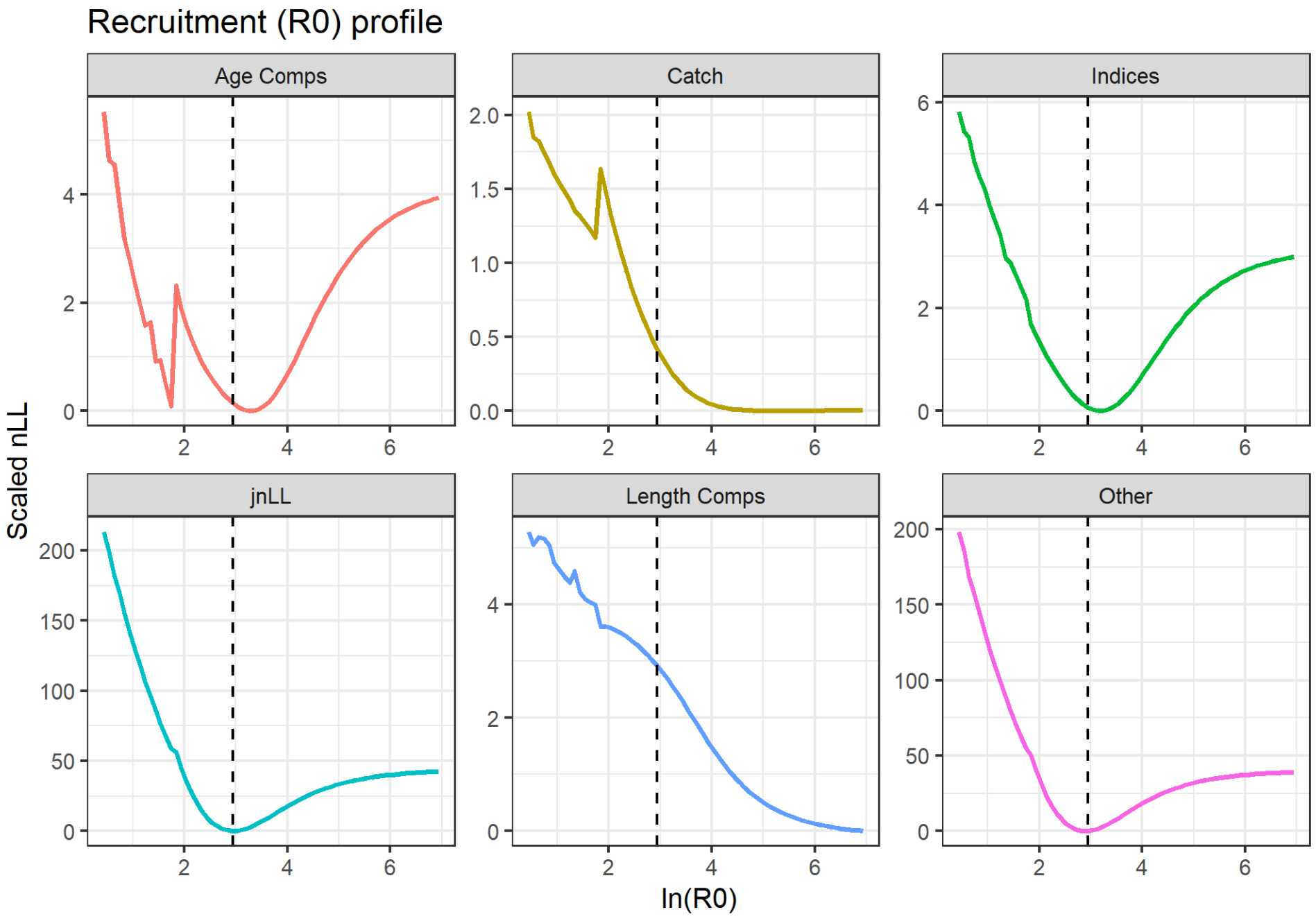
Years since cohort was last estimated

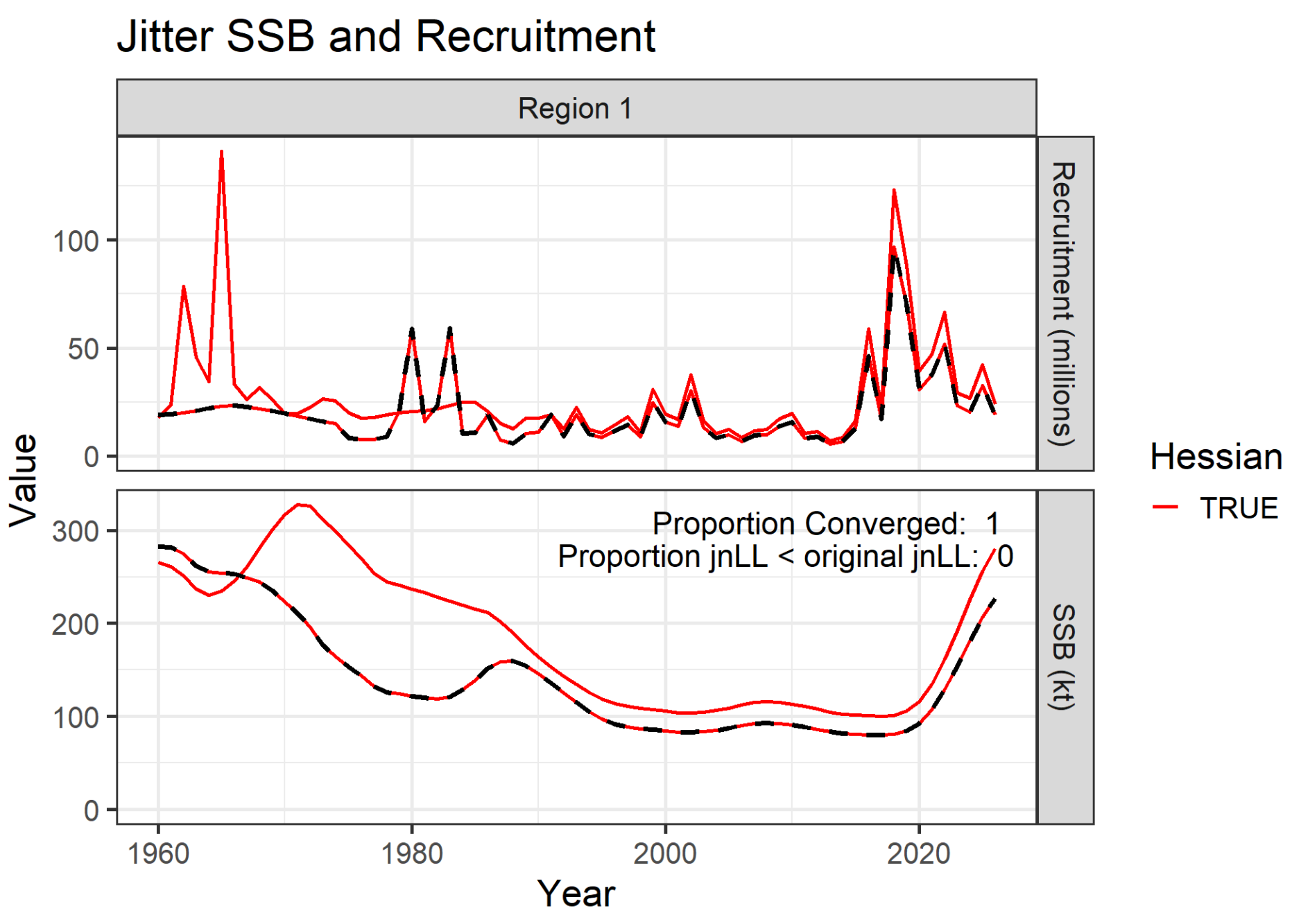
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

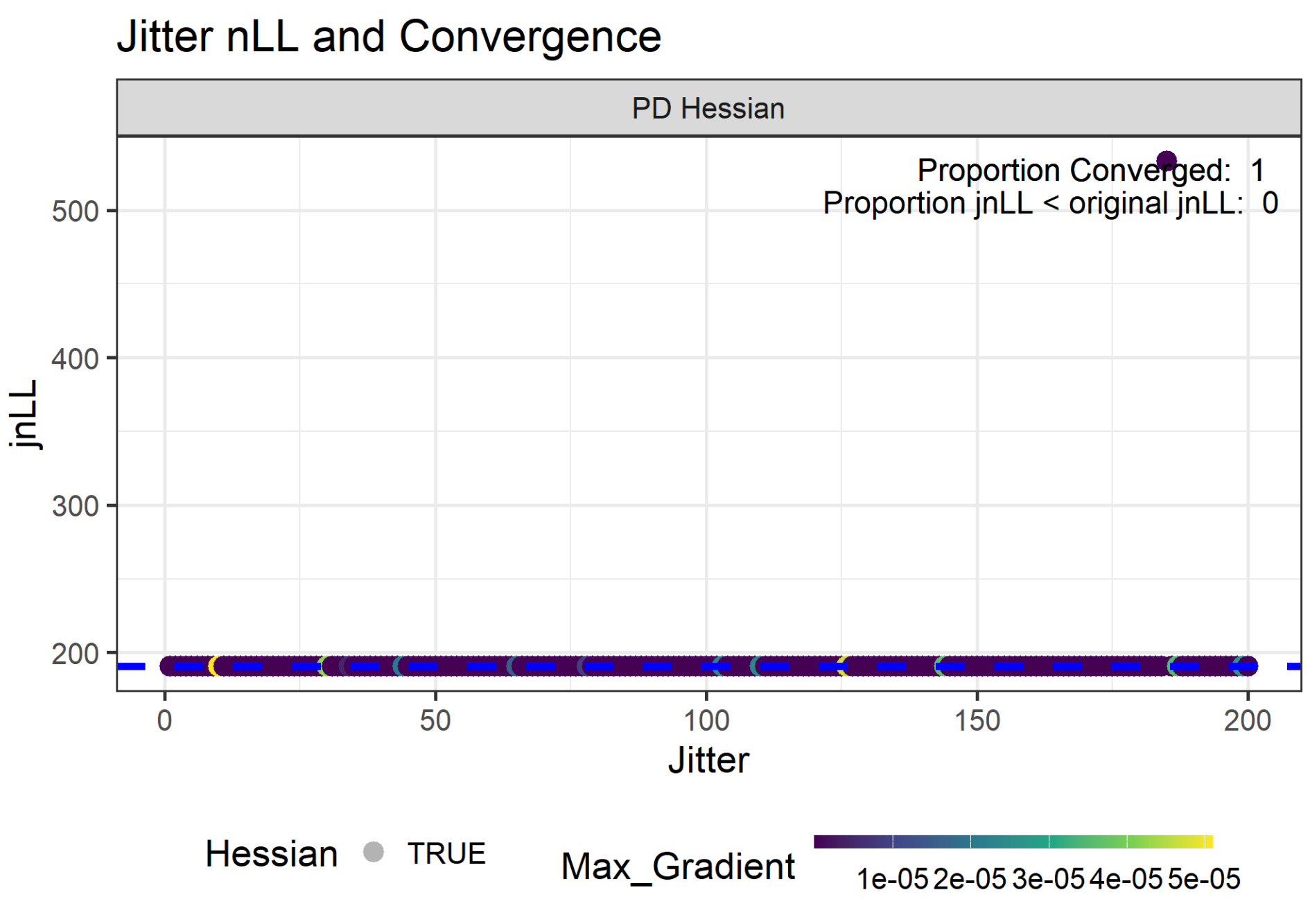

Natural mortality profile
Scaled nLL
750
500
250
0
-5
-4
-3
-2
-1
ln(M)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

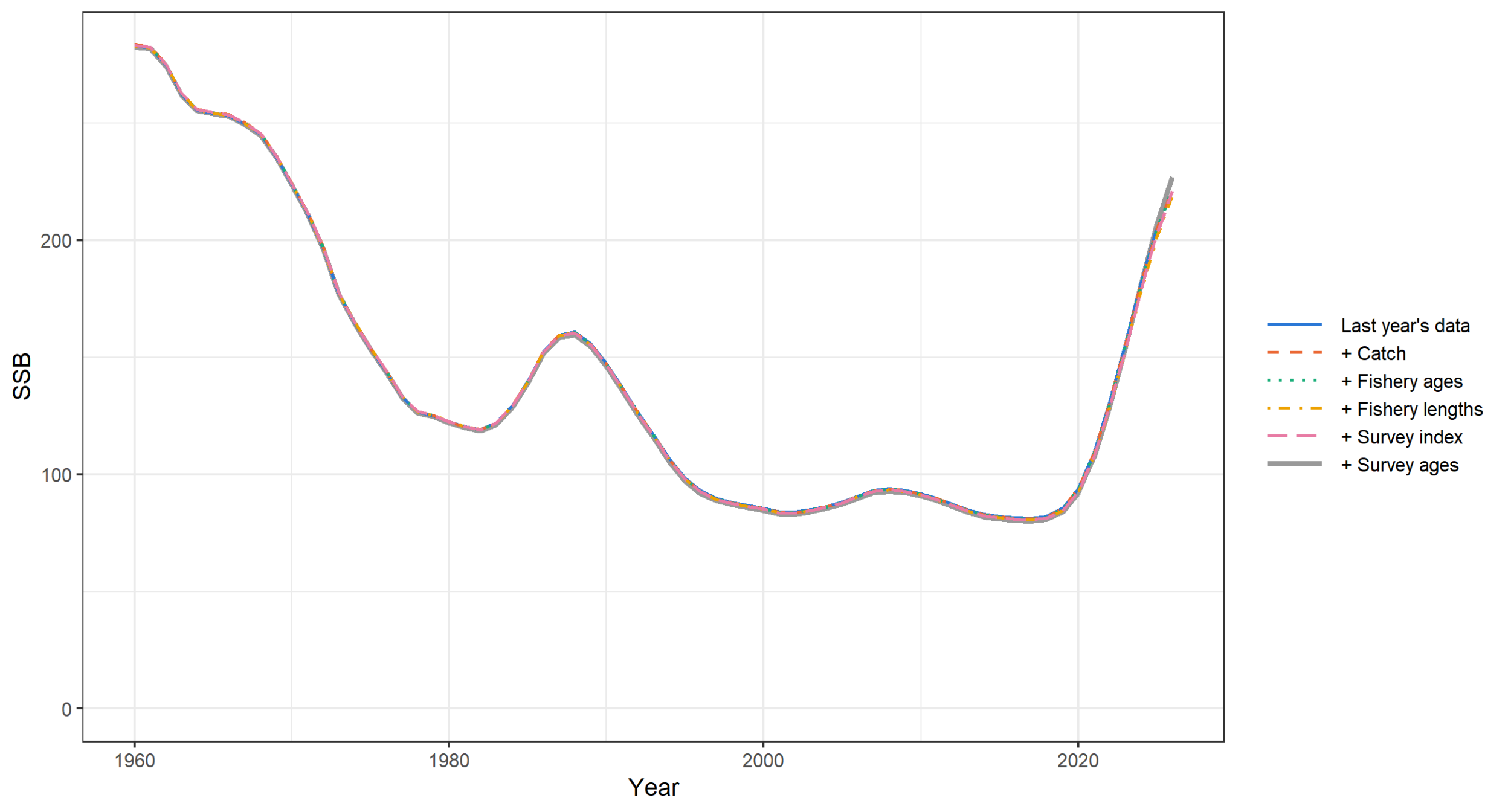

Natural mortality profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(M)

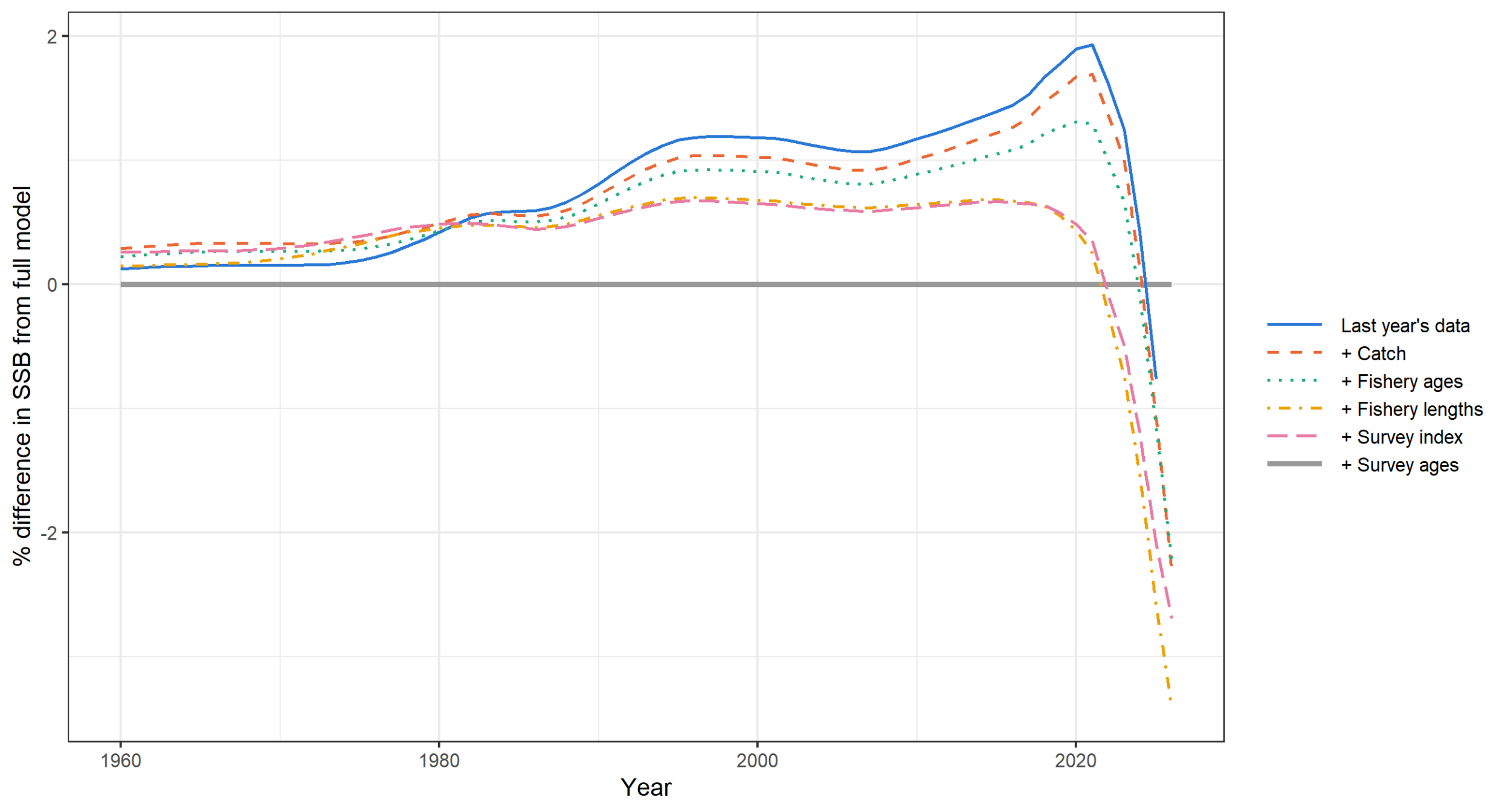

Recruitment (R0) profile
Scaled nLL
200
150
100
50
0
2
4
6
ln(R0)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

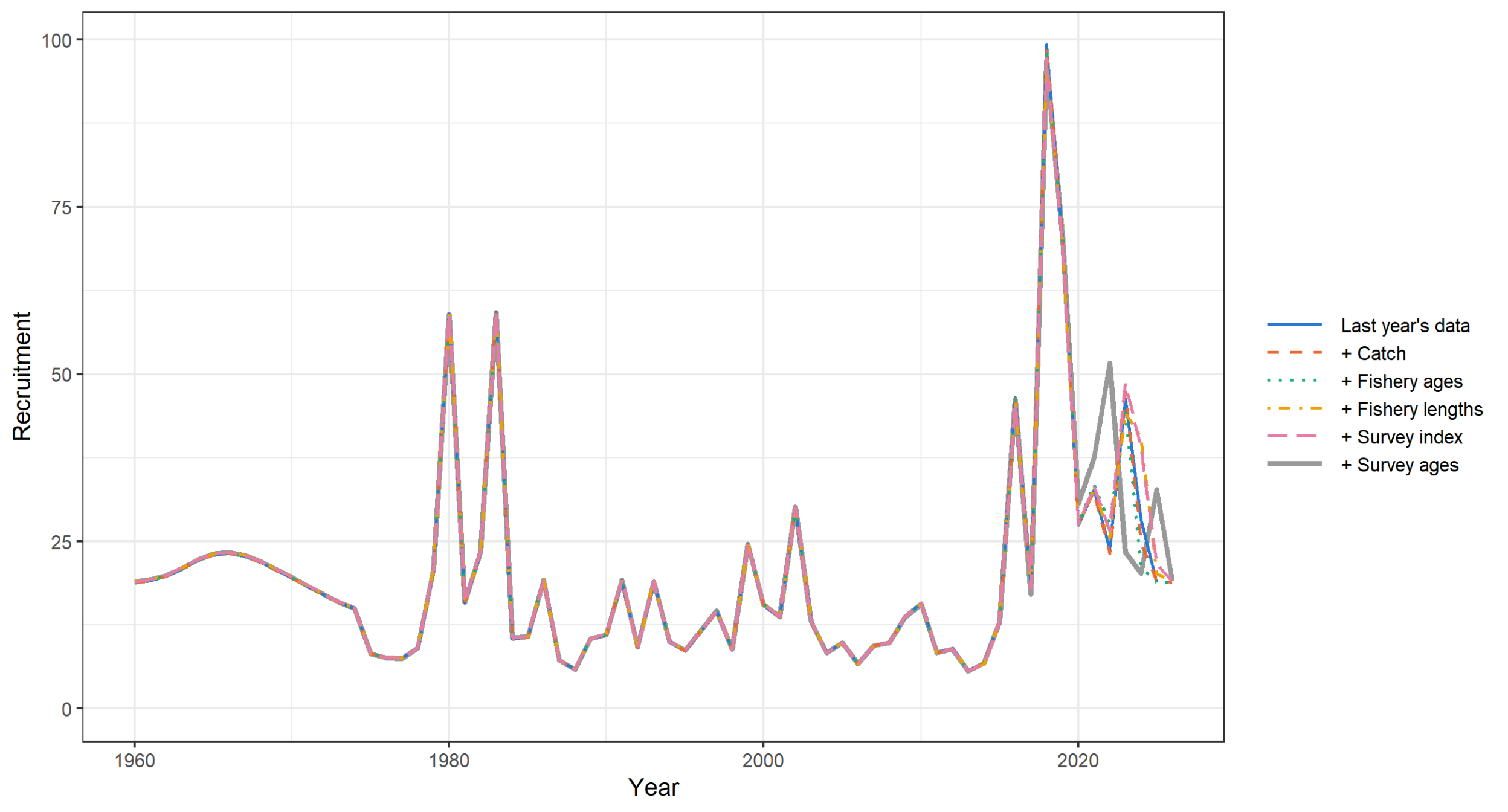

Recruitment (R0) profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(R0)

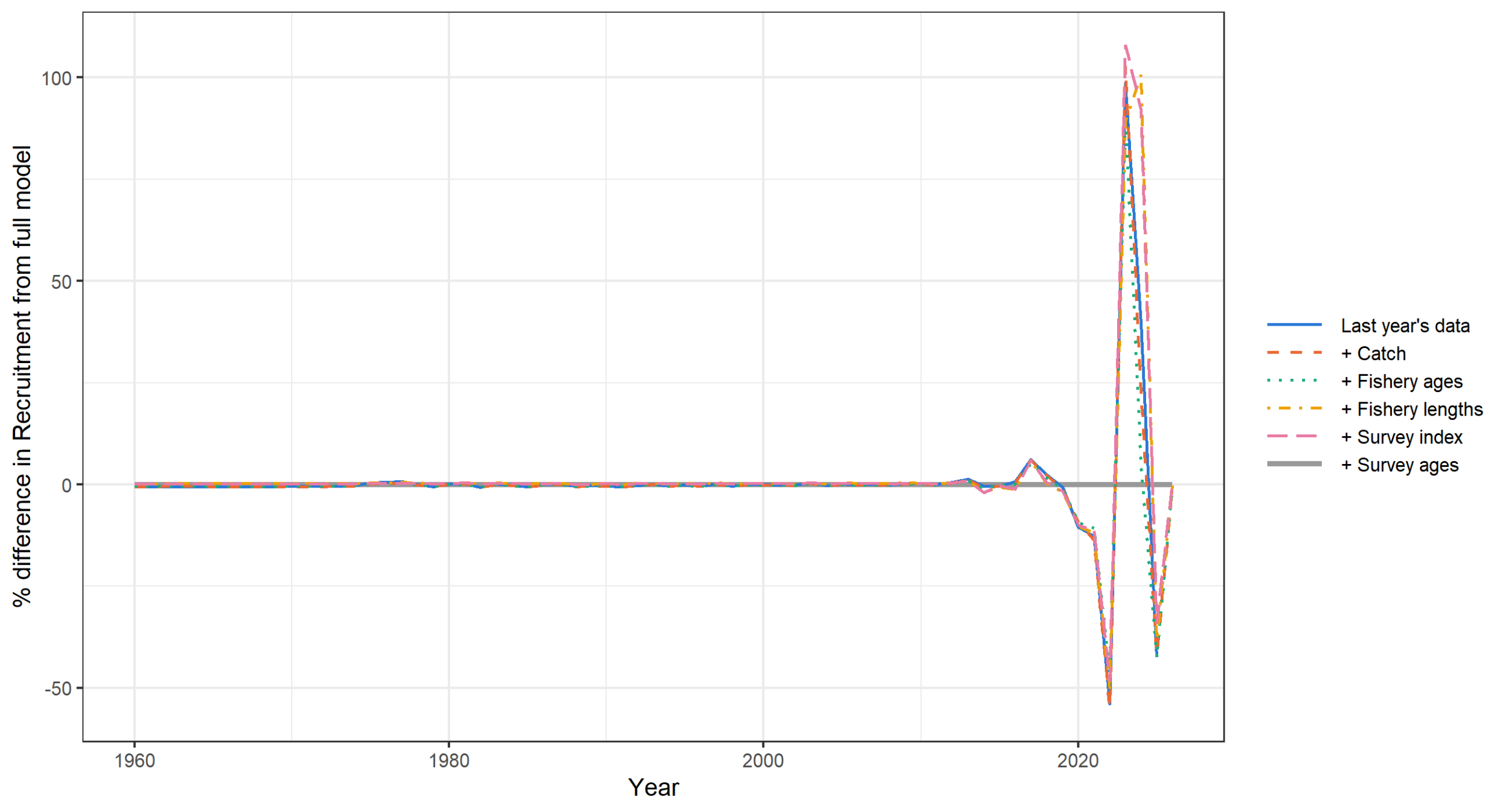

Jitter SSB and Recruitment
Region 1
Recruitment (millions)
SSB (kt)
Value
Year
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
Hessian
TRUE
0
50
100
0
100
200
300
1960
1980
2000
2020

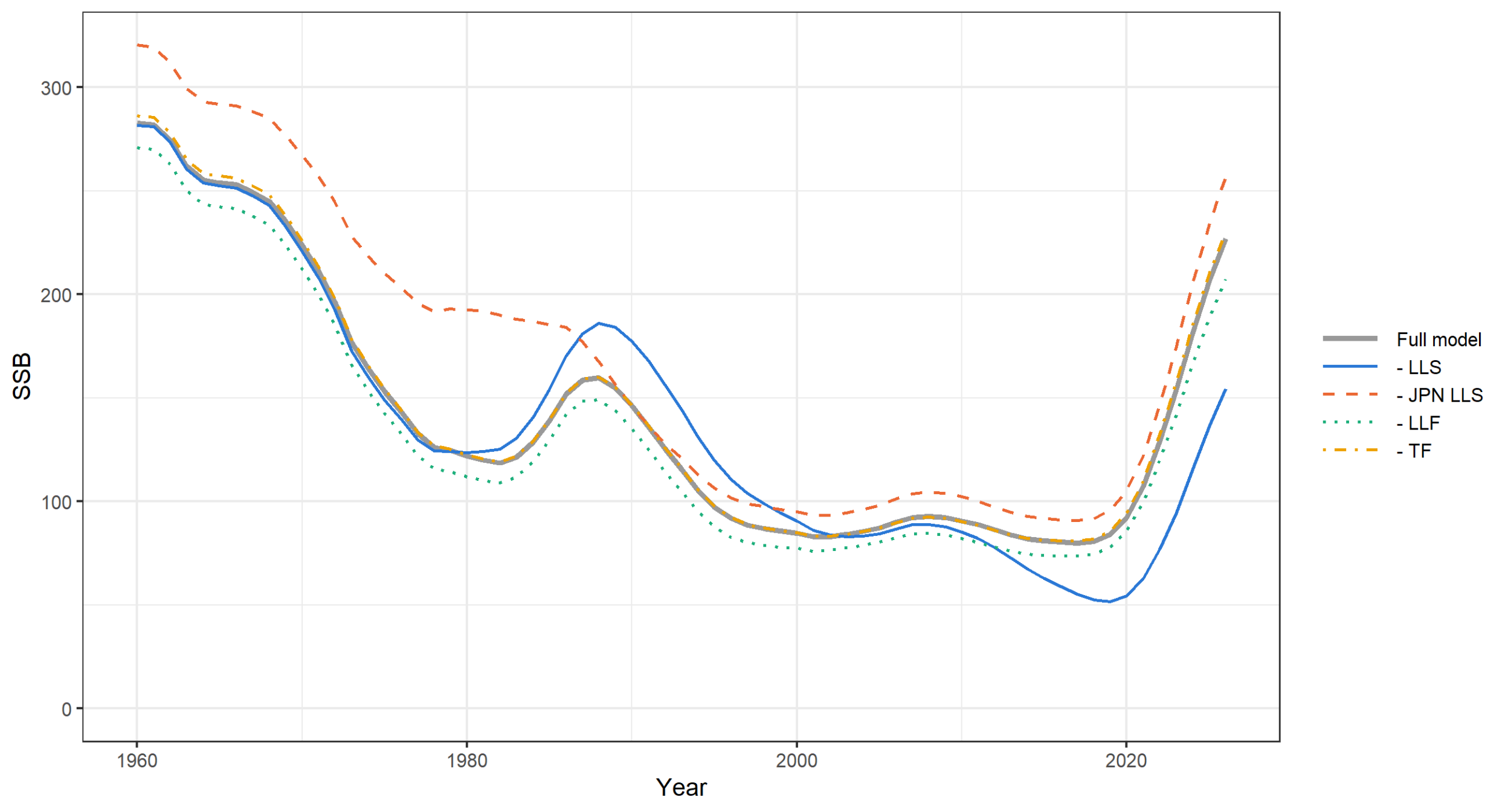

Jitter nLL and Convergence
PD Hessian
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
jnLL
500
400
300
200
0
50
100
150
200
Jitter
Hessian
TRUE
Max_Gradient
1e-05
2e-05
3e-05
4e-05
5e-05

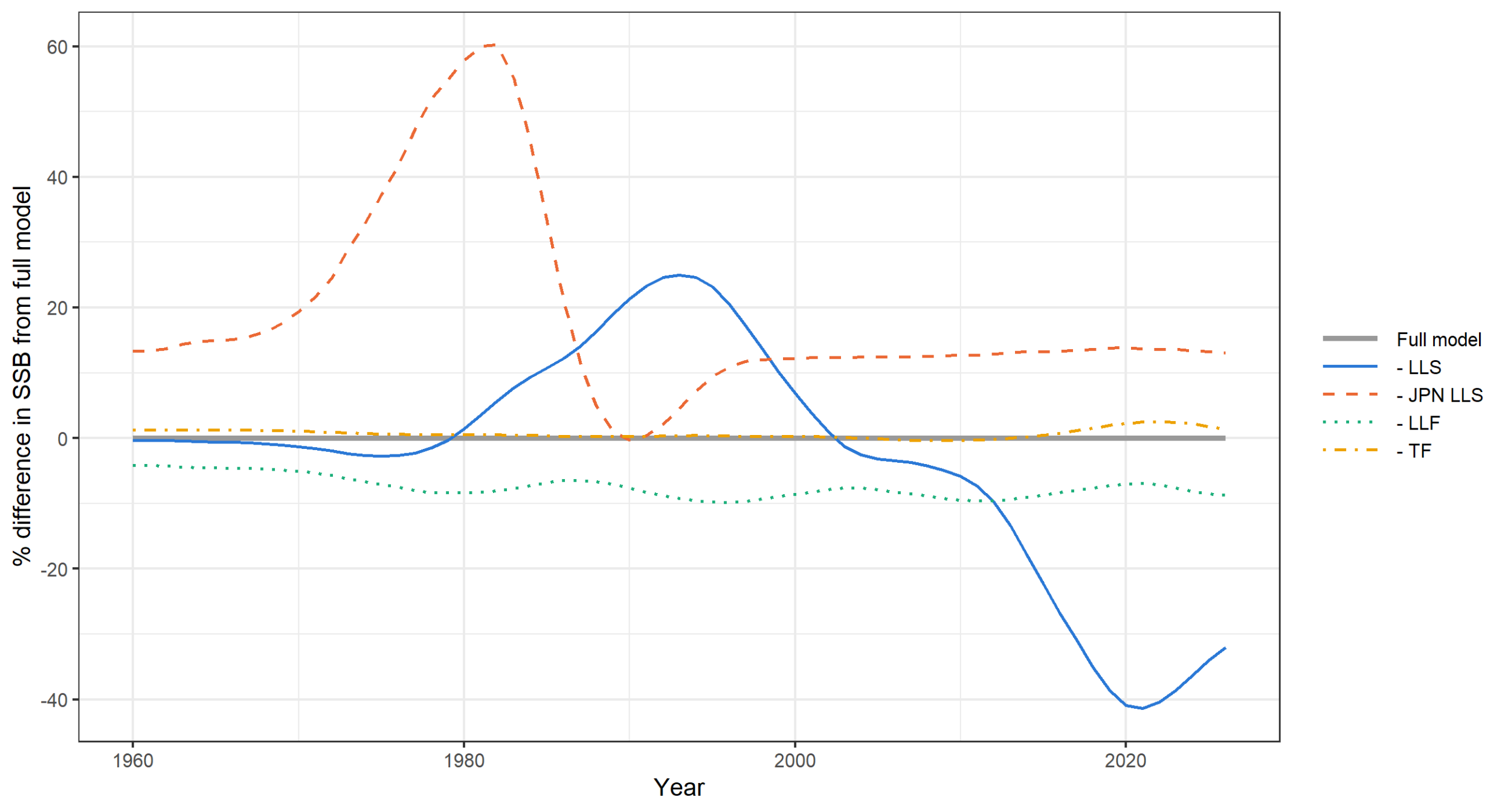

SSB
200
100
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

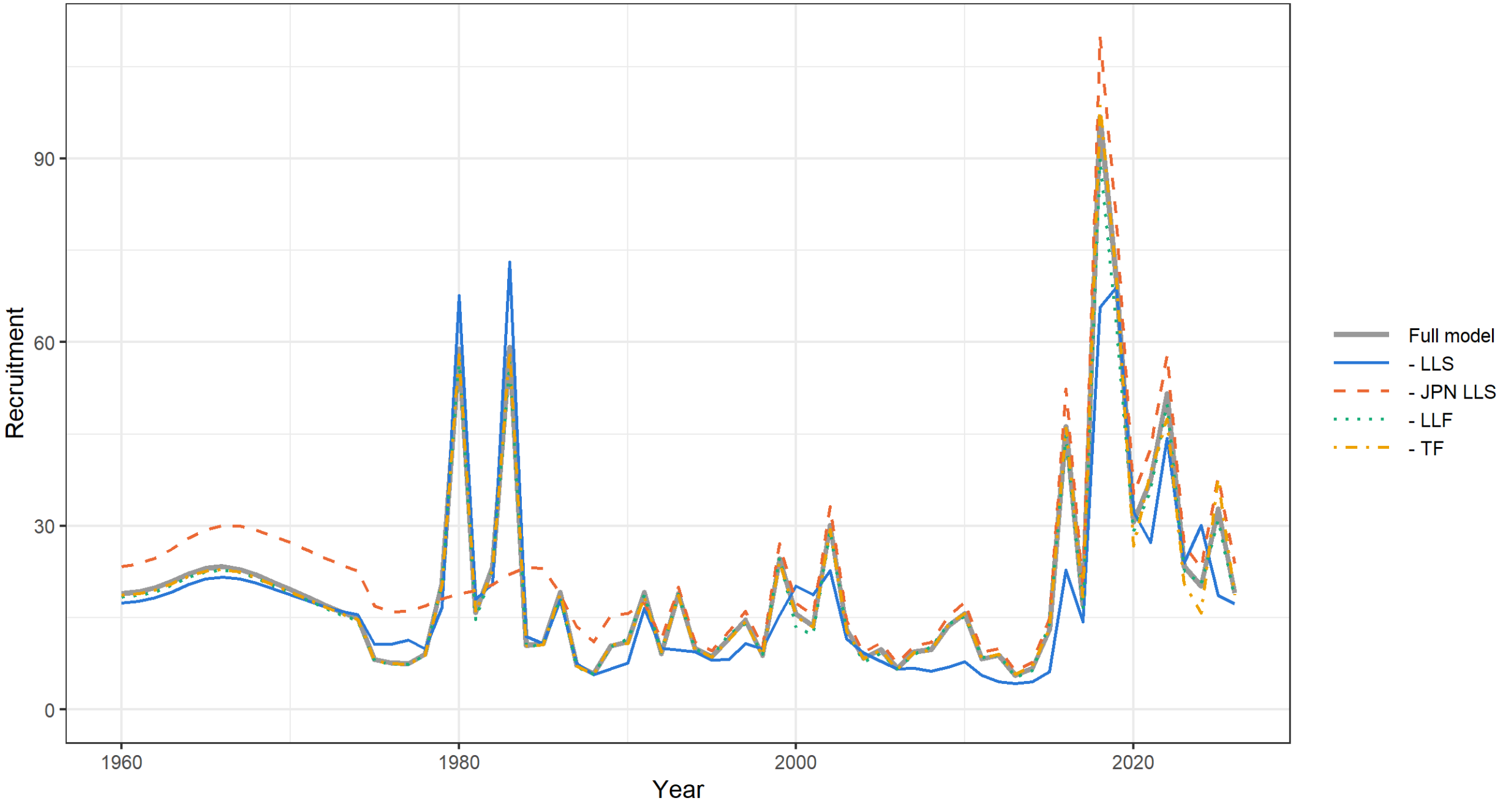

2
0
-2
% difference in SSB from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

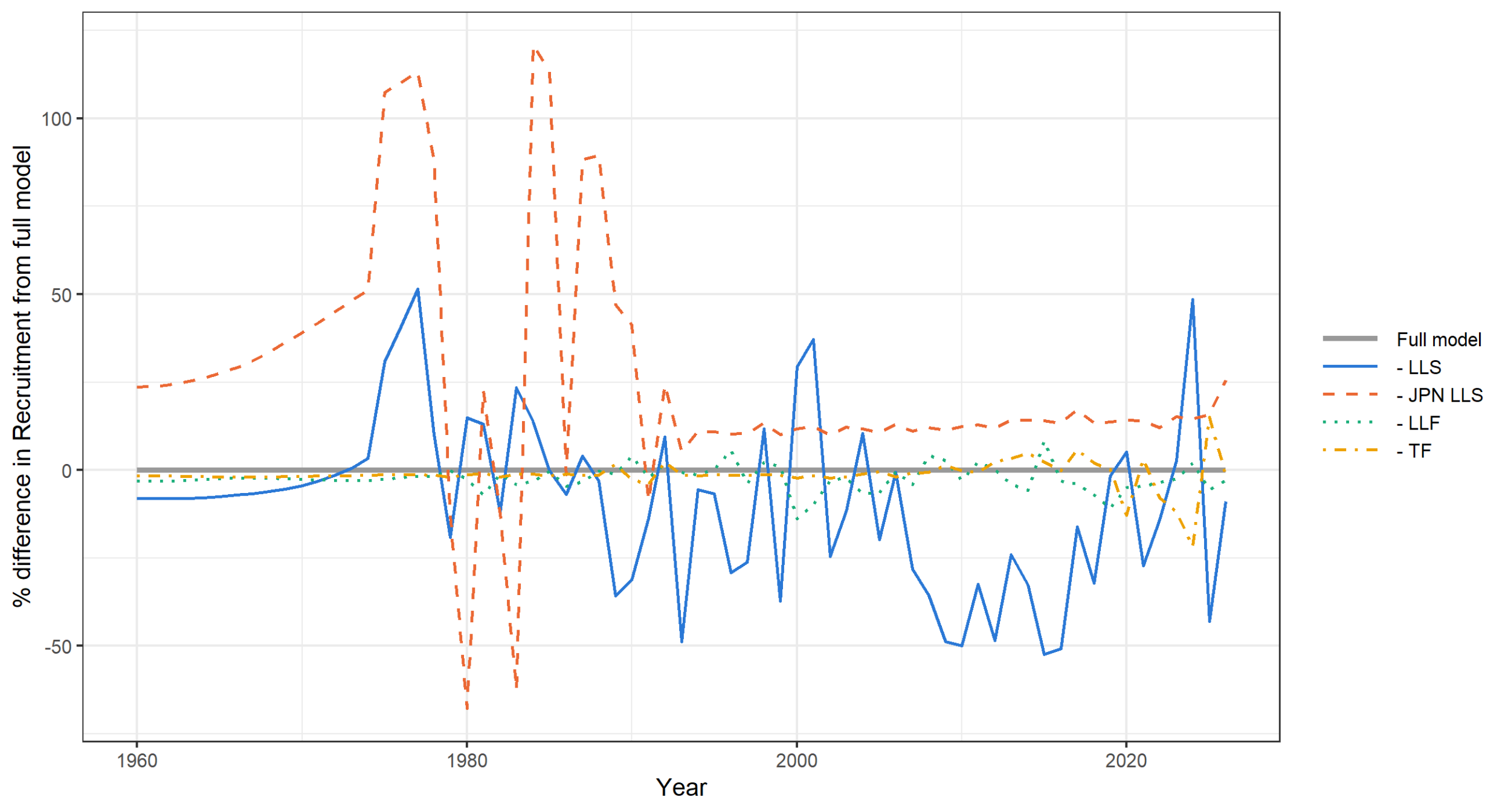

Recruitment
100
75
50
25
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

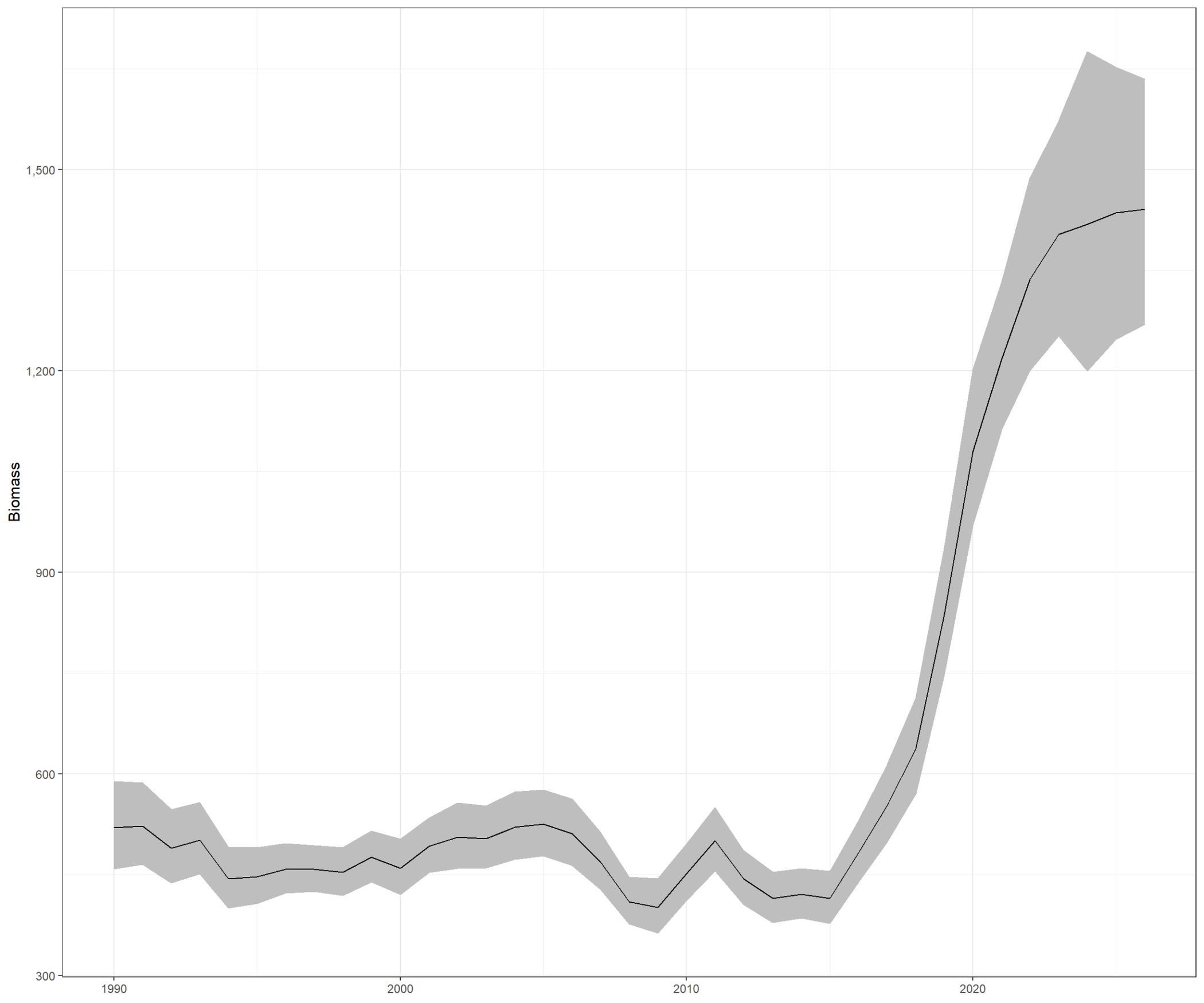

% difference in Recruitment from full model
100
50
0
-50
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


SSB
300
200
100
0
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF


60
40
20
0
-20
-40
% difference in SSB from full model
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF

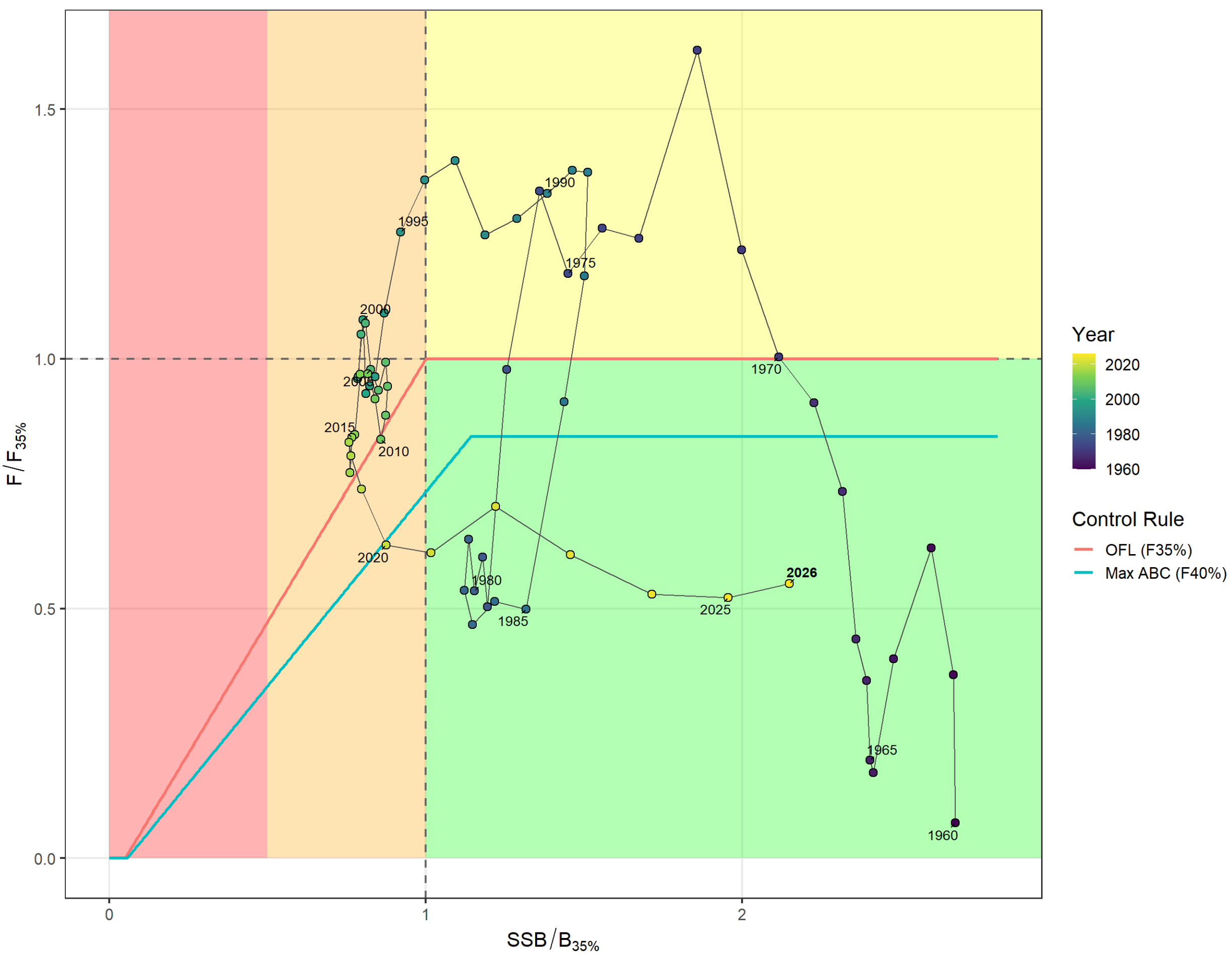

Recruitment
90
60
30
0
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF


% difference in Recruitment from full model
100
50
0
-50
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF


Biomass
1,500
1,200
900
600
300
1990
2000
2010
2020

Source	SE	Data_Wt	ISS	ESS
LL Survey Index	0.07	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
LL Survey Ages	NA	0.33	164.21	54.34
JPN LL Survey Lengths	NA	0.33	54.62	18.27
JPN LL Survey Ages	NA	0.18	83.33	15.01
Fixed Gear Fishery Ages	NA	0.11	142.85	15.45
Trawl Fishery Lengths	NA	0.12	47.62	5.86

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.1_1Reg_REMA	1	227.0554	393.8825	0.04667029	38.52187	138.7196	0.07168	1.63679	0.57645	0.65107


1.5
1.0
0.5
0.0
F/F35%
0
1
2
SSB/B35%
1990
1995
1975
2000
2005
2015
2010
2020
1980
1985
2025
2026
1970
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0
100
200
300
400

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0
100
200
300
400

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
250
500
750

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
250
500
750

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
25
50
75
100

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
25
50
75
100

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.02

0.04

0.06

0.08

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
20
40

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

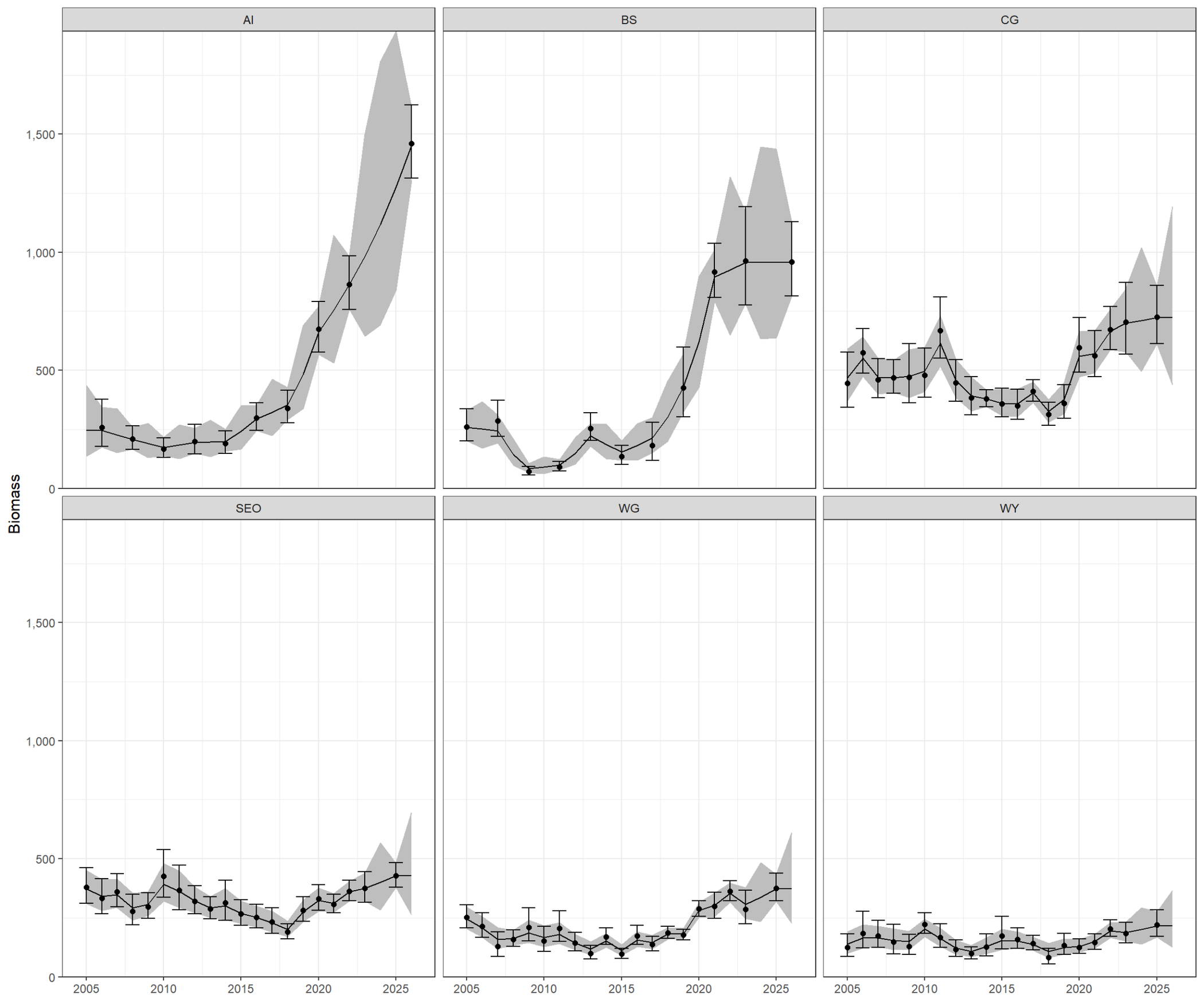
0
10
20
30
40

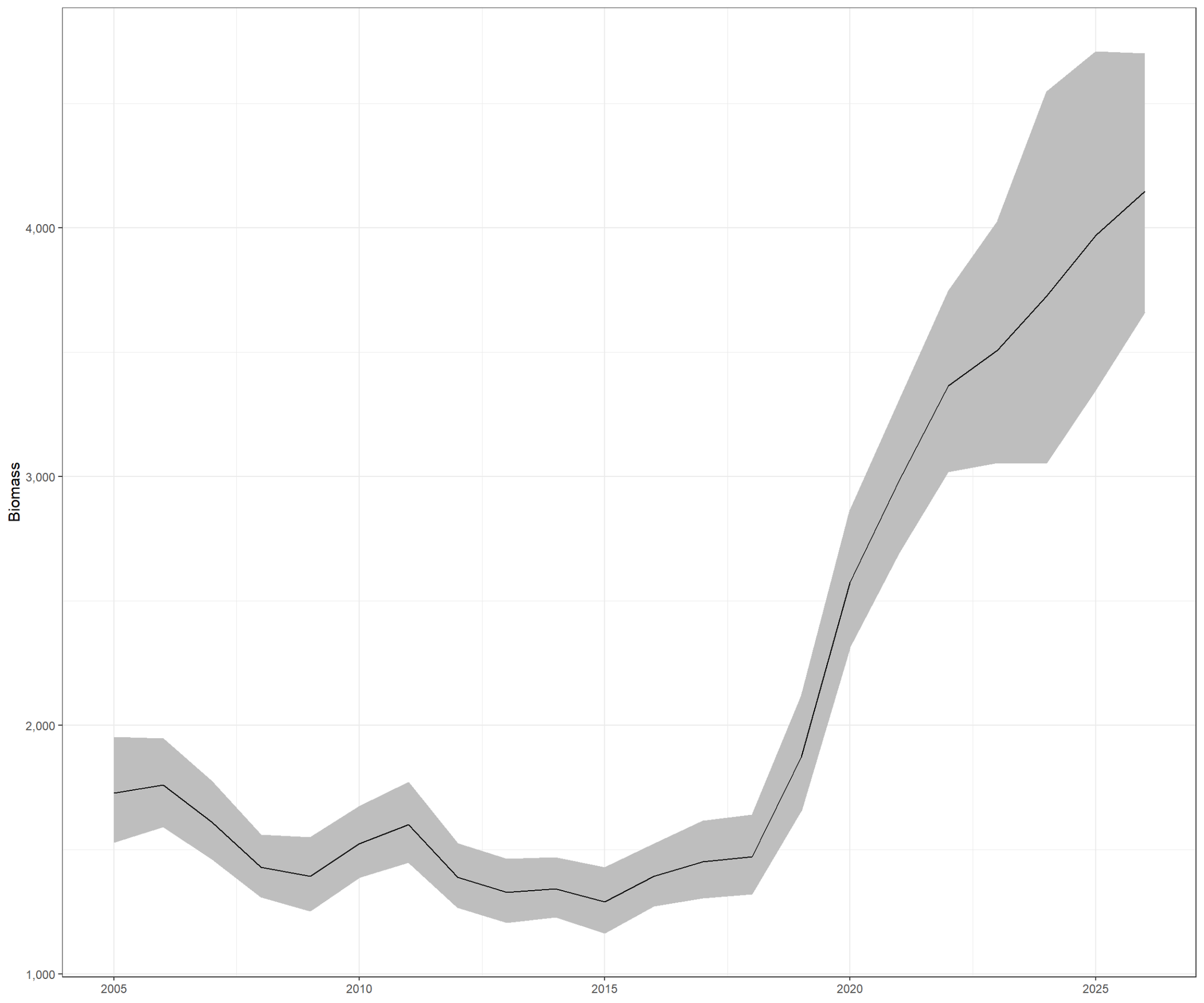
2010
2020
2030
2040

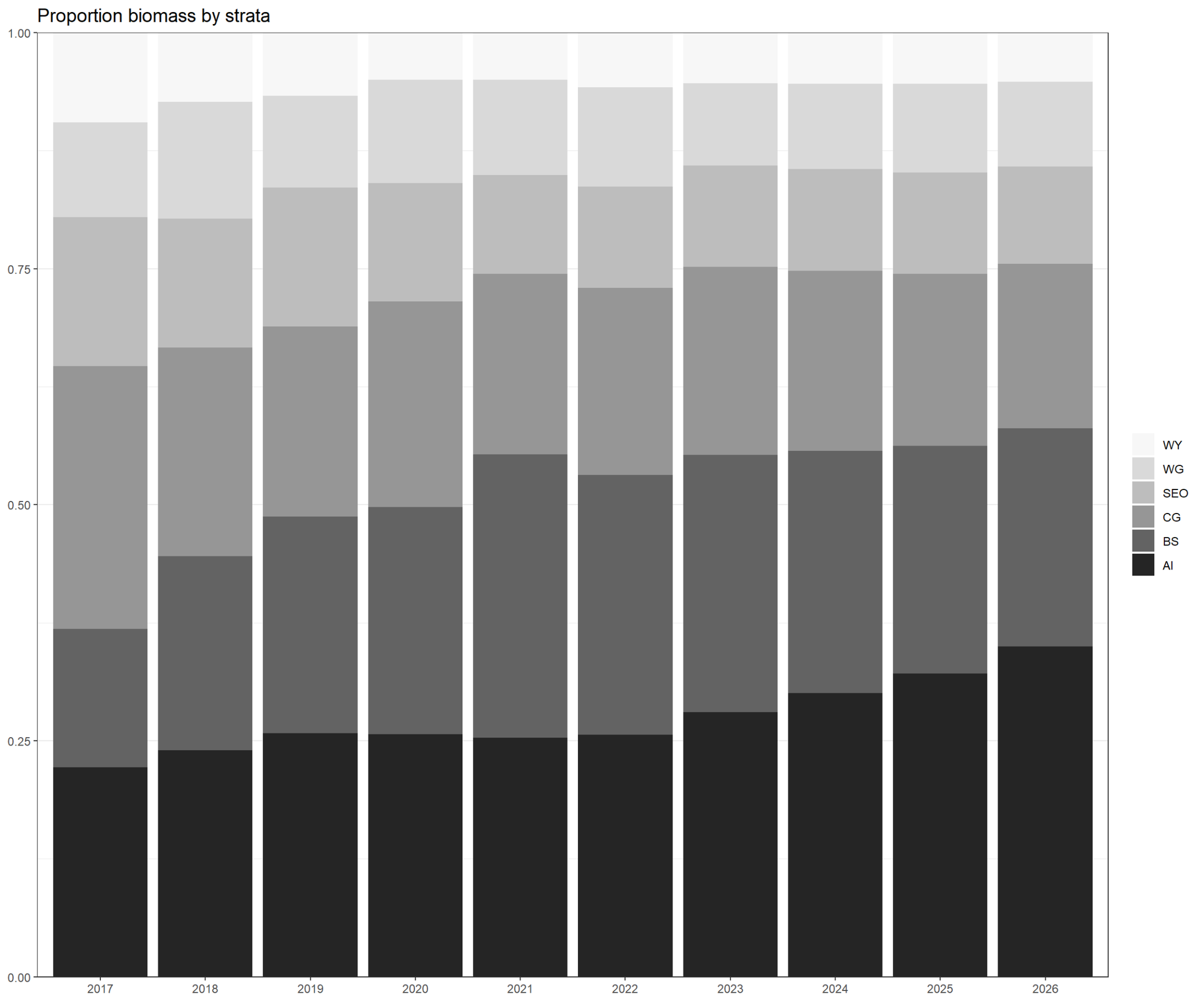
Year

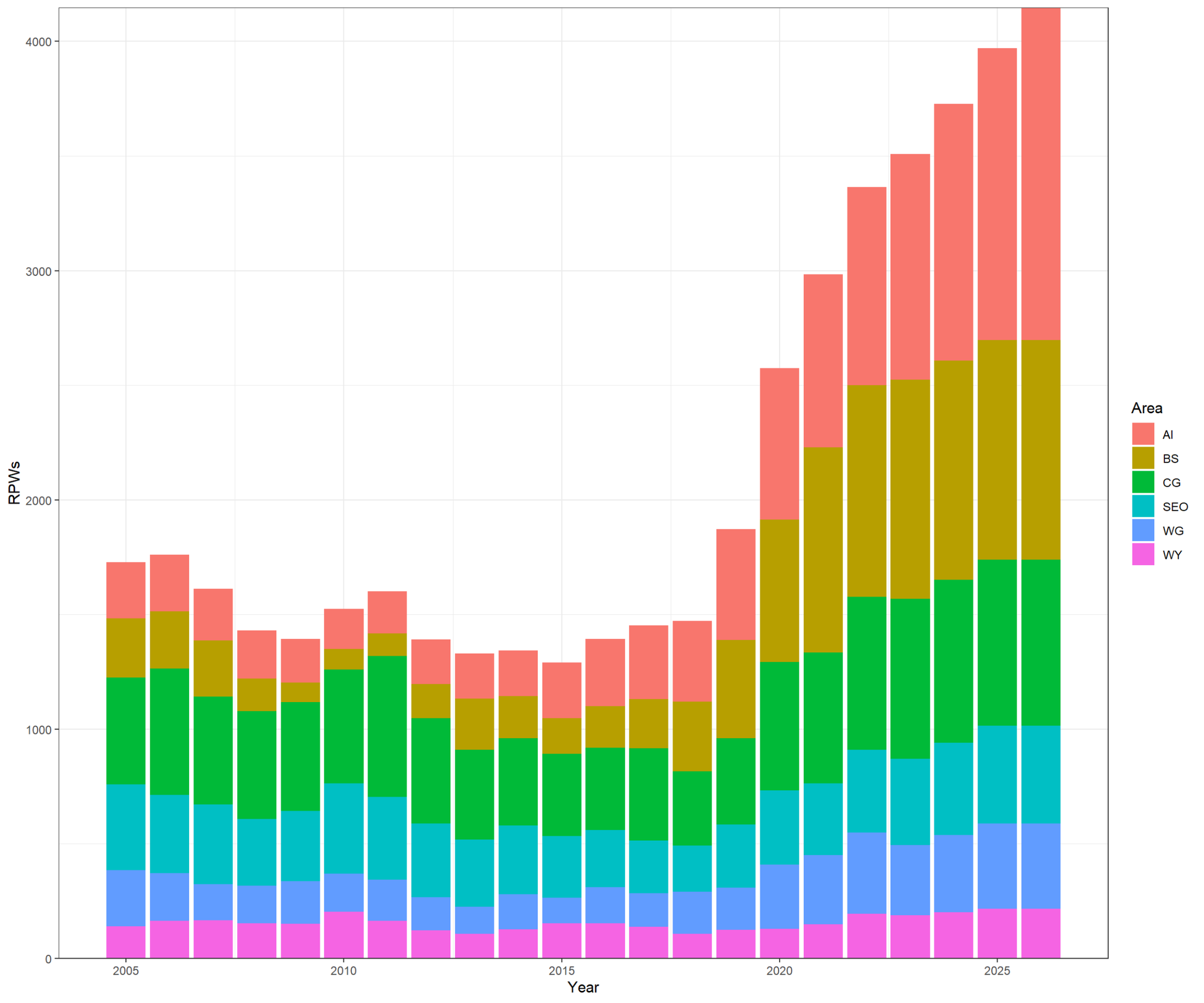
Fleet
LLF
TF

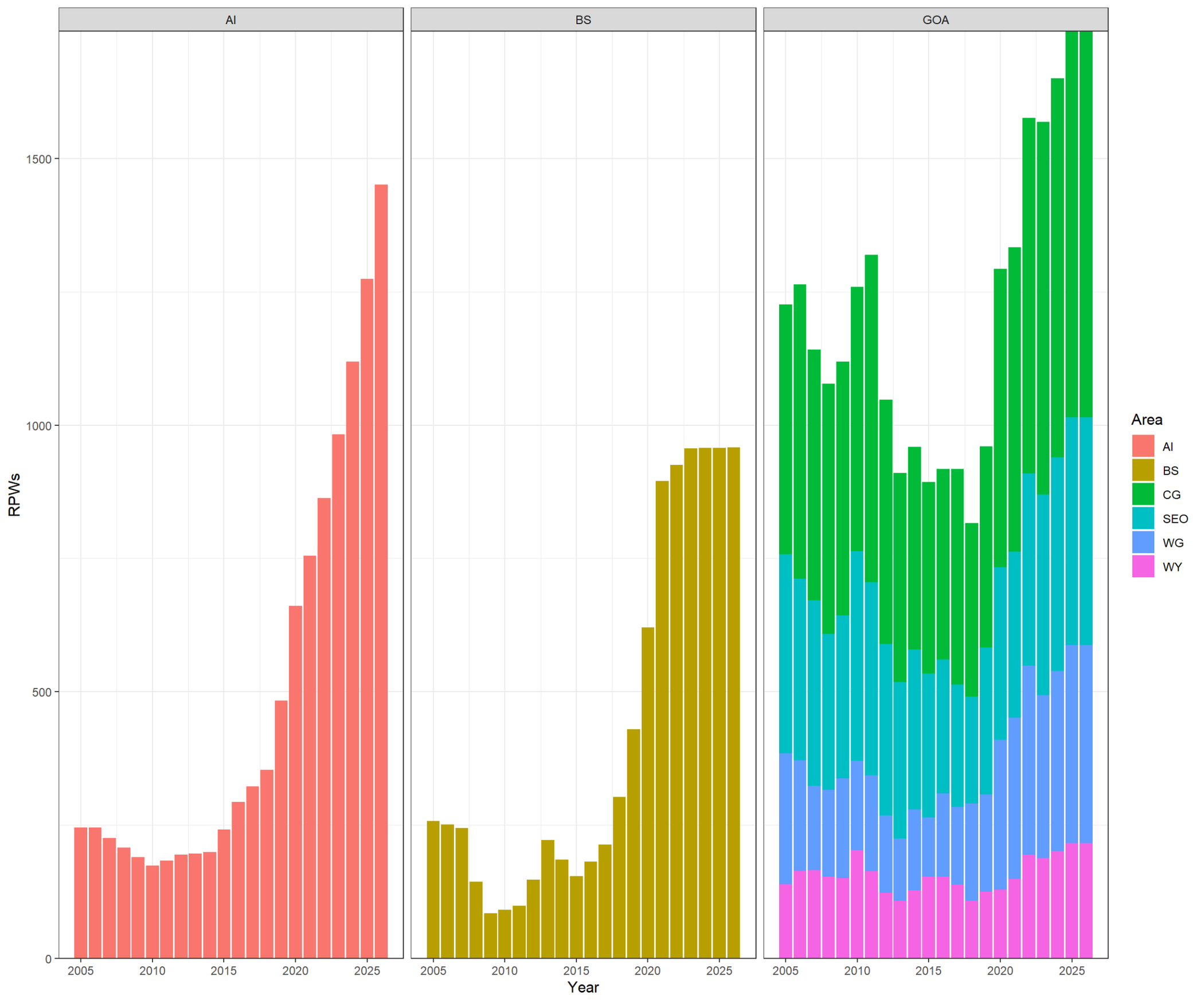
Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.091	0.091
M Male (estimated)	0.114	0.114	0.091	0.091
Average Recruitment (millions)	25.9	25.9	20.9	20.9
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	622063	619247
Projected Female Spawning Biomass (t)	219,714	241,217	241271	252096
B100% (t)	302,672	302,672	346799	346799
B40% (t)	121,069	121,069	138720	138720
B35% (t)	105,935	105,935	121380	121380
FOFL	0.102	0.102	0.085	0.085
max FABC	0.087	0.087	0.072	0.072
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	45310	45114
OFLw (t)	58,532	57,797	45179	44983
max ABC (t)	50,283	49,651	38519	38353
ABC (t)	50,283	49,651	38519	38353
ABCw (t)	50,111	49,482	38388	38222
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

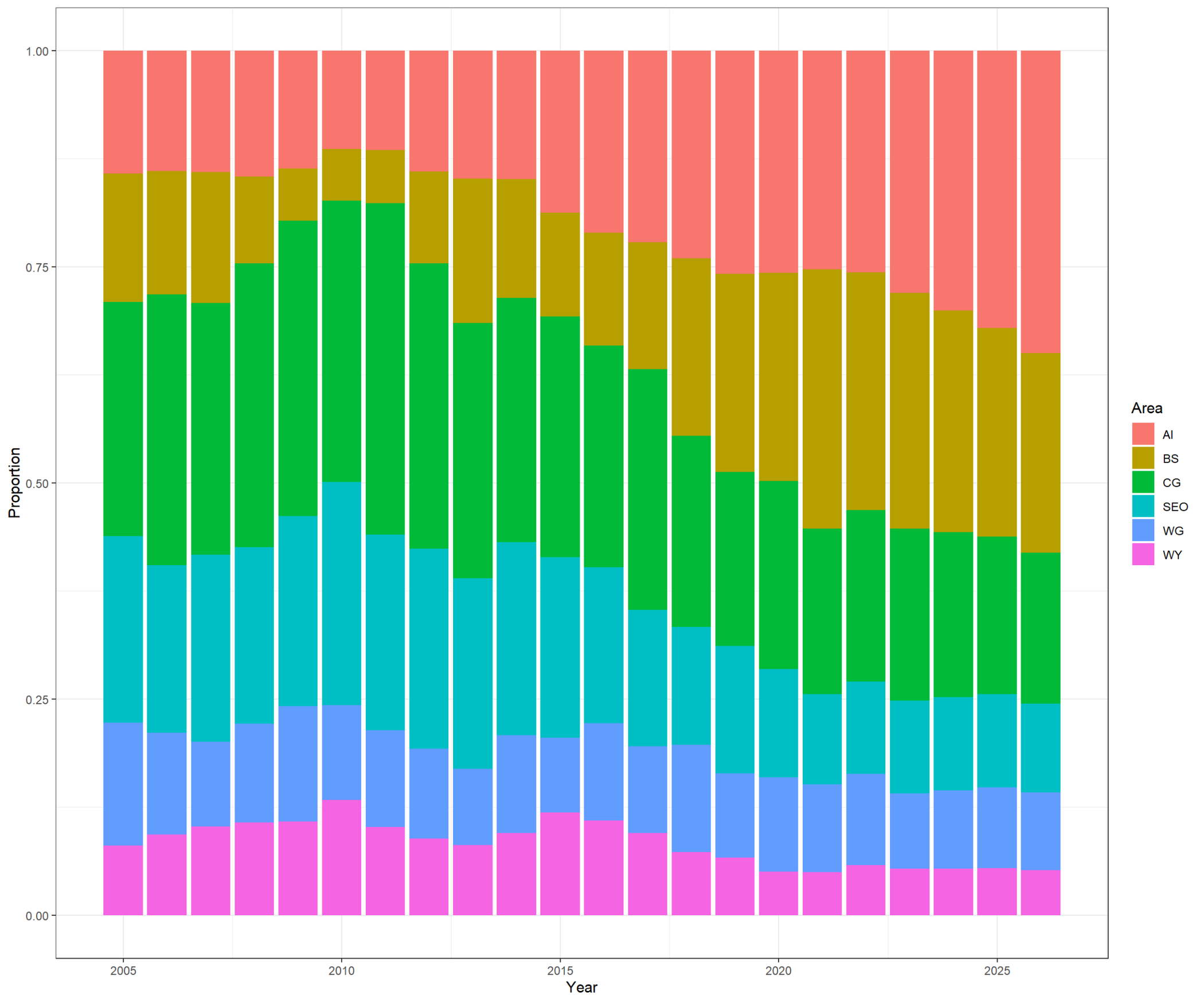

AI
BS
CG
SEO
WG
WY
Biomass
0
500
1,000
1,500
2005
2010
2015
2020
2025

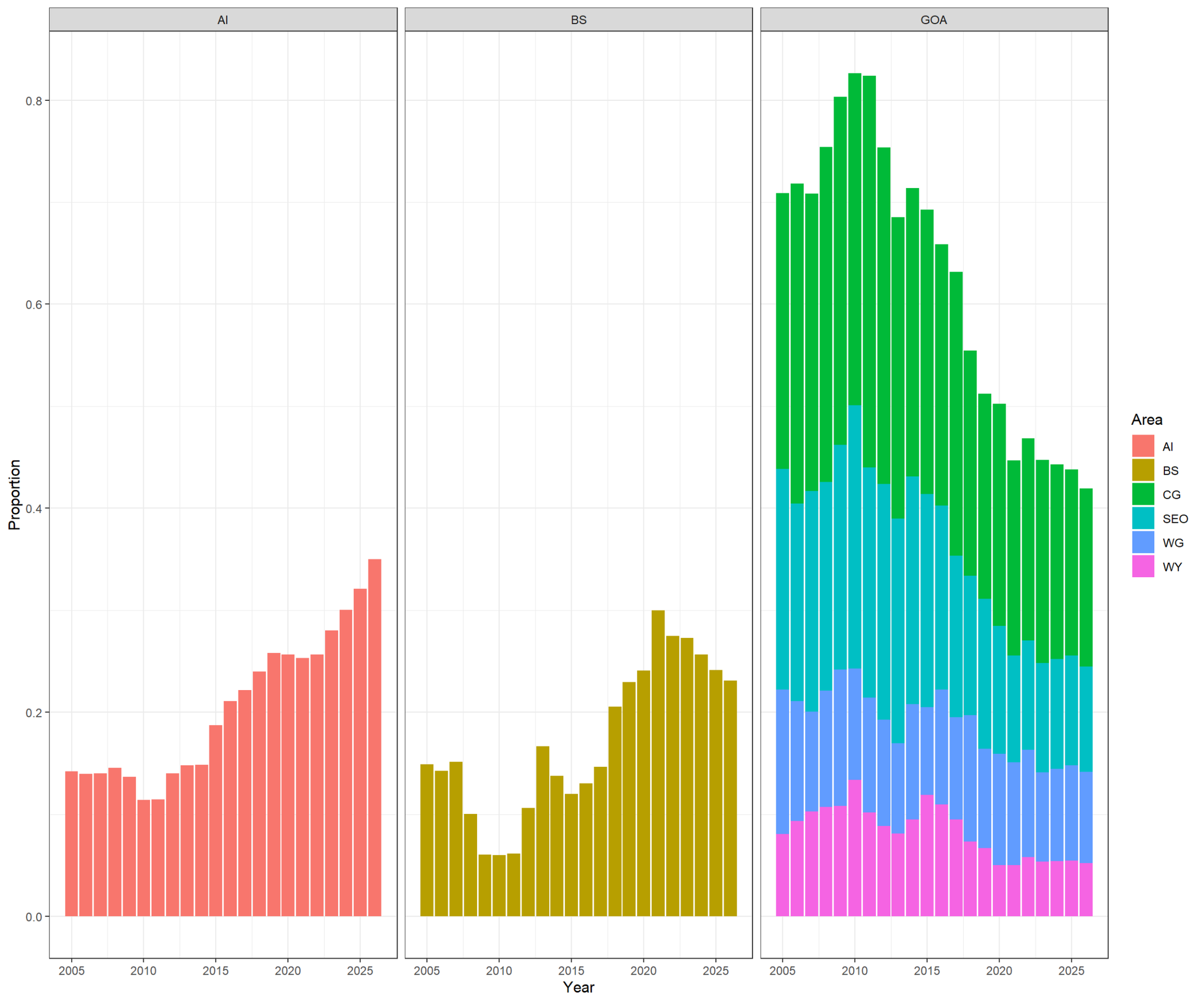

Biomass
4,000
3,000
2,000
1,000
2005
2010
2015
2020
2025

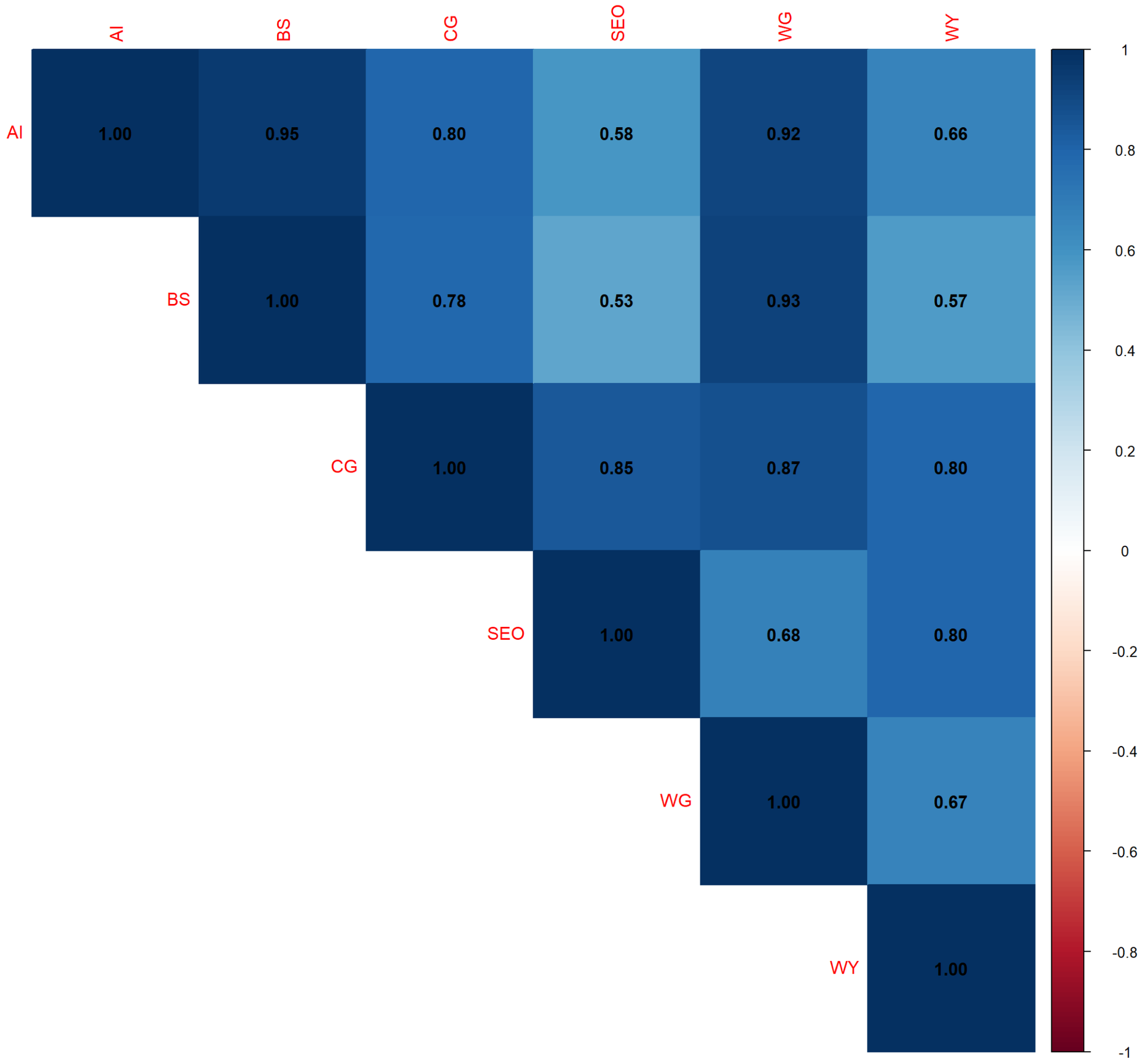

Proportion biomass by strata
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
WY
WG
SEO
CG
BS
AI

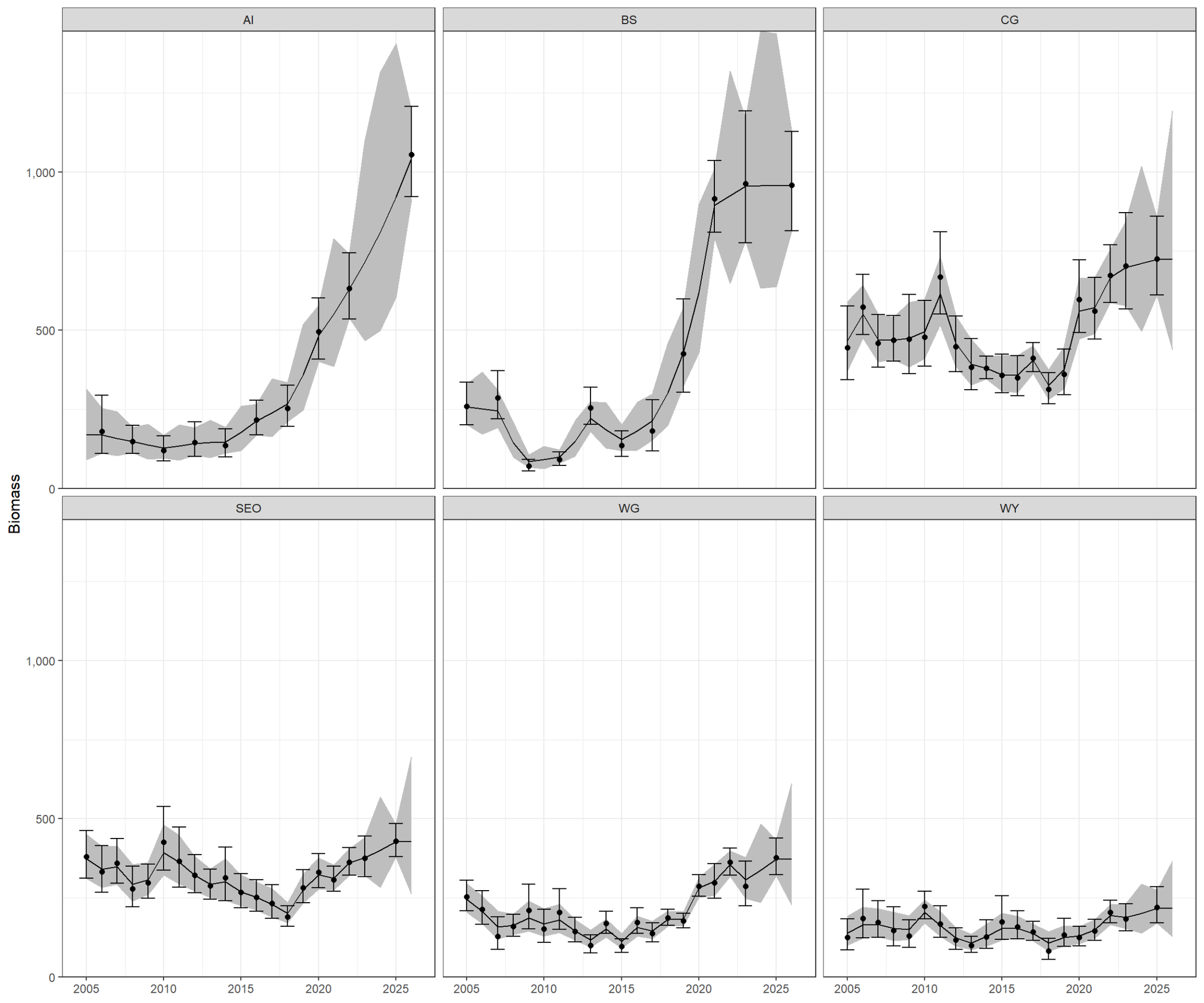

4000
3000
2000
1000
0
RPWs
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

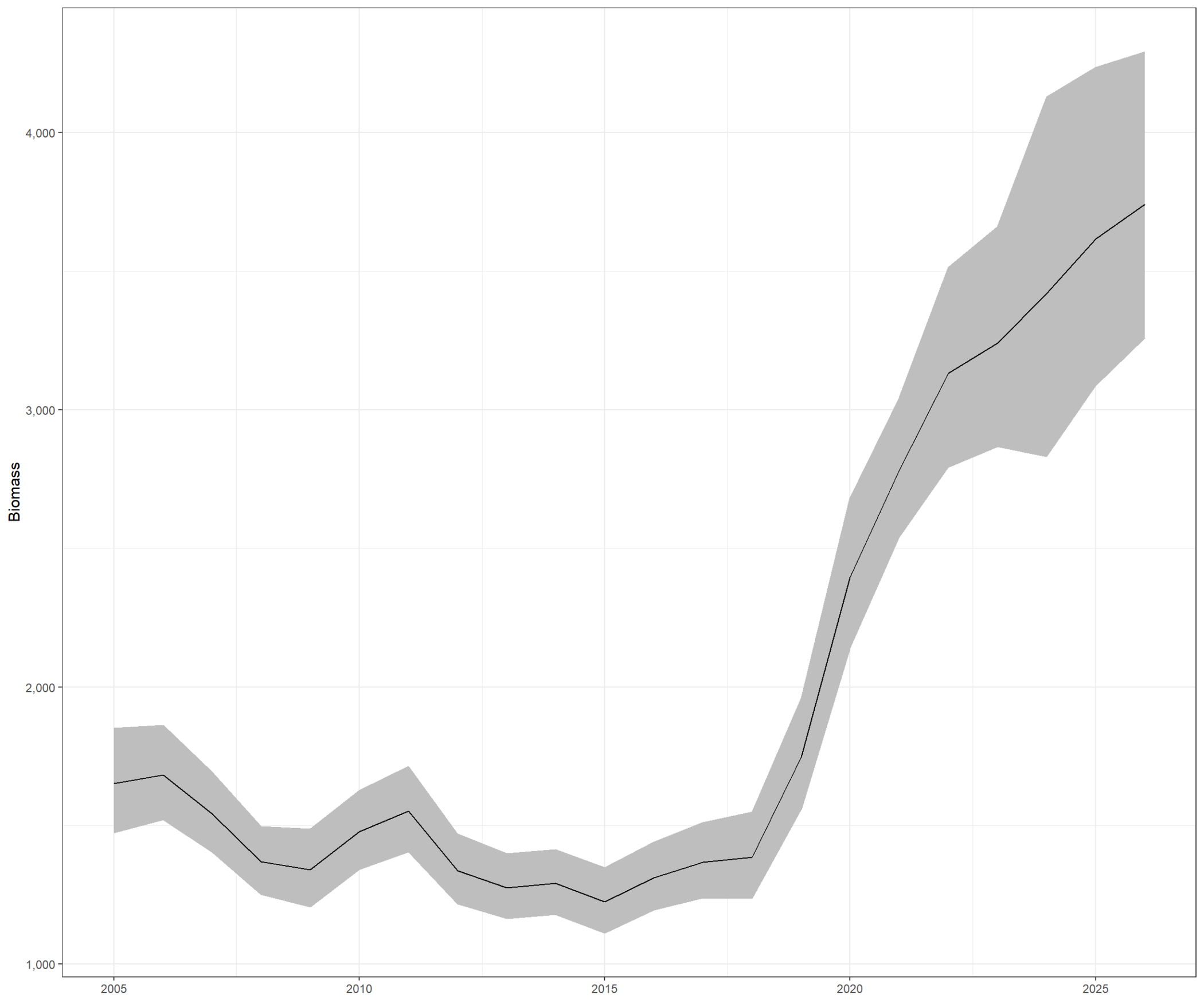

AI
BS
GOA
RPWs
0
500
1000
1500
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

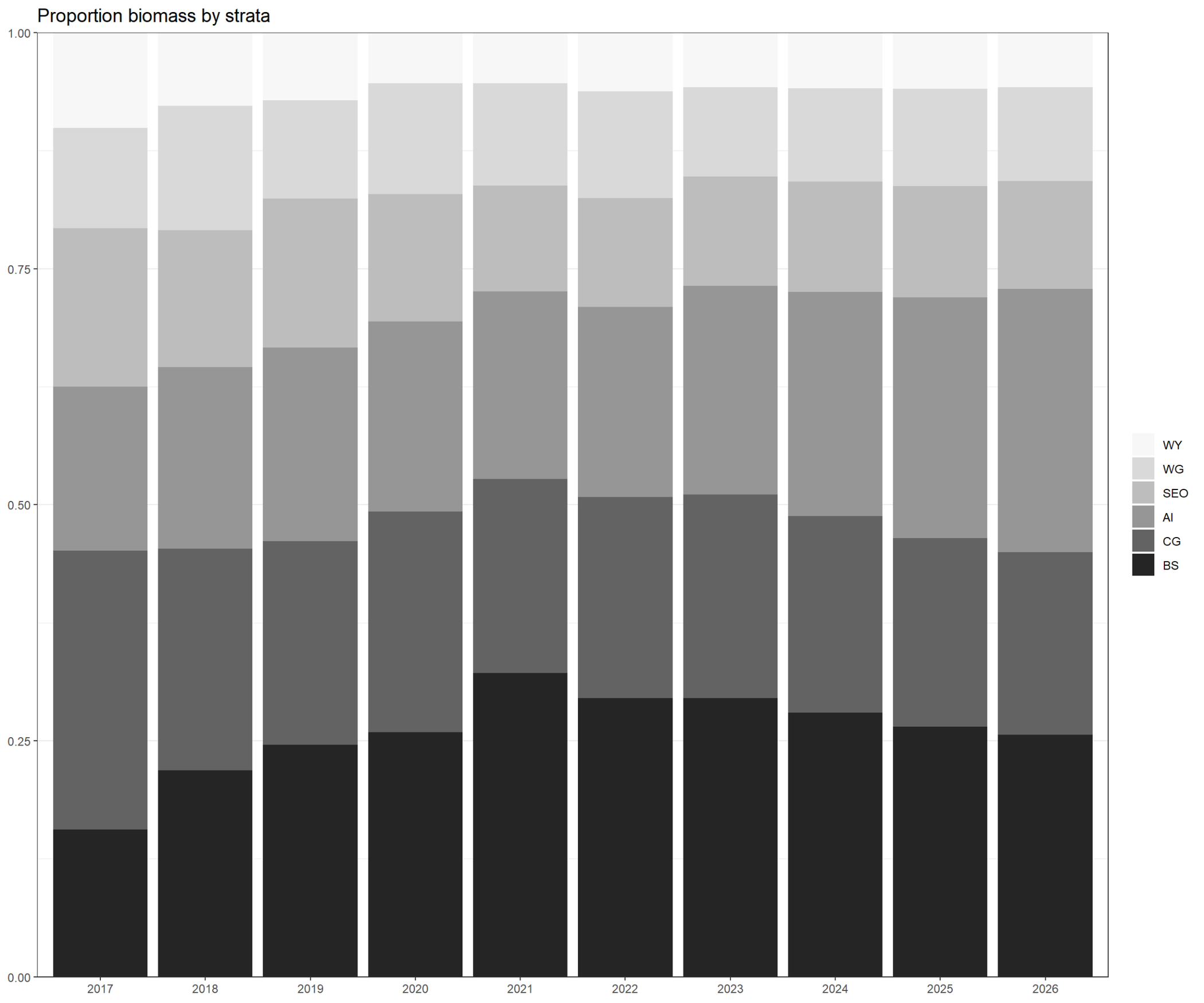

1.00
0.75
0.50
0.25
0.00
Proportion
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

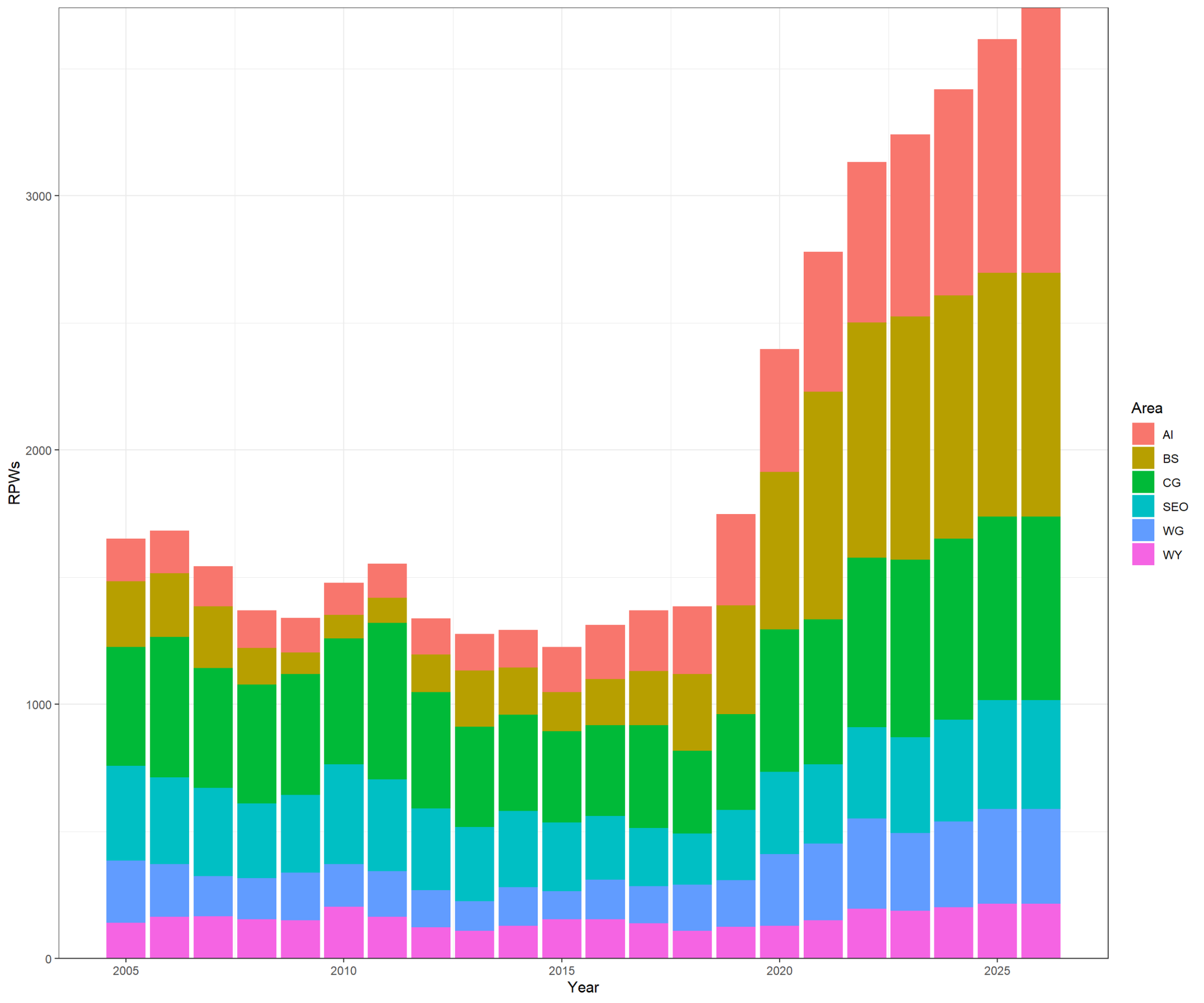

AI
BS
GOA
Proportion
0.0
0.2
0.4
0.6
0.8
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

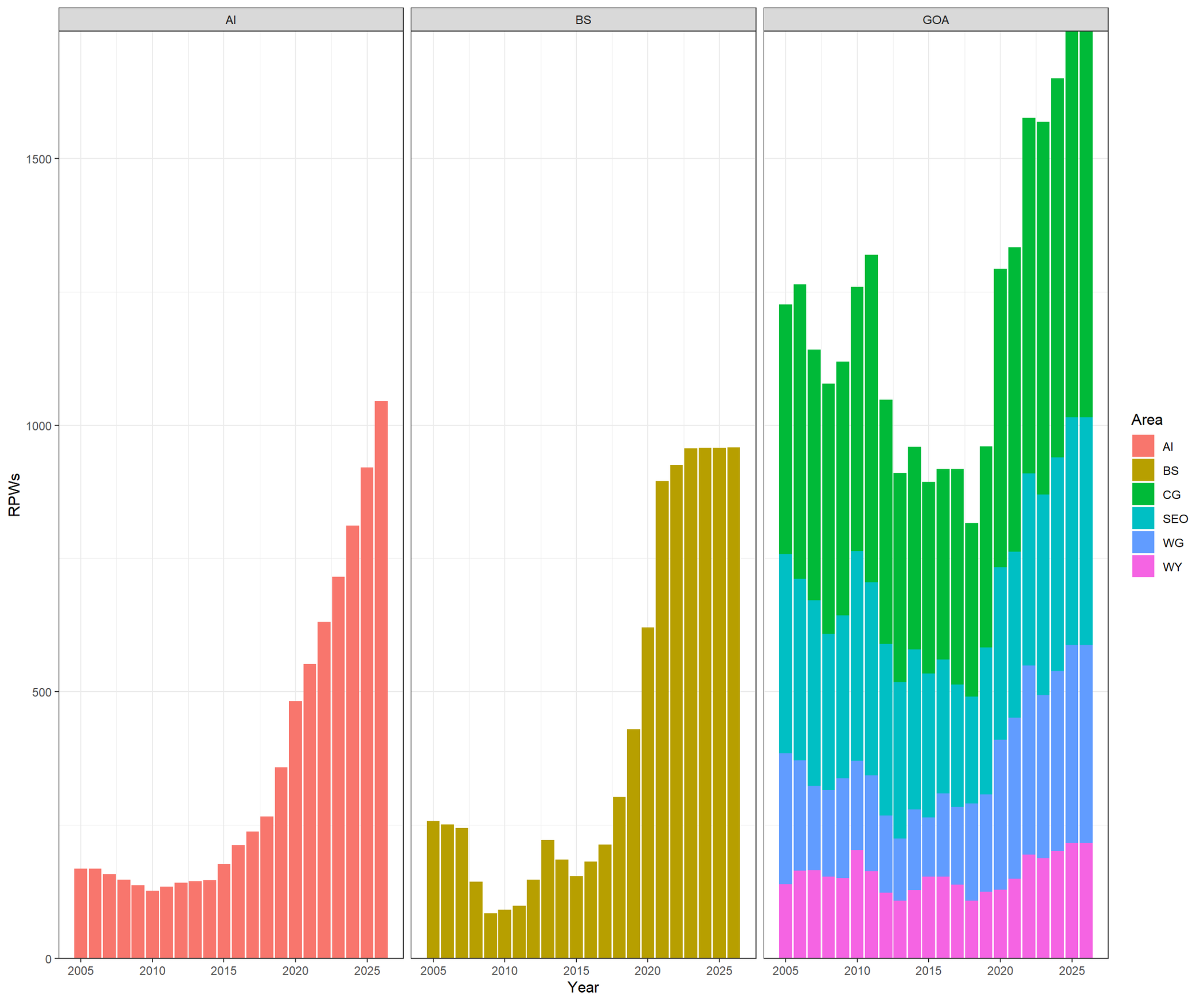

AI
BS
CG
SEO
WG
WY
1.00
0.95
0.80
0.58
0.92
0.66
1.00
0.78
0.53
0.93
0.57
1.00
0.85
0.87
0.80
1.00
0.68
0.80
1.00
0.67
1.00
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

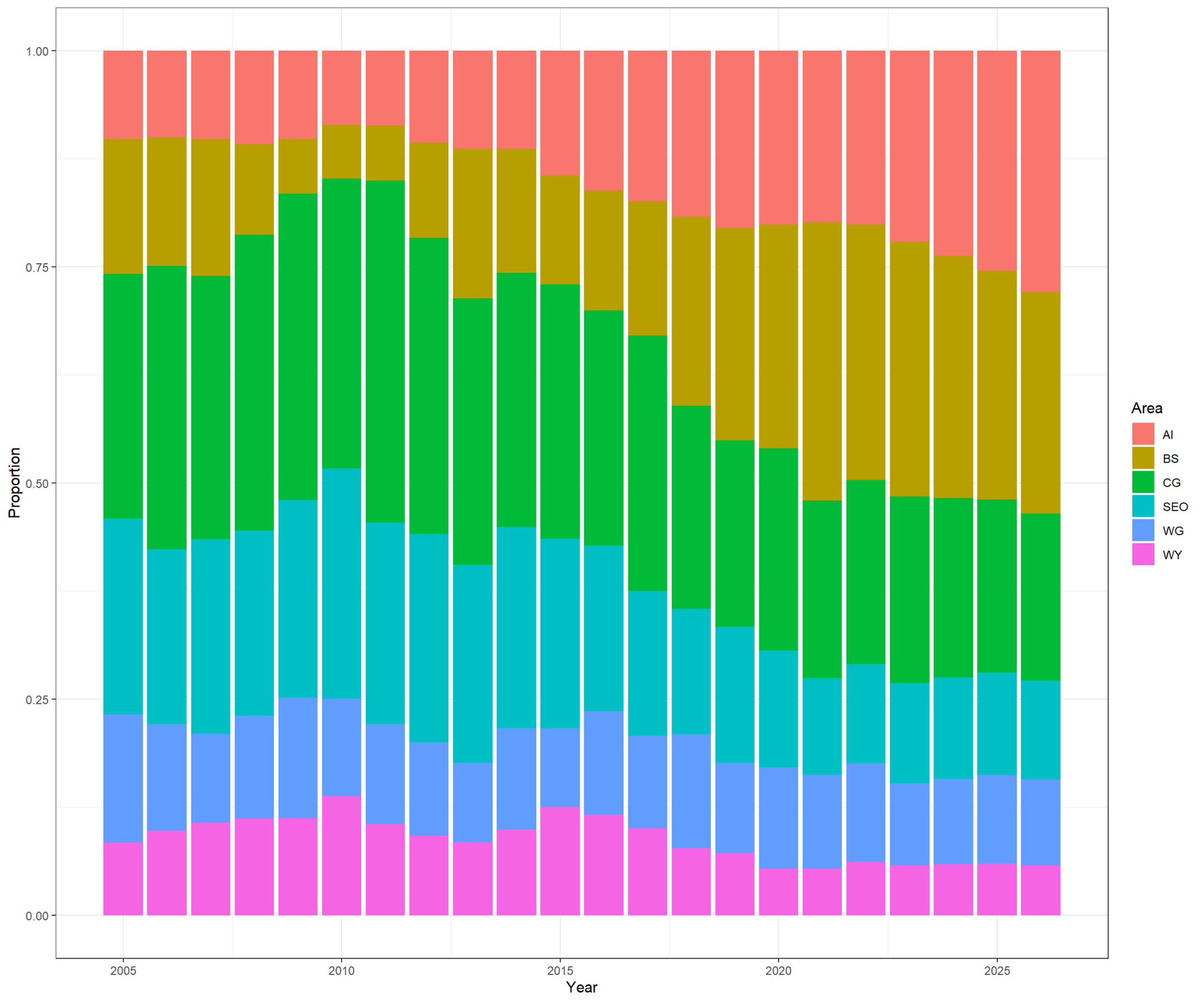

AI
BS
CG
SEO
WG
WY
Biomass
0
500
1,000
2005
2010
2015
2020
2025

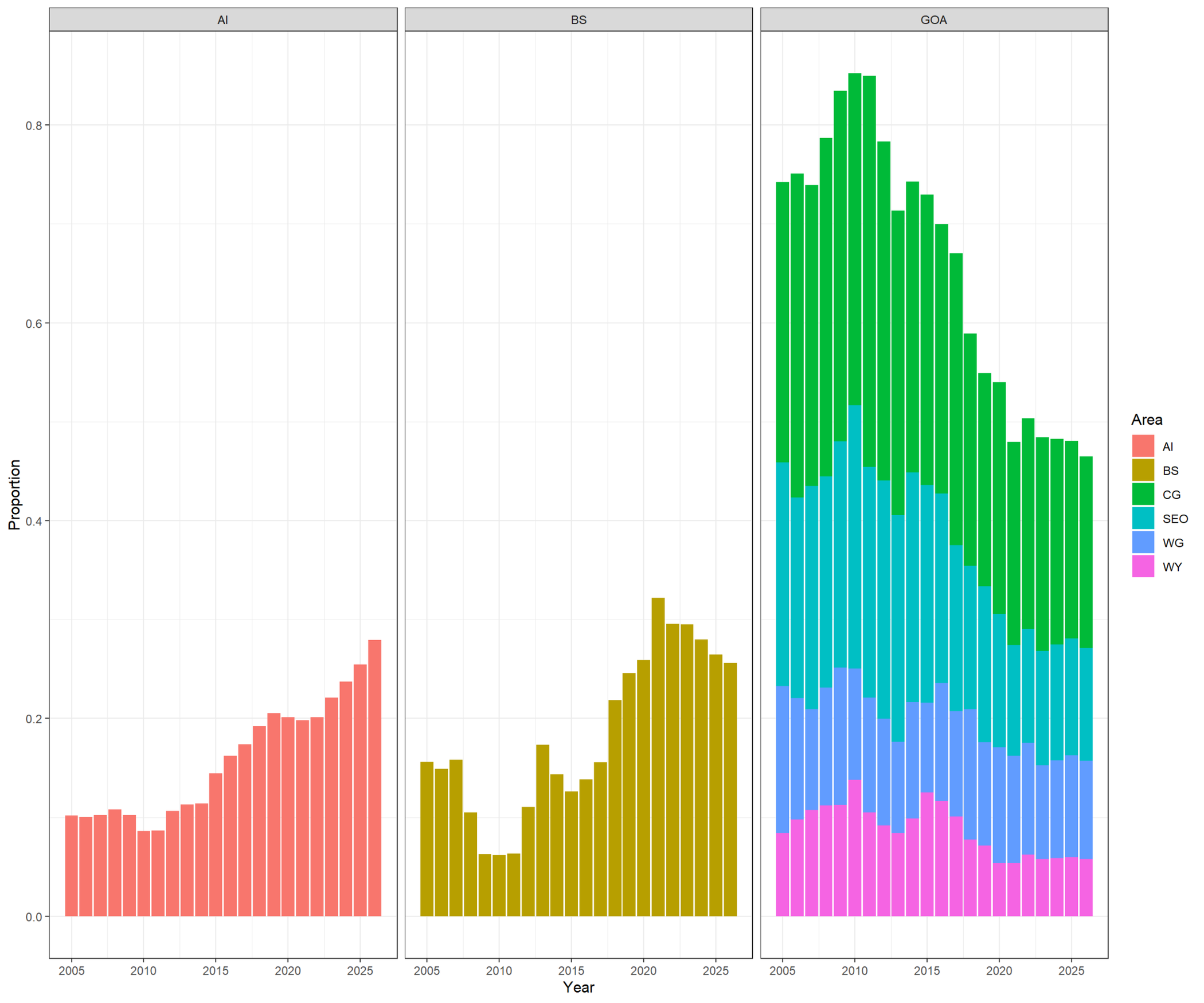

Biomass
4,000
3,000
2,000
1,000
2005
2010
2015
2020
2025

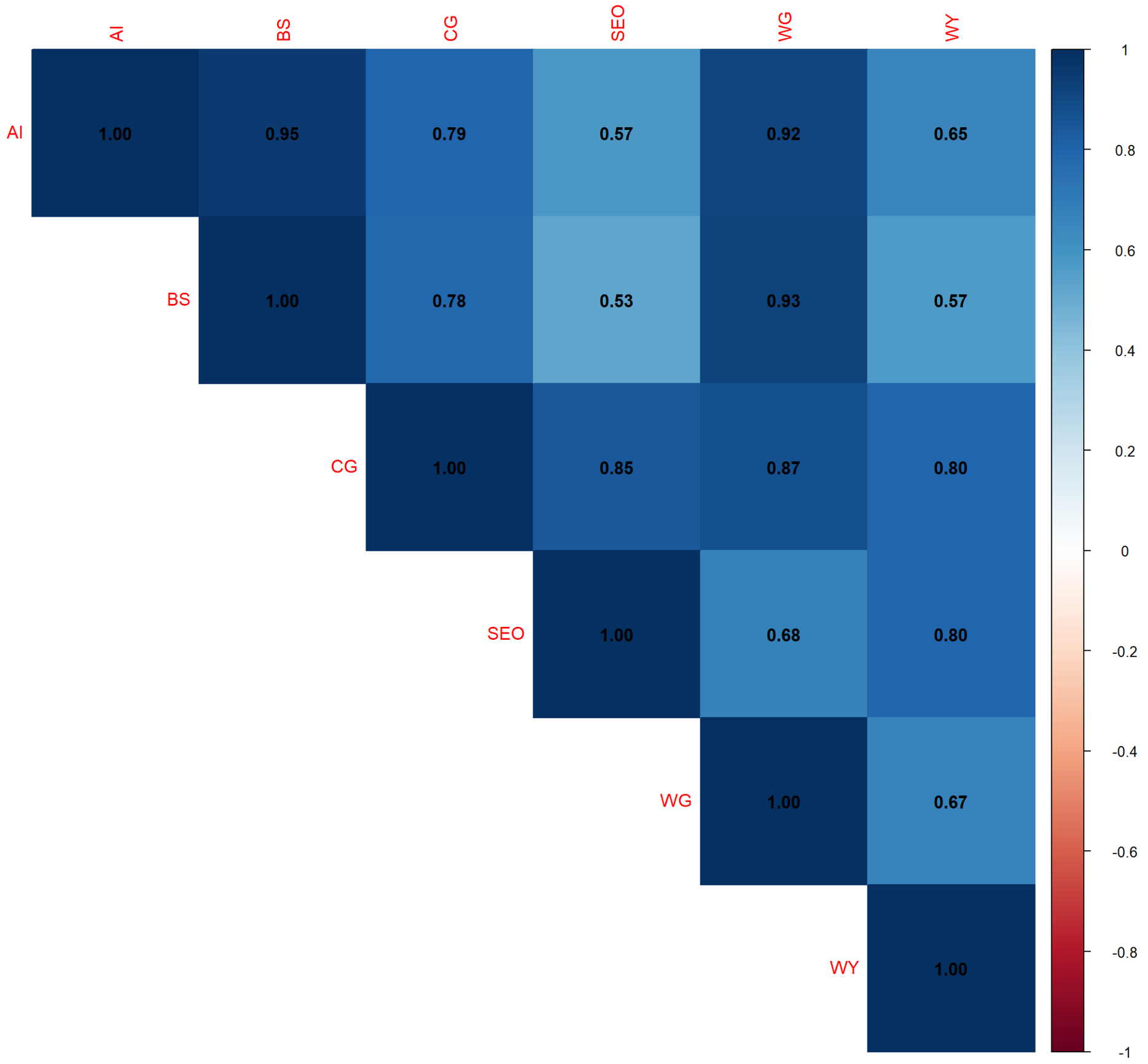

Proportion biomass by strata
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
WY
WG
SEO
AI
CG
BS


RPWs
3000
2000
1000
0
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


AI
BS
GOA
RPWs
0
500
1000
1500
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


1.00
0.75
0.50
0.25
0.00
Proportion
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


AI
BS
GOA
Proportion
0.0
0.2
0.4
0.6
0.8
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


AI
BS
CG
SEO
WG
WY
1.00
0.95
0.79
0.57
0.92
0.65
1.00
0.78
0.53
0.93
0.57
1.00
0.85
0.87
0.80
1.00
0.68
0.80
1.00
0.67
1.00
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

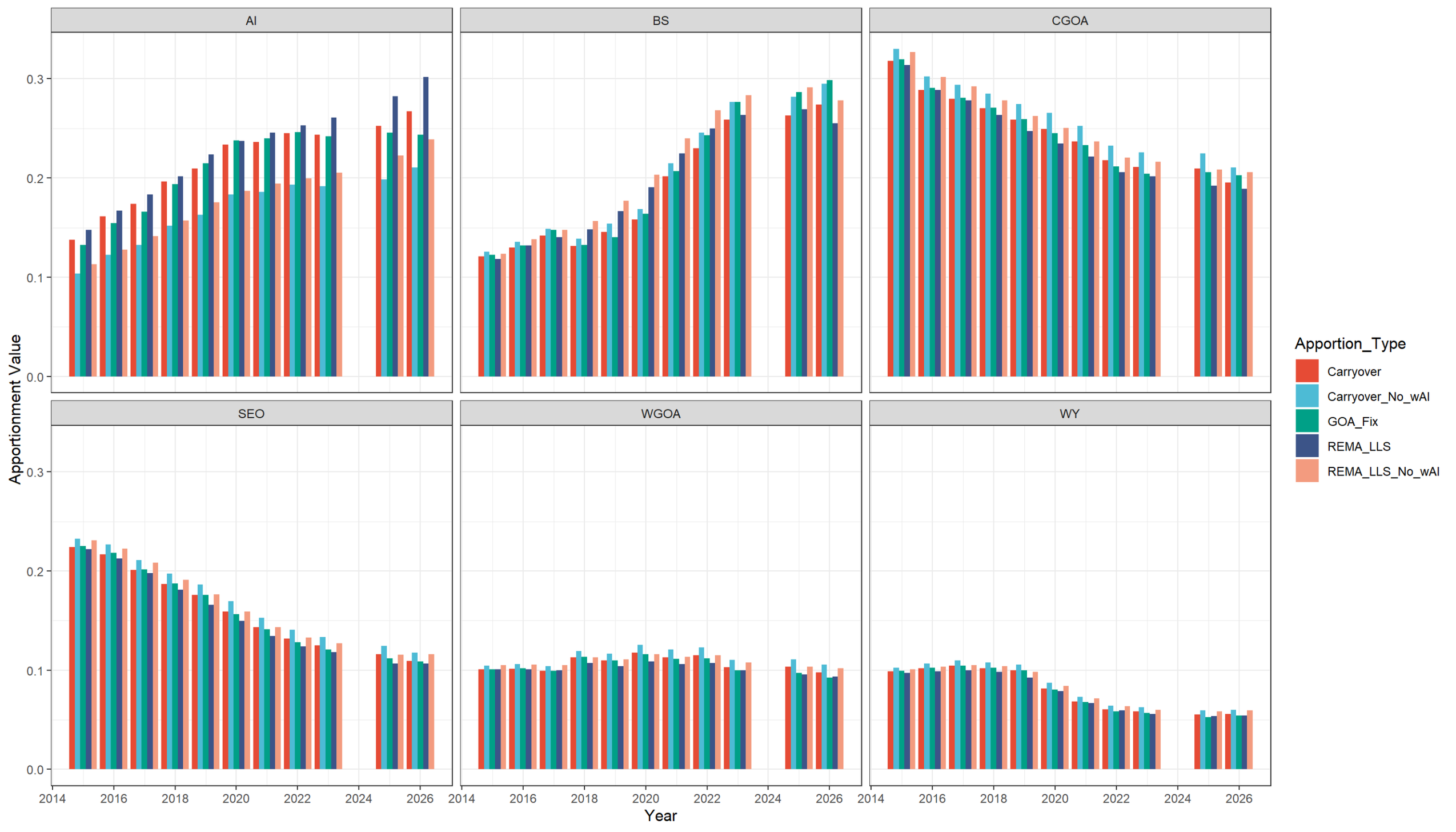
Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

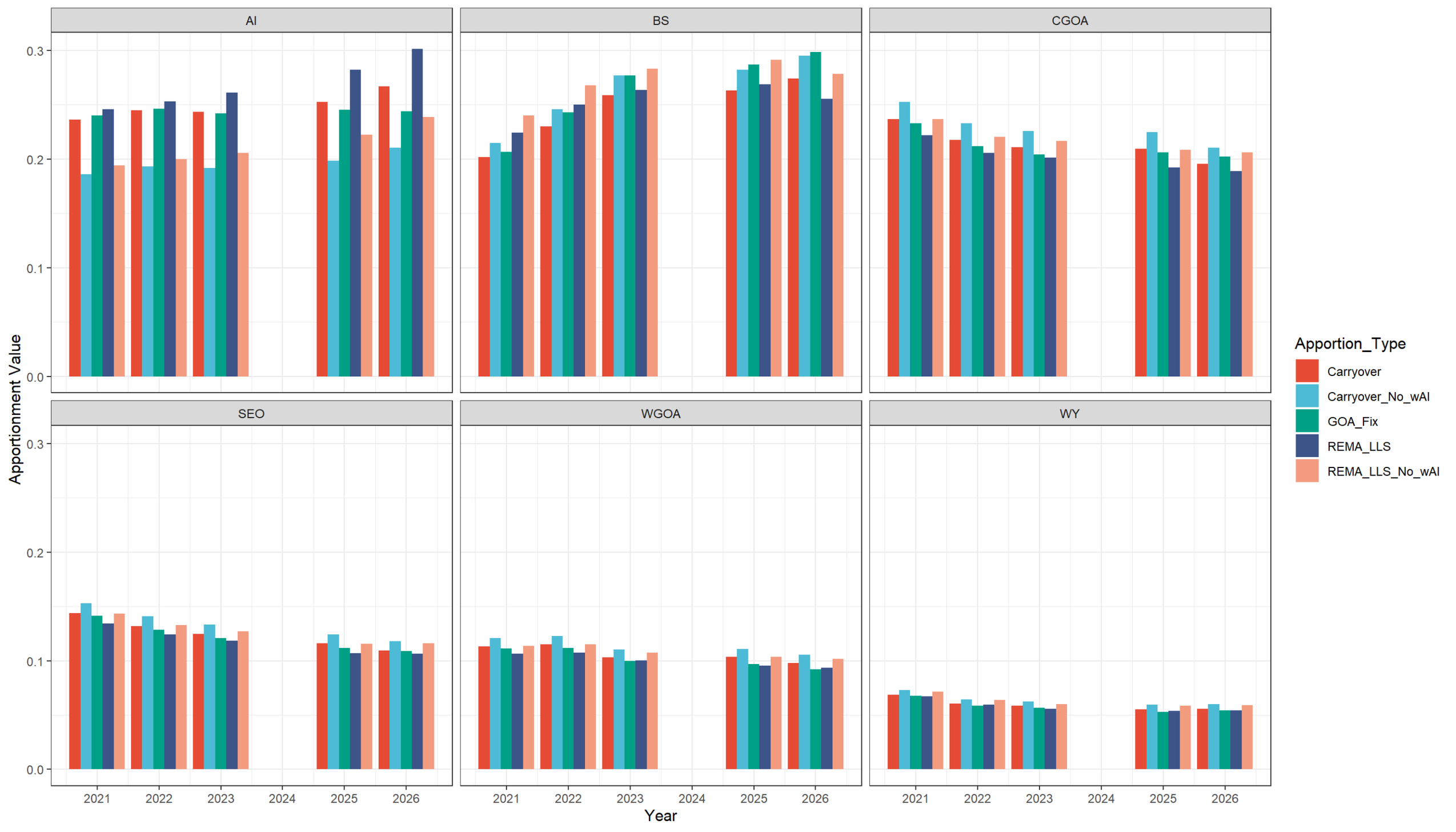
Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	11516	7678	19194	4223	8061	2303	4607	38388
2027	REMA_LLS_No_wAI	10749	8829	18810	3839	8061	2303	4607	38388
2027	Carryover	10365	9980	18042	3839	7678	2303	4223	38388
2027	REMA_LLS	9981	11516	16891	3455	7294	1919	4223	38388
2027	GOA_Fix	11516	9213	17658	3455	7678	1919	4607	38388
2028	Carryover_No_wAI	11467	7644	19111	4204	8027	2293	4587	38222
2028	REMA_LLS_No_wAI	10702	8791	18729	3822	8027	2293	4587	38222
2028	Carryover	10320	9938	17964	3822	7644	2293	4205	38222
2028	REMA_LLS	9938	11467	16818	3440	7262	1911	4204	38222
2028	GOA_Fix	11467	9173	17582	3440	7644	1911	4587	38222

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	-12	-33	-15	-10	-16	-13	-18	-18
2027	REMA_LLS_No_wAI	-18	-23	-17	-18	-16	-13	-18	-18
2027	Carryover	-20	-13	-20	-18	-20	-13	-24	-18
2027	REMA_LLS	-23	1	-25	-26	-24	-28	-24	-18
2027	GOA_Fix	-12	-19	-22	-26	-20	-28	-18	-18
2028	Carryover_No_wAI	-12	-33	-15	-10	-17	-14	-18	-19
2028	REMA_LLS_No_wAI	-18	-23	-17	-18	-17	-14	-18	-19
2028	Carryover	-21	-13	-20	-18	-21	-14	-25	-19
2028	REMA_LLS	-24	0	-25	-27	-25	-28	-25	-19
2028	GOA_Fix	-12	-20	-22	-27	-21	-28	-18	-19

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	11516	2028	BS	NA	11467
2026	AI	NA	11421	7440	632	2027	AI	NA	7678	2028	AI	NA	7644
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	19194	2028	GOA	NA	19111
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	4223	2028	WGOA	NA	4204
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	8061	2028	CGOA	NA	8027
2026	WY	NA	2652	2652	2271	2027	WY	NA	2303	2028	WY	NA	2293
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	4607	2028	SEO	NA	4587
2026	Total	57797	47008	38986	22161	2027	Total	45179	38388	2028	Total	44983	38222

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	267748	NA	19194	NA	NA
GOA	2028	265325	NA	19111	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	146613	NA	11516	NA	NA
BS	2028	145286	NA	11467	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	161408	NA	7678	NA	NA
AI	2028	159948	NA	7644	NA	NA


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI